Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r")
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as f
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: ['r_NH3_wl_2h_D2_energy', 'r_NH3_wl_2h_D2_energy_rel', 'r_NH3_wl_4h_D3_energy', 'r_NH3_wl_4h_D3_energy_rel', 'r_temperature_wl_2h_D2_energy', 'r_temperature_wl_2h_D2_energy_rel', 'r_temperature_wl_4h_D3_energy', 'r_temperature_wl_4h_D3_energy_rel', 'r_umidity_wl_2h_D2_energy', 'r_umidity_wl_2h_D2_energy_rel', 'r_umidity_wl_4h_D3_energy', 'r_umidity_wl_4h_D3_energy_rel']

1.2 Metadata

Nombre de lignes df	7953
Nombre de lignes df clean	7953
Nombre de features	62
% de NaN	0.0
Taille dataset entraînement	5567
Taille dataset test	2385
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_NH3_wl_2h_A_energy_rel, r_NH3_wl_3h_D1_energy, r_NH3_wl_3h_D2_energy, r_NH3_wl_3h_A_energy, r_NH3_wl_3h_A_rms, r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_D2_energy_rel, r_NH3_wl_3h_A_energy_rel, r_NH3_wl_4h_D1_energy, r_NH3_wl_4h_D2_energy, r_NH3_wl_4h_A_energy, r_NH3_wl_4h_A_rms, r_NH3_wl_4h_D1_energy_rel, r_NH3_wl_4h_D2_energy_rel, r_NH3_wl_4h_A_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy_rel, r_temperature_wl_3h_D1_energy, r_temperature_wl_3h_D2_energy, r_temperature_wl_3h_A_energy, r_temperature_wl_3h_A_rms, r_temperature_wl_3h_D1_energy_rel, r_temperature_wl_3h_D2_energy_rel, r_temperature_wl_3h_A_energy_rel, r_temperature_wl_4h_D1_energy, r_temperature_wl_4h_D2_energy, r_temperature_wl_4h_A_energy, r_temperature_wl_4h_A_rms, r_temperature_wl_4h_D1_energy_rel, r_temperature_wl_4h_D2_energy_rel, r_temperature_wl_4h_A_energy_rel, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_3h_D1_energy, r_umidity_wl_3h_D2_energy, r_umidity_wl_3h_A_energy, r_umidity_wl_3h_A_rms, r_umidity_wl_3h_D1_energy_rel, r_umidity_wl_3h_D2_energy_rel, r_umidity_wl_3h_A_energy_rel, r_umidity_wl_4h_D1_energy, r_umidity_wl_4h_D2_energy, r_umidity_wl_4h_A_energy, r_umidity_wl_4h_A_rms, r_umidity_wl_4h_D1_energy_rel, r_umidity_wl_4h_D2_energy_rel, r_umidity_wl_4h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

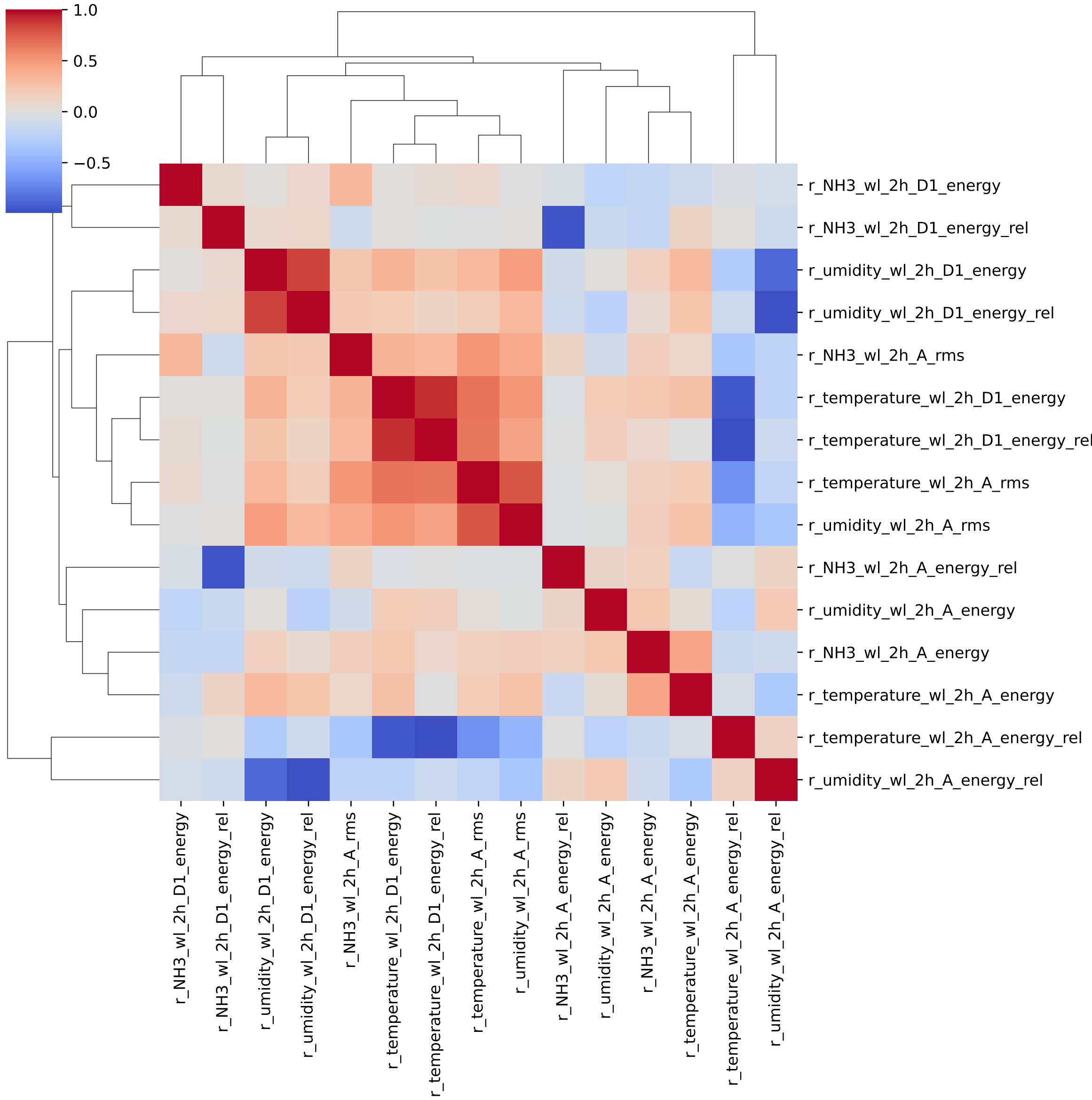
cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_wl_2h_D1_energy ; **cluster 7:** r_NH3_wl_2h_A_energy ; **cluster 8:** r_NH3_wl_2h_A_rms ; **cluster 9:** r_NH3_wl_2h_A_energy_rel, r_NH3_wl_2h_D1_energy_rel ; **cluster 10:** r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy_rel ; **cluster 11:** r_temperature_wl_2h_A_energy ; **cluster 12:** r_temperature_wl_2h_A_rms ; **cluster 13:** r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_2h_D1_energy_rel ; **cluster 14:** r_umidity_wl_2h_A_energy ; **cluster 15:** r_umidity_wl_2h_A_rms ; **cluster**

16: r_NH3_wl_3h_D1_energy ; **cluster 17:** r_NH3_wl_3h_D2_energy ; **cluster 18:** r_NH3_wl_3h_A_energy ; **cluster 19:** r_NH3_wl_3h_A_rms ; **cluster 20:** r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_A_energy_rel ; **cluster 21:** r_NH3_wl_3h_D2_energy_rel ; **cluster 22:** r_temperature_wl_3h_A_energy_rel, r_temperature_wl_3h_D1_energy_rel, r_temperature_wl_3h_D2_energy_rel, r_temperature_wl_3h_D1_energy, r_temperature_wl_3h_D2_energy ; **cluster 23:** r_umidity_wl_3h_A_energy, r_temperature_wl_3h_A_energy ; **cluster 24:** r_temperature_wl_3h_A_rms, r_umidity_wl_3h_A_rms ; **cluster 25:** r_umidity_wl_3h_A_energy_rel, r_umidity_wl_3h_D1_energy_rel, r_umidity_wl_3h_D1_energy ; **cluster 26:** r_umidity_wl_3h_D2_energy, r_umidity_wl_3h_D2_energy_rel ; **cluster 27:** r_NH3_wl_4h_D1_energy ; **cluster 28:** r_NH3_wl_4h_D2_energy ; **cluster 29:** r_NH3_wl_4h_A_energy ; **cluster 30:** r_NH3_wl_4h_A_rms ; **cluster 31:** r_NH3_wl_4h_D1_energy_rel, r_NH3_wl_4h_A_energy_rel, r_NH3_wl_4h_D2_energy_rel ; **cluster 32:** r_temperature_wl_4h_D1_energy, r_temperature_wl_4h_D2_energy, r_temperature_wl_4h_A_energy_rel, r_temperature_wl_4h_D1_energy_rel, r_temperature_wl_4h_D2_energy_rel ; **cluster 33:** r_temperature_wl_4h_A_energy ; **cluster 34:** r_umidity_wl_4h_A_rms, r_temperature_wl_4h_A_rms ; **cluster 35:** r_umidity_wl_4h_D1_energy ; **cluster 36:** r_umidity_wl_4h_D2_energy, r_umidity_wl_4h_D2_energy_rel ; **cluster 37:** r_umidity_wl_4h_A_energy ; **cluster 38:** r_umidity_wl_4h_D1_energy_rel, r_umidity_wl_4h_A_energy_rel ;

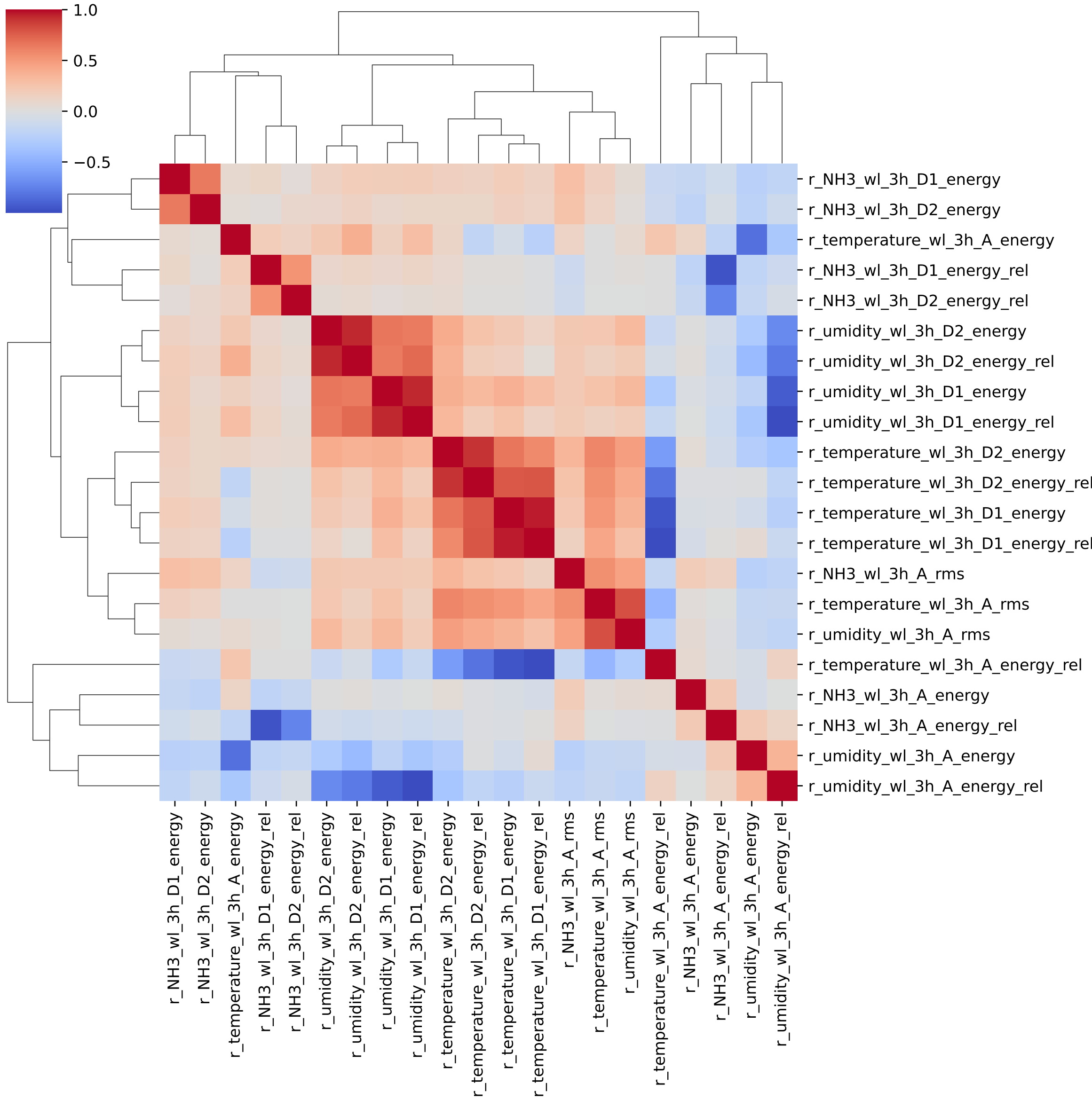
2.2 Features selection

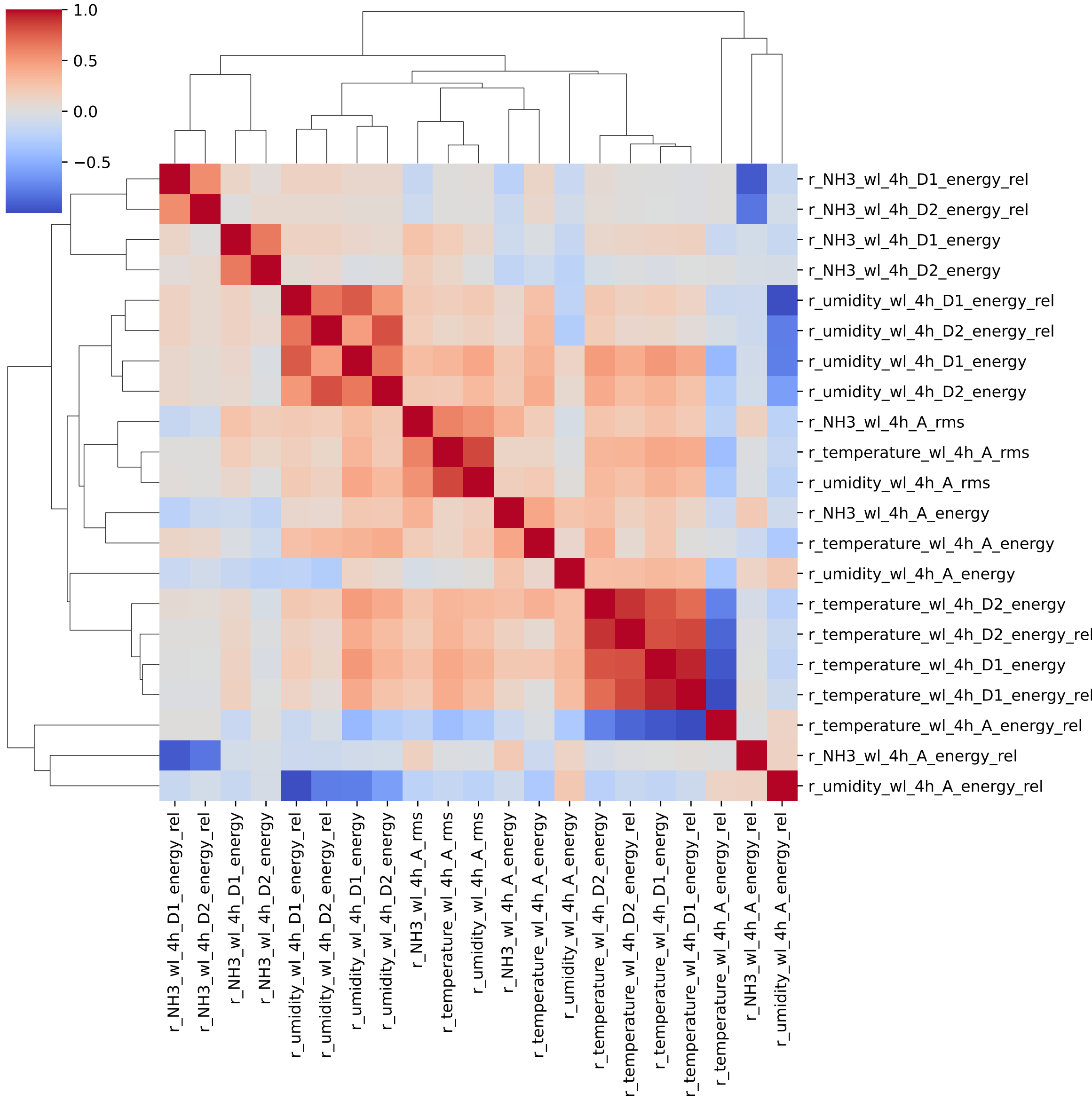
Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_3h_D1_energy, r_NH3_wl_3h_D2_energy, r_NH3_wl_3h_A_energy, r_NH3_wl_3h_A_rms, r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_D2_energy_rel, r_temperature_wl_3h_A_energy_rel, r_umidity_wl_3h_A_energy, r_temperature_wl_3h_A_rms, r_umidity_wl_3h_A_energy_rel, r_umidity_wl_3h_D2_energy, r_NH3_wl_4h_D1_energy, r_NH3_wl_4h_D2_energy, r_NH3_wl_4h_A_energy, r_NH3_wl_4h_A_rms, r_NH3_wl_4h_D1_energy_rel, r_temperature_wl_4h_D1_energy, r_temperature_wl_4h_A_energy, r_umidity_wl_4h_A_rms, r_umidity_wl_4h_D1_energy, r_umidity_wl_4h_D2_energy, r_umidity_wl_4h_A_energy, r_umidity_wl_4h_D1_energy_rel



1.0
0.5
0.0
−0.5
r_NH3_wl_2h_D1_energy
r_NH3_wl_2h_D1_energy_rel
r_umidity_wl_2h_D1_energy
r_umidity_wl_2h_D1_energy_rel
r_NH3_wl_2h_A_rms
r_temperature_wl_2h_D1_energy
r_temperature_wl_2h_D1_energy_rel
r_temperature_wl_2h_A_rms
r_umidity_wl_2h_A_rms
r_NH3_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy
r_NH3_wl_2h_A_energy
r_temperature_wl_2h_A_energy
r_temperature_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy_rel


1.0
0.5
0.0
−0.5
r_NH3_wl_3h_D1_energy
r_NH3_wl_3h_D2_energy
r_temperature_wl_3h_A_energy
r_NH3_wl_3h_D1_energy_rel
r_NH3_wl_3h_D2_energy_rel
r_umidity_wl_3h_D2_energy
r_umidity_wl_3h_D2_energy_rel
r_umidity_wl_3h_D1_energy
r_umidity_wl_3h_D1_energy_rel
r_temperature_wl_3h_D2_energy
r_temperature_wl_3h_D2_energy_rel
r_temperature_wl_3h_D1_energy
r_temperature_wl_3h_D1_energy_rel
r_NH3_wl_3h_A_rms
r_temperature_wl_3h_A_rms
r_umidity_wl_3h_A_rms
r_temperature_wl_3h_A_energy_rel
r_NH3_wl_3h_A_energy
r_NH3_wl_3h_A_energy_rel
r_umidity_wl_3h_A_energy
r_umidity_wl_3h_A_energy_rel



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_3h_D1_energy, r_NH3_wl_3h_D2_energy, r_NH3_wl_3h_A_energy, r_NH3_wl_3h_A_rms, r_NH3_wl_3h_D1_energy_rel, r_NH3_wl_3h_D2_energy_rel, r_temperature_wl_3h_A_energy_rel, r_umidity_wl_3h_A_energy, r_temperature_wl_3h_A_rms, r_umidity_wl_3h_A_energy_rel, r_umidity_wl_3h_D2_energy, r_NH3_wl_4h_D1_energy, r_NH3_wl_4h_D2_energy, r_NH3_wl_4h_A_energy, r_NH3_wl_4h_A_rms, r_NH3_wl_4h_D1_energy_rel, r_temperature_wl_4h_D1_energy, r_temperature_wl_4h_A_energy, r_umidity_wl_4h_A_rms, r_umidity_wl_4h_D1_energy, r_umidity_wl_4h_D2_energy, r_umidity_wl_4h_A_energy, r_umidity_wl_4h_D1_energy_rel

Len(Features_X): 38

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

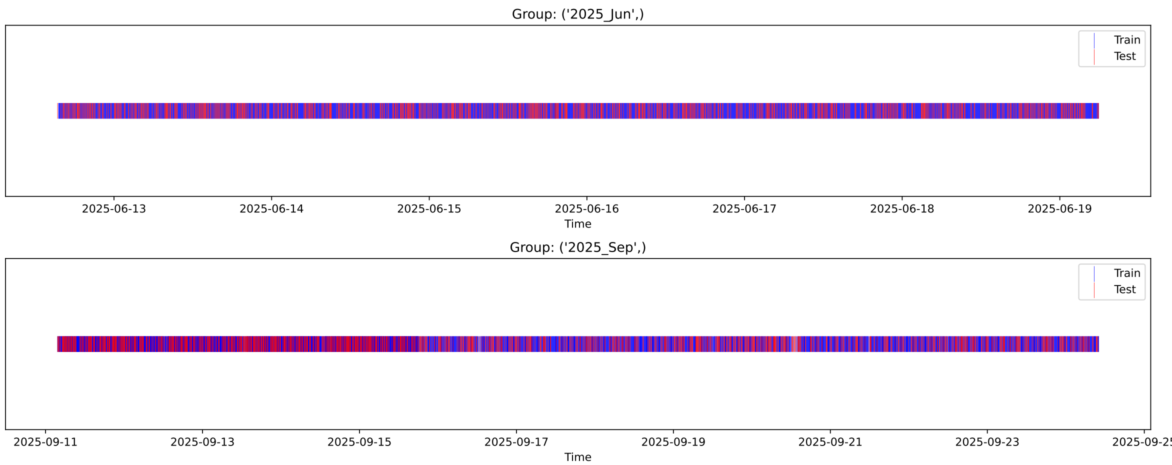
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

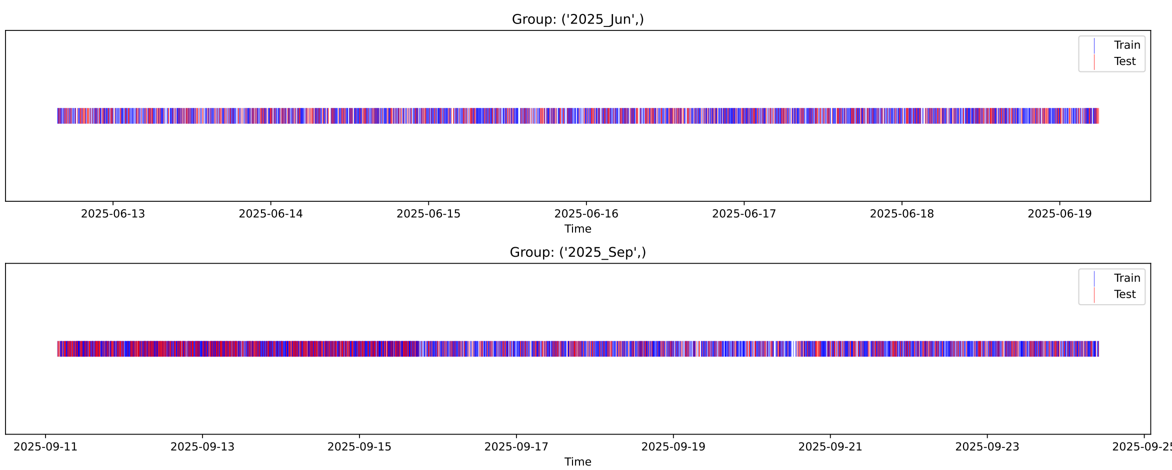
External Split: Train: 5567 samples Test: 2386 samples



3.3.2 Internal CV Folds

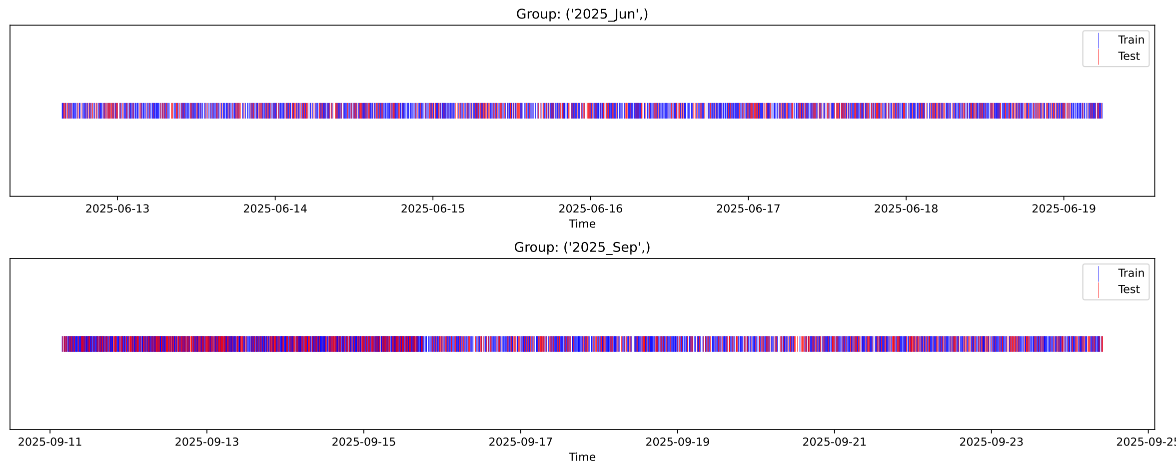
Fold 1/5

Train: 3896 samples Test: 1671 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3896 samples Test: 1671 samples Excluded: 0 samples Window exclusion: 0h

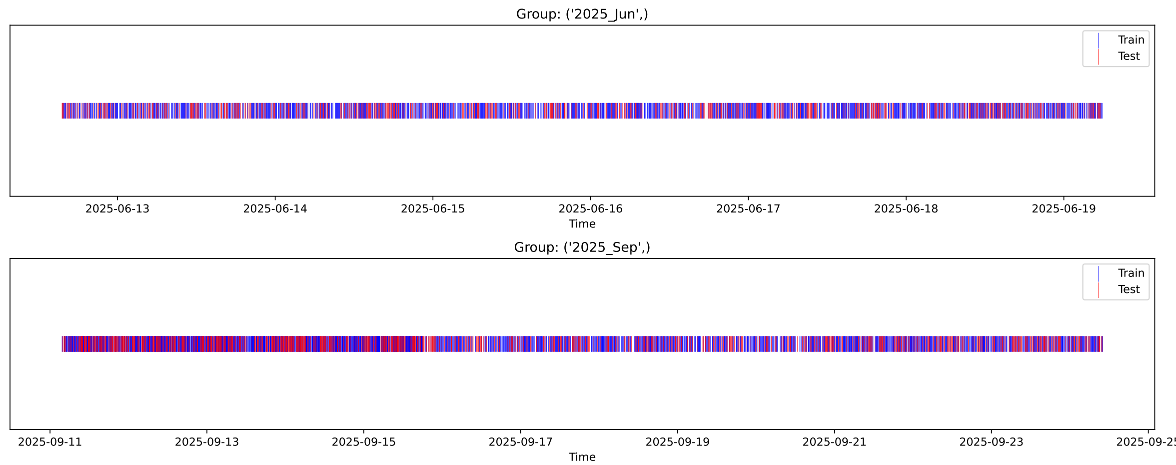


Fold 3/5

Train: 3896 samples Test: 1671 samples Excluded: 0 samples Window exclusion: 0h

Fold 4/5

Train: 3896 samples Test: 1671 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3896 samples Test: 1671 samples Excluded: 0 samples Window exclusion: 0h

3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.351, std=0.790, min=0.959, max=8.682 y_test: mean=2.340, std=0.815, min=1.041, max=13.845

Fold 0 — y_test: mean=2.33, std=0.78, y_train: mean=2.36, std=0.79

Fold 1 — y_test: mean=2.34, std=0.80, y_train: mean=2.35, std=0.79

Fold 2 — y_test: mean=2.34, std=0.78, y_train: mean=2.36, std=0.79

Fold 3 — y_test: mean=2.35, std=0.78, y_train: mean=2.35, std=0.80

Fold 4 — y_test: mean=2.34, std=0.76, y_train: mean=2.35, std=0.80

train scores CV: [0.61277995 0.59556947 0.61514206 0.59338475 0.59765303]

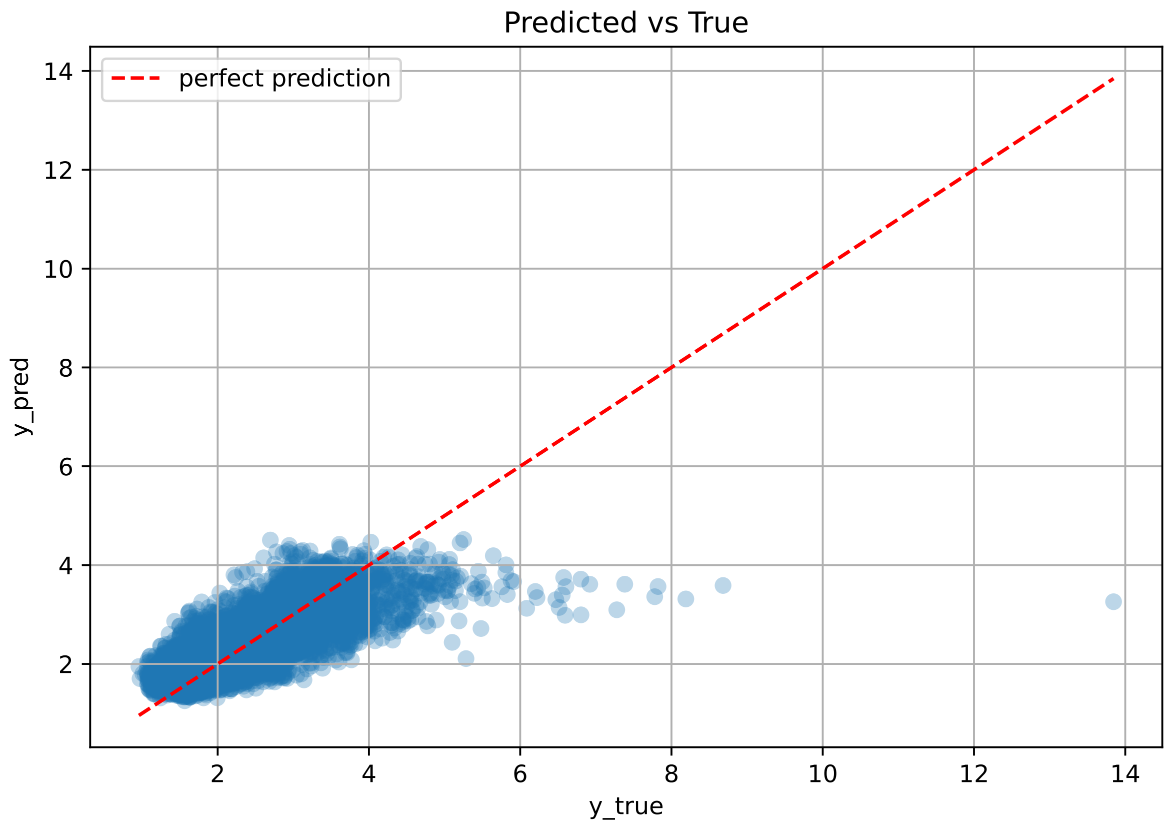
test scores CV: [0.55894624 0.59754957 0.55283465 0.60753109 0.59501571]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.6029	0.5824	0.0221	0.5426	0.5434	-0.039	0.9331	1.0353

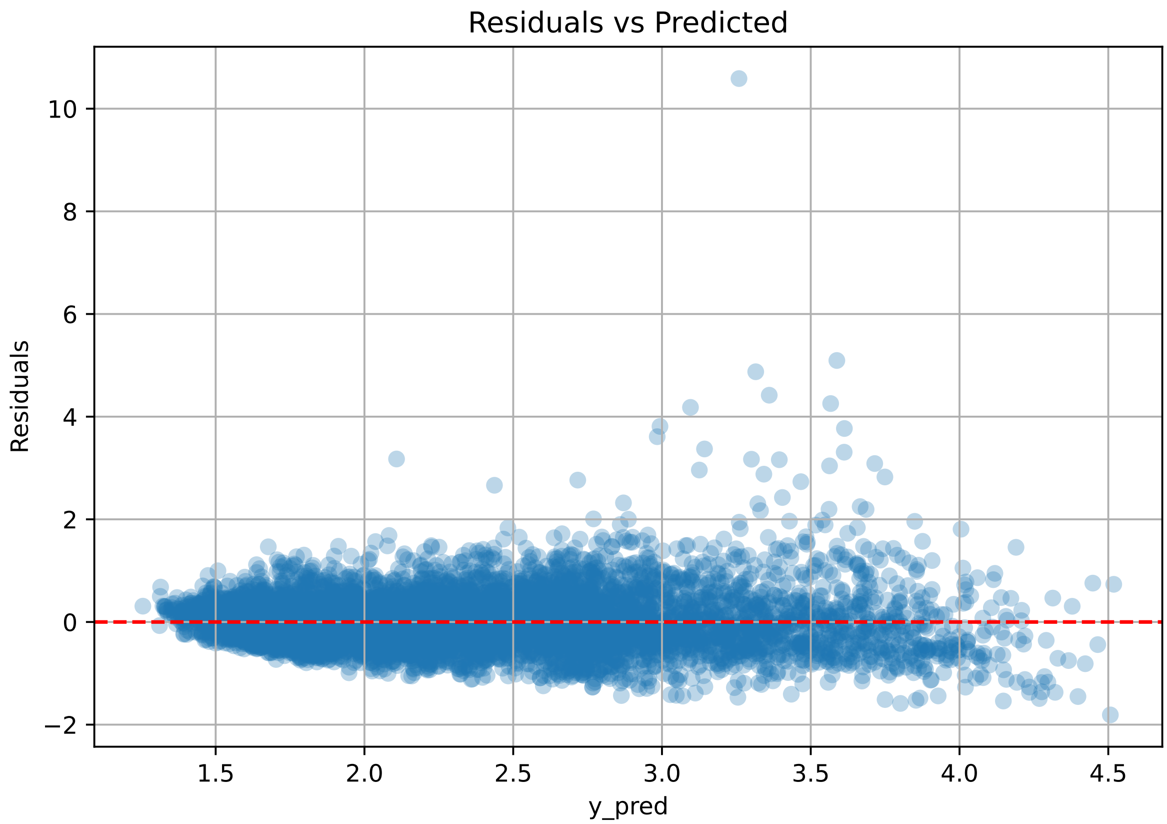
Coefficients:

coef_r_CO2 = -0.0519, coef_r_NH3 = 0.3223, coef_r_NH3_wl_2h_A_energy = -0.1809, coef_r_NH3_wl_2h_A_energy_rel = -0.0231, coef_r_NH3_wl_2h_A_rms = 0.0084, coef_r_NH3_wl_2h_D1_energy = 0.0125, coef_r_NH3_wl_3h_A_energy = -0.1563, coef_r_NH3_wl_3h_A_rms = 0.0610, coef_r_NH3_wl_3h_D1_energy = 0.0510, coef_r_NH3_wl_3h_D1_energy_rel = -0.0321, coef_r_NH3_wl_3h_D2_energy = -0.0158, coef_r_NH3_wl_3h_D2_energy_rel = -0.0263, coef_r_NH3_wl_4h_A_energy = -0.0808, coef_r_NH3_wl_4h_A_rms = -0.0809, coef_r_NH3_wl_4h_D1_energy = 0.0567, coef_r_NH3_wl_4h_D1_energy_rel = -0.0064, coef_r_NH3_wl_4h_D2_energy = -0.0468, coef_r_THI = 0.7120, coef_r_temperature = -0.8212, coef_r_temperature_wl_2h_A_energy = 0.1222, coef_r_temperature_wl_2h_A_rms = -0.0394, coef_r_temperature_wl_2h_D1_energy_rel = 0.0093, coef_r_temperature_wl_3h_A_energy_rel = -0.0349, coef_r_temperature_wl_3h_A_rms = 0.1827, coef_r_temperature_wl_4h_A_energy = -0.0199, coef_r_temperature_wl_4h_D1_energy = 0.0325, coef_r_umidity = -0.9526, coef_r_umidity_wl_2h_A_energy = -0.0153, coef_r_umidity_wl_2h_A_rms = 0.0387, coef_r_umidity_wl_2h_D1_energy = 0.0148, coef_r_umidity_wl_3h_A_energy = 0.4511, coef_r_umidity_wl_3h_A_energy_rel = 0.0198, coef_r_umidity_wl_3h_D2_energy = 0.0100, coef_r_umidity_wl_4h_A_energy = 0.0371, coef_r_umidity_wl_4h_A_rms = -0.0785, coef_r_umidity_wl_4h_D1_energy = 0.0339, coef_r_umidity_wl_4h_D1_energy_rel = -0.0384, coef_r_umidity_wl_4h_D2_energy = 0.0446, intercept = 2.3506,

3.4.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

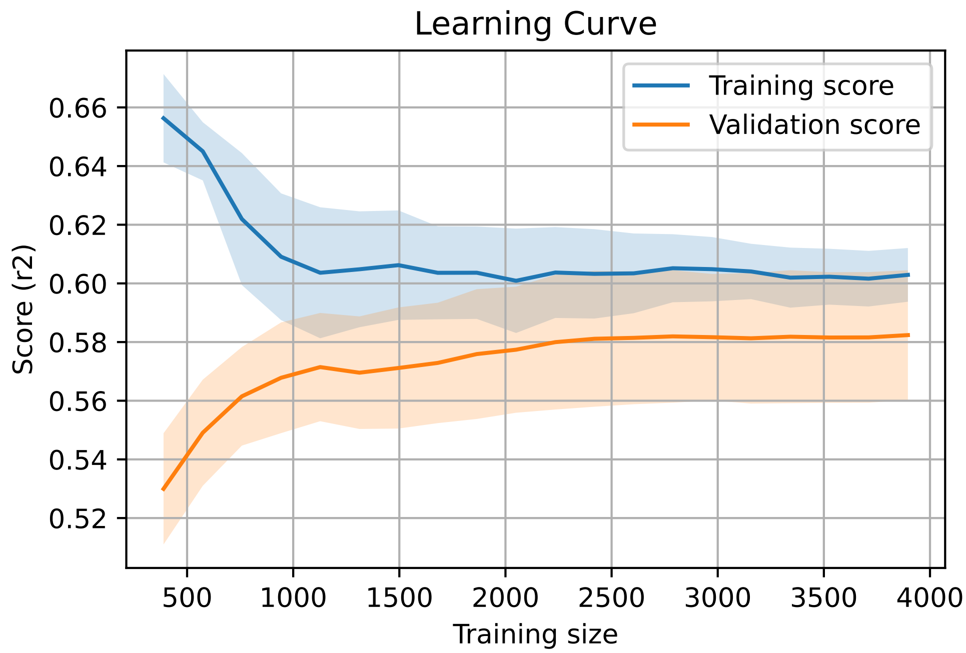
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7953, **Len(training)** : 5568,

Len(training_real) : 3898, **Len(test_of_training)** : 1670, **Len(test)** : 2385

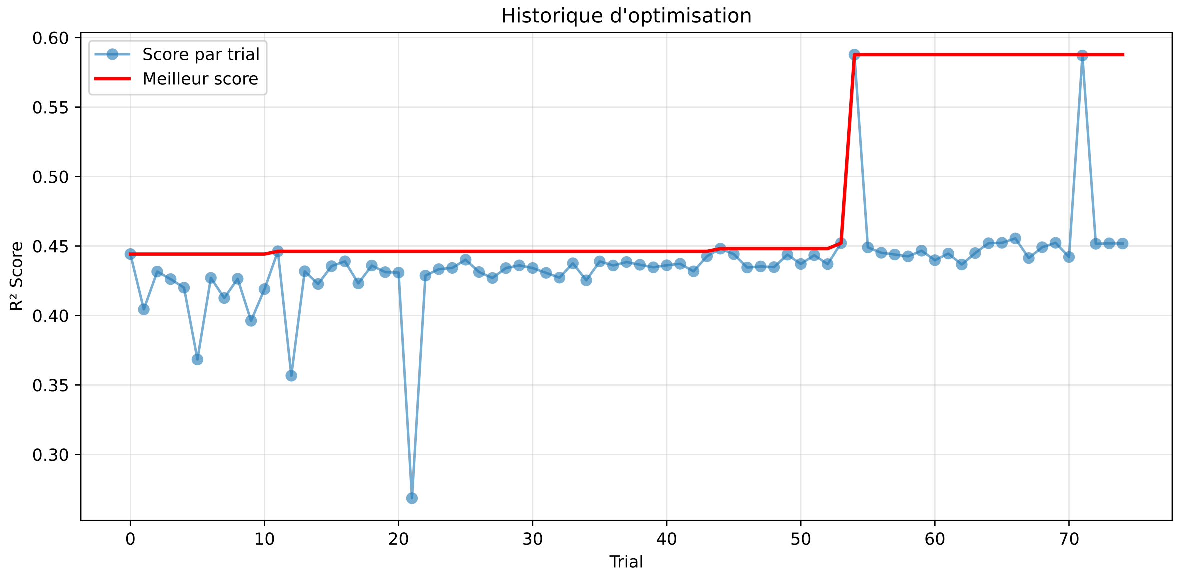


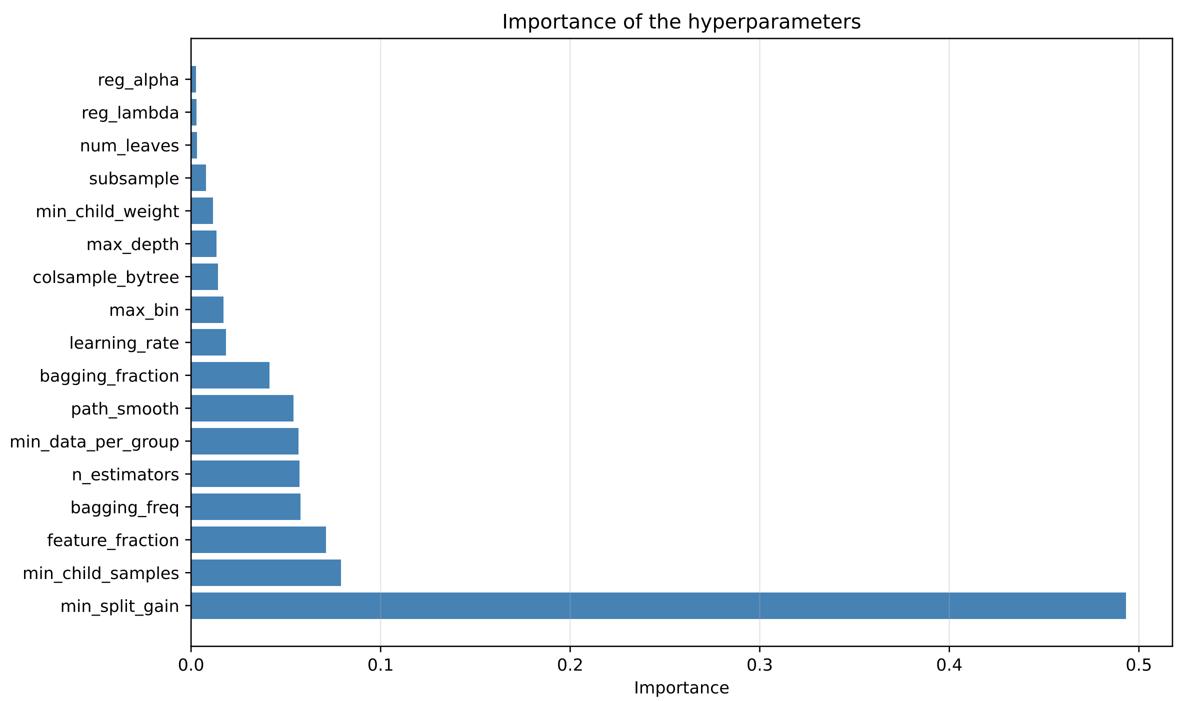
3.5 Model : LGBM

3.5.1 Gridsearch

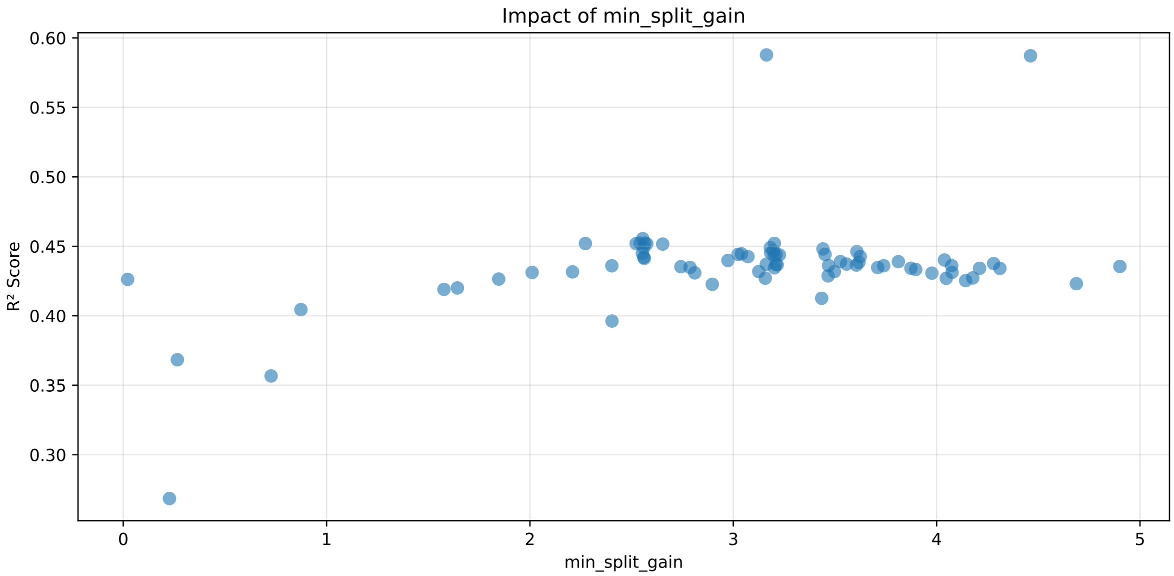
Distribution y_train vs y_test: y_train: mean=2.351, std=0.790, min=0.959, max=8.682 y_test: mean=2.340, std=0.815, min=1.041, max=13.845

==================== TOP Importance FEATURES ==================== feature importance r_CO2 188 r_umidity 168 r_temperature_wl_3h_A_rms 111 r_temperature_wl_2h_A_rms 107 r_temperature_wl_3h_A_energy_rel 102 r_temperature 95 r_NH3_wl_4h_D2_energy 88 r_temperature_wl_2h_D1_energy_rel 68 r_NH3_wl_4h_D1_energy 63 r_NH3_wl_4h_A_energy 61 r_THI 48 r_NH3_wl_3h_D1_energy 42 r_NH3_wl_3h_D2_energy 38 r_umidity_wl_3h_A_energy_rel 37 r_NH3_wl_2h_D1_energy 30 r_NH3_wl_3h_A_energy 29 r_umidity_wl_4h_D1_energy 28 r_umidity_wl_4h_A_energy 28 r_NH3_wl_2h_A_energy 28 r_umidity_wl_2h_A_rms 26 r_NH3 23 r_umidity_wl_3h_A_energy 20 r_temperature_wl_4h_A_energy 20 r_NH3_wl_3h_D2_energy_rel 19 r_umidity_wl_2h_D1_energy 18 r_umidity_wl_3h_D2_energy 16 r_umidity_wl_4h_D2_energy 16 r_NH3_wl_3h_A_rms 15 r_umidity_wl_2h_A_energy 14 r_NH3_wl_4h_A_rms 13 r_temperature_wl_2h_A_energy 12 r_temperature_wl_4h_D1_energy 9 r_umidity_wl_4h_A_rms 8 r_NH3_wl_3h_D1_energy_rel 6 r_NH3_wl_4h_D1_energy_rel 5 r_NH3_wl_2h_A_rms 3 r_umidity_wl_4h_D1_energy_rel 3 r_NH3_wl_2h_A_energy_rel 0

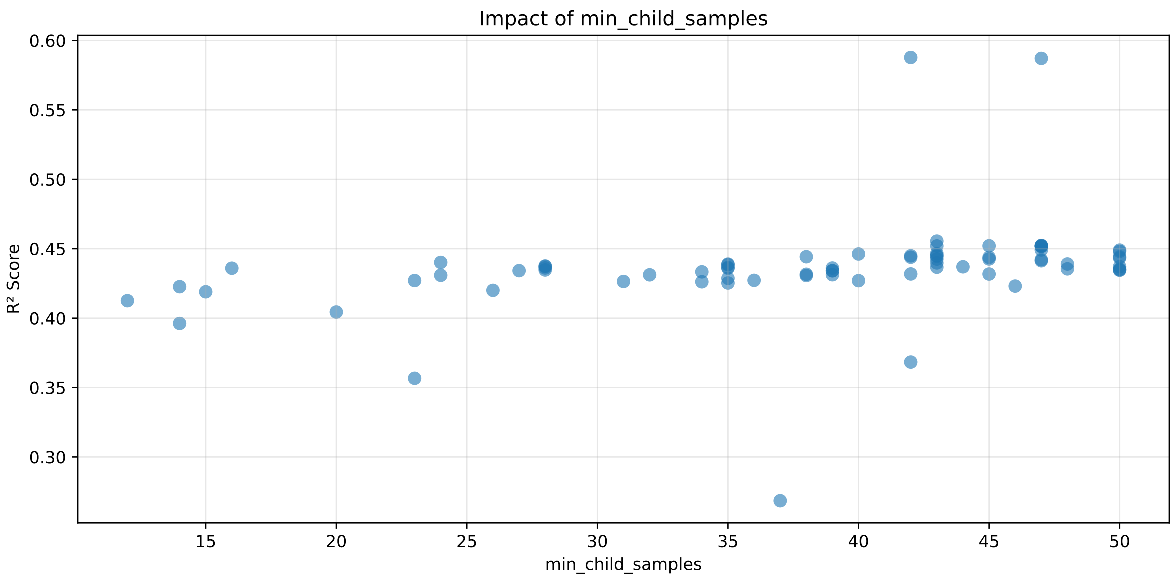




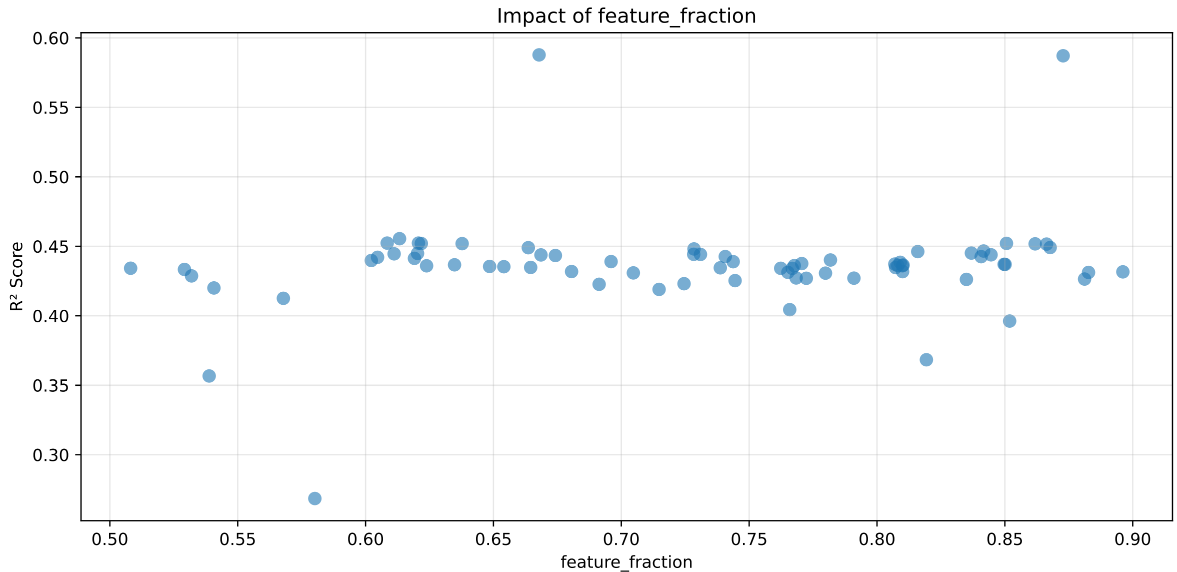
Paramètres avec >5% d'importance : ['min_split_gain', 'min_child_samples', 'feature_fraction', 'bagging_freq', 'n_estimators', 'min_data_per_group', 'path_smooth']

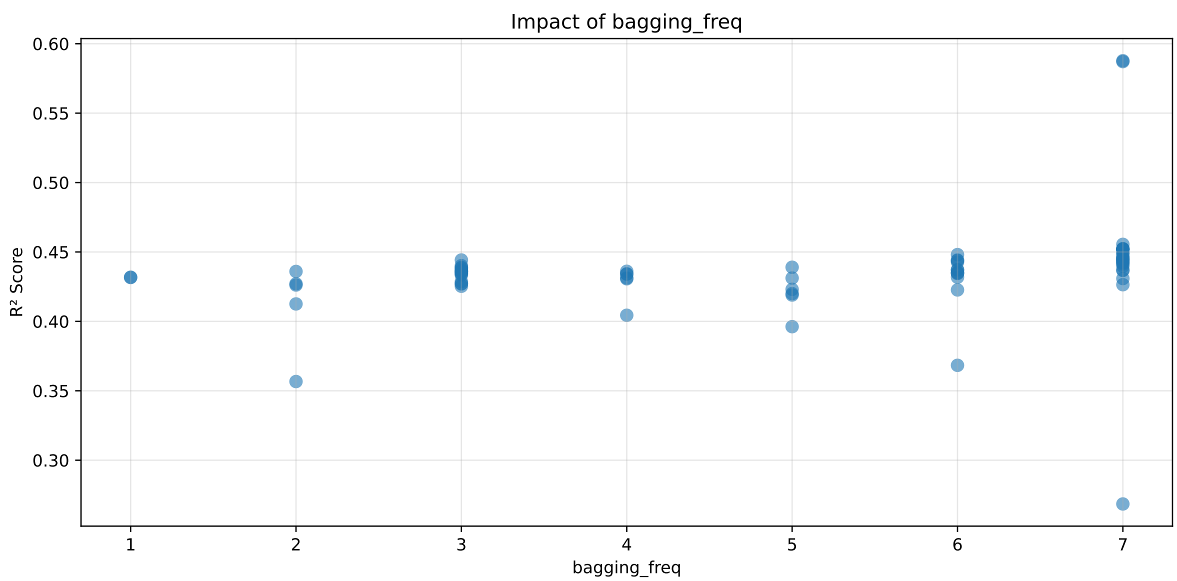


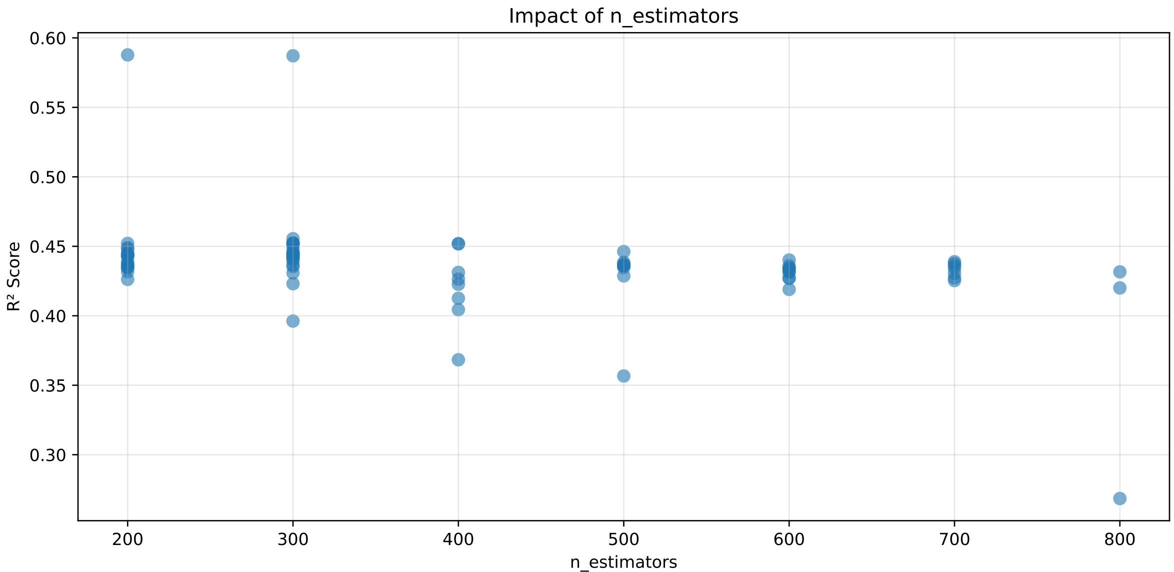
Impact of min_split_gain
R² Score
min_split_gain

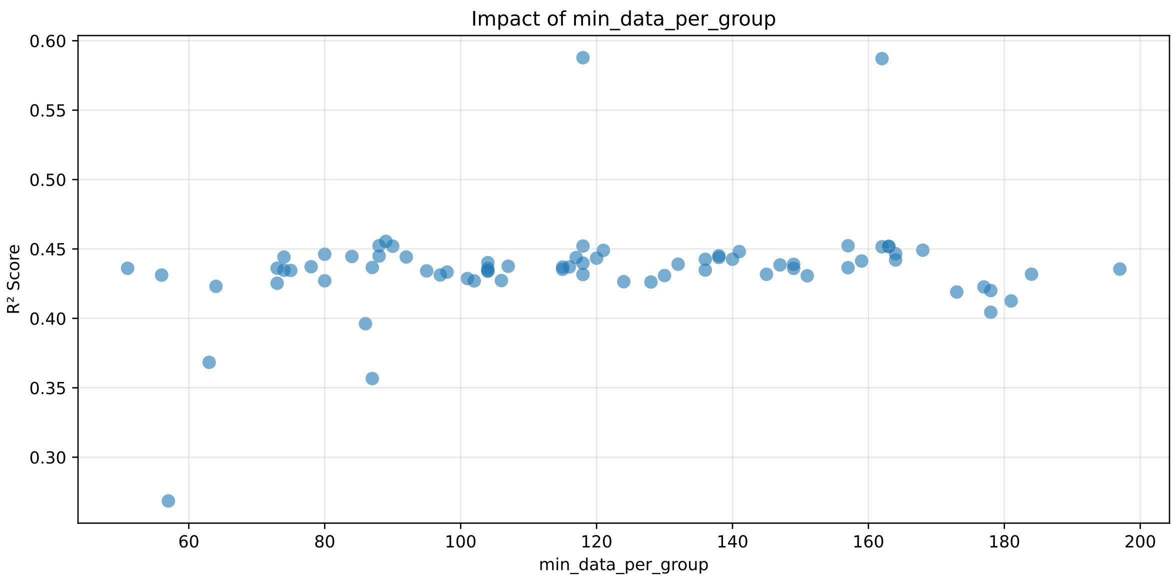


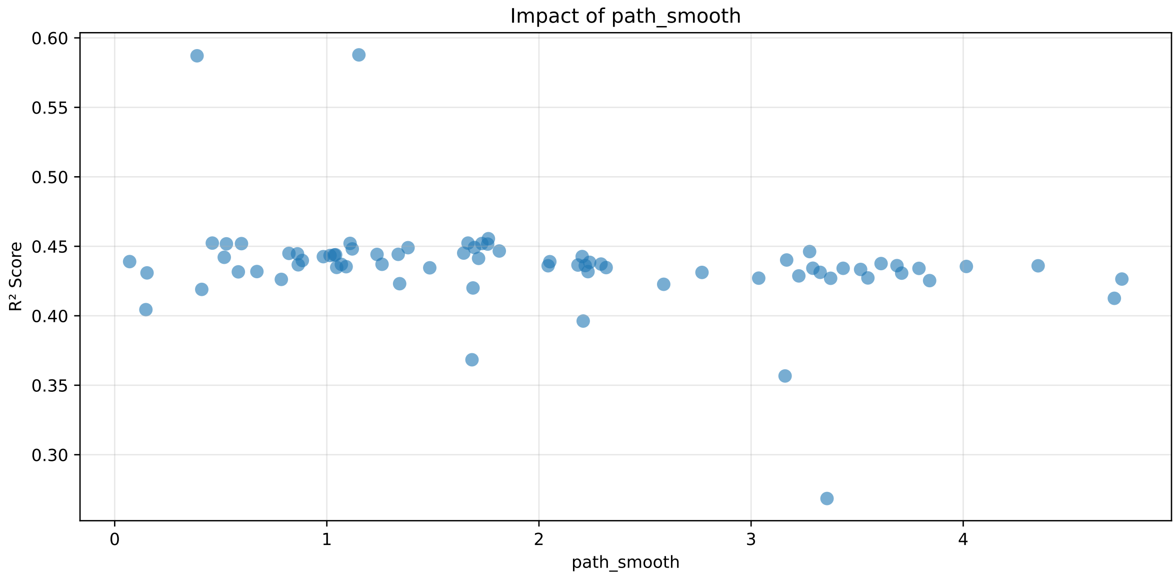
Impact of min_child_samples
R² Score
min_child_samples


Impact of feature_fraction
R² Score
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
feature_fraction


Impact of bagging_freq
R² Score
0.60
0.55
0.50
0.45
0.40
0.35
0.30
1
2
3
4
5
6
7
bagging_freq

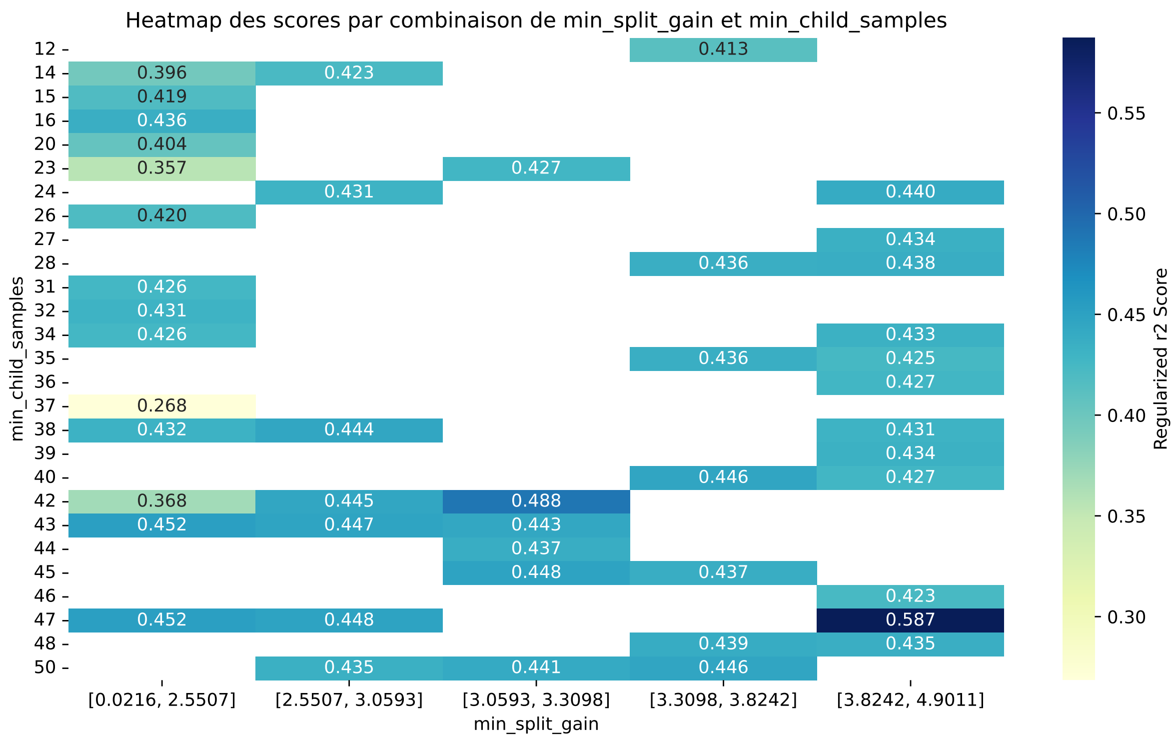

Impact of n_estimators
R² Score
n_estimators
0.60
0.55
0.50
0.45
0.40
0.35
0.30
200
300
400
500
600
700
800


Impact of min_data_per_group
R² Score
min_data_per_group
0.60
0.55
0.50
0.45
0.40
0.35
0.30
60
80
100
120
140
160
180
200



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6142	0.5878	0.5444	0.0184	0.5878	0.5652	-0.0226	0.9615	1.045
2	0.6155	0.5871	0.5451	0.0167	0.5871	0.5676	-0.0196	0.9667	1.0483
3	0.6465	0.6147	0.572	0.0207	0.4555	0.5942	-0.0205	0.9667	1.0518
4	0.6353	0.6048	0.5614	0.0204	0.4523	0.575	-0.0298	0.9507	1.0504
5	0.6466	0.6142	0.5725	0.0207	0.4523	0.5948	-0.0194	0.9684	1.0527
6	0.6485	0.6158	0.5766	0.0228	0.4521	0.5986	-0.0172	0.9721	1.0532
7	0.641	0.6095	0.5658	0.0209	0.452	0.5779	-0.0315	0.9482	1.0517
8	0.6407	0.6092	0.5668	0.0203	0.4519	0.582	-0.0272	0.9553	1.0516
9	0.645	0.6128	0.5715	0.0214	0.4518	0.5865	-0.0262	0.9572	1.0525
10	0.6504	0.6172	0.5769	0.0204	0.4516	0.5963	-0.021	0.966	1.0537
---	---	---	---	---	---	---	---	---	---
75	0.8379	0.743	0.6987	0.0169	0.2685	0.7133	-0.0297	0.96	1.1277

Hyperparameters:

Rank 1: {’reg___n_estimators’: 200, ’reg___max_depth’: 5, ’reg___learning_rate’: 0.012708716770208633, ’reg___num_leaves’: 51, ’reg___subsample’: 0.6837032951233722, ’reg___colsample_bytree’: 0.6856912877886208, ’reg___min_child_samples’: 42, ’reg___reg_alpha’: 5.619459414935131, ’reg___reg_lambda’: 45.44352415586509, ’reg___min_split_gain’: 3.1633220996690903, ’reg___path_smooth’: 1.151189134916482, ’reg___min_child_weight’: 0.8773955096590228, ’reg___feature_fraction’: 0.6678474883795271, ’reg___bagging_fraction’: 0.5629787583274474, ’reg___bagging_freq’: 7, ’reg___max_bin’: 77, ’reg___min_data_per_group’: 118}

Rank 2: {’reg___n_estimators’: 300, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.033107733056533646, ’reg___num_leaves’: 31, ’reg___subsample’: 0.7268034076957587, ’reg___colsample_bytree’: 0.5779489036557587, ’reg___min_child_samples’: 47, ’reg___reg_alpha’: 6.525794488005516, ’reg___reg_lambda’: 45.46621884312047, ’reg___min_split_gain’: 4.461689294984225, ’reg___path_smooth’: 0.3882940480958885, ’reg___min_child_weight’: 0.2711163318355096, ’reg___feature_fraction’: 0.8727890003398957, ’reg___bagging_fraction’: 0.5317590250408128, ’reg___bagging_freq’: 7, ’reg___max_bin’: 117, ’reg___min_data_per_group’: 162}

Rank 3: {’reg___n_estimators’: 300, ’reg___max_depth’: 5, ’reg___learning_rate’: 0.013307420607149527, ’reg___num_leaves’: 56, ’reg___subsample’: 0.6835870715128085, ’reg___colsample_bytree’: 0.5024873703117909, ’reg___min_child_samples’: 43, ’reg___reg_alpha’: 4.601293663158325, ’reg___reg_lambda’: 45.39055304975639, ’reg___min_split_gain’: 2.5549048640078538, ’reg___path_smooth’: 1.7616778527320125, ’reg___min_child_weight’: 0.4217869186631017, ’reg___feature_fraction’: 0.6132900450046801, ’reg___bagging_fraction’: 0.5313080385186028, ’reg___bagging_freq’: 7, ’reg___max_bin’: 79, ’reg___min_data_per_group’: 89}

Rank 4: {’reg___n_estimators’: 300, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.012316329047413989, ’reg___num_leaves’: 40, ’reg___subsample’: 0.7199623663350587, ’reg___colsample_bytree’: 0.5822611264531139, ’reg___min_child_samples’: 47, ’reg___reg_alpha’: 4.5693569765813224, ’reg___reg_lambda’: 45.72547331769405, ’reg___min_split_gain’: 2.5651322716171414, ’reg___path_smooth’: 0.4601480692897981, ’reg___min_child_weight’: 0.38079801901432836, ’reg___feature_fraction’: 0.6206700495252182, ’reg___bagging_fraction’: 0.5309970900449348, ’reg___bagging_freq’: 7, ’reg___max_bin’: 118, ’reg___min_data_per_group’: 157}

Rank 5: {’reg___n_estimators’: 300, ’reg___max_depth’: 5, ’reg___learning_rate’: 0.013004474681195416, ’reg___num_leaves’: 42, ’reg___subsample’: 0.6799056907949013, ’reg___colsample_bytree’: 0.6902681240043461, ’reg___min_child_samples’: 47, ’reg___reg_alpha’: 4.248616081387235, ’reg___reg_lambda’: 44.90251239961726, ’reg___min_split_gain’: 2.541329788318677, ’reg___path_smooth’: 1.6655186166885483, ’reg___min_child_weight’: 0.4444072738767472, ’reg___feature_fraction’: 0.6084470986125599, ’reg___bagging_fraction’: 0.5344179898714233, ’reg___bagging_freq’: 7, ’reg___max_bin’: 81, ’reg___min_data_per_group’: 88}

Rank 6: {’reg___n_estimators’: 200, ’reg___max_depth’: 4, ’reg___learning_rate’: 0.019083397173492722, ’reg___num_leaves’: 52, ’reg___subsample’: 0.6469671441105554, ’reg___colsample_bytree’: 0.5333268362602273, ’reg___min_child_samples’: 45, ’reg___reg_alpha’: 5.634302330538, ’reg___reg_lambda’: 1.216120020807658, ’reg___min_split_gain’: 3.201899586621438, ’reg___path_smooth’: 1.1096539602886901, ’reg___min_child_weight’: 0.8723453132134453, ’reg___feature_fraction’: 0.8506309219818213, ’reg___bagging_fraction’: 0.5036594622592507, ’reg___bagging_freq’: 7, ’reg___max_bin’: 78, ’reg___min_data_per_group’: 118}

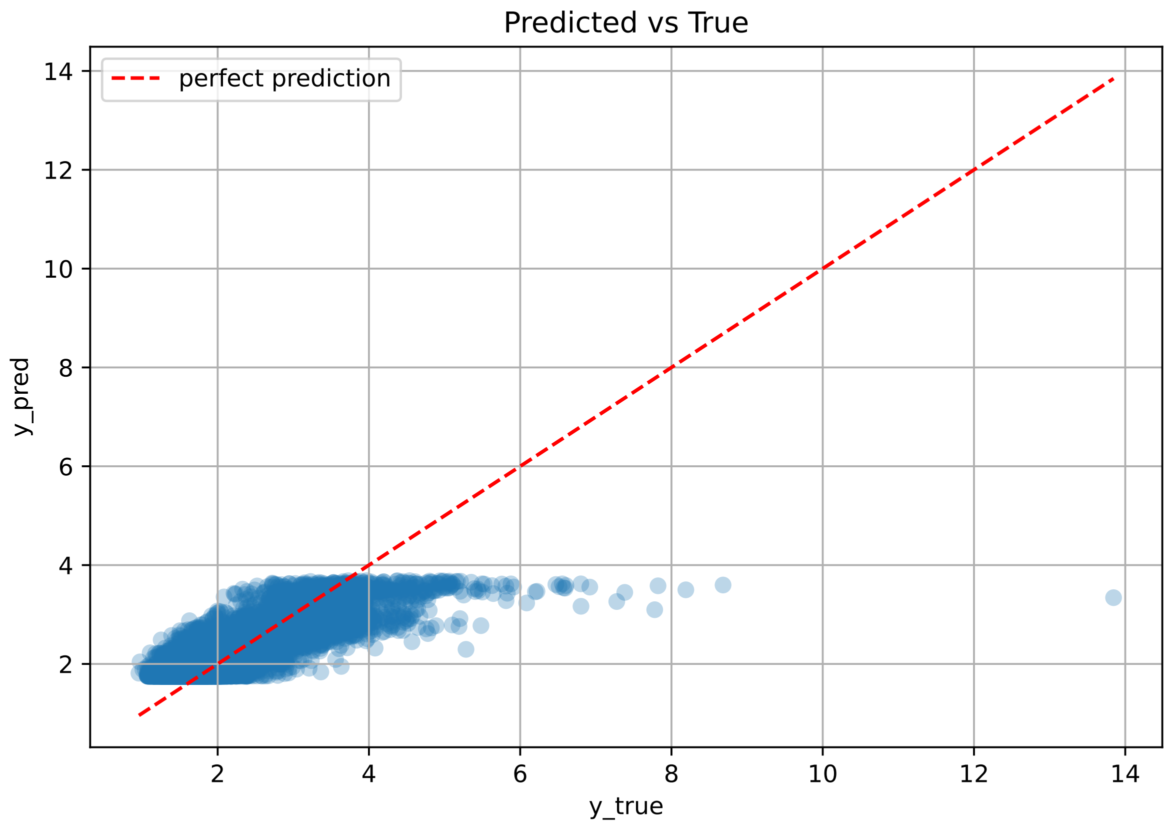
Rank 7: {’reg___n_estimators’: 300, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.012674907881512334, ’reg___num_leaves’: 42, ’reg___subsample’: 0.6795789657392113, ’reg___colsample_bytree’: 0.5765707486572611, ’reg___min_child_samples’: 43, ’reg___reg_alpha’: 4.555357628177499, ’reg___reg_lambda’: 45.900688184403236, ’reg___min_split_gain’: 2.2726683645315306, ’reg___path_smooth’: 1.7307423549315053, ’reg___min_child_weight’: 0.5350802593313151, ’reg___feature_fraction’: 0.6217937668480293, ’reg___bagging_fraction’: 0.5271302740332776, ’reg___bagging_freq’: 7, ’reg___max_bin’: 254, ’reg___min_data_per_group’: 90}

Rank 8: {’reg___n_estimators’: 400, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.011411954108706742, ’reg___num_leaves’: 33, ’reg___subsample’: 0.7143418962510466, ’reg___colsample_bytree’: 0.5779209605514466, ’reg___min_child_samples’: 47, ’reg___reg_alpha’: 6.422605543098348, ’reg___reg_lambda’: 45.5294442647805, ’reg___min_split_gain’: 2.5222172492394845, ’reg___path_smooth’: 0.5976616867718998, ’reg___min_child_weight’: 1.3859976339209985, ’reg___feature_fraction’: 0.6377575161715494, ’reg___bagging_fraction’: 0.5223341246360196, ’reg___bagging_freq’: 7, ’reg___max_bin’: 116, ’reg___min_data_per_group’: 163}

Rank 9: {'reg___n_estimators': 400, 'reg___max_depth': 3, 'reg___learning_rate': 0.01155335316094138, 'reg___num_leaves': 33, 'reg___subsample': 0.7086330645973696, 'reg___colsample_bytree': 0.5643843051586667, 'reg___min_child_samples': 47, 'reg___reg_alpha': 6.406339466169777, 'reg___reg_lambda': 45.615854144248296, 'reg___min_split_gain': 2.5745483960956346, 'reg___path_smooth': 0.5266302877830386, 'reg___min_child_weight': 0.18390211560003003, 'reg___feature_fraction': 0.8618378836682439, 'reg___bagging_fraction': 0.5450597133046967, 'reg___bagging_freq': 7, 'reg___max_bin': 113, 'reg___min_data_per_group': 163}

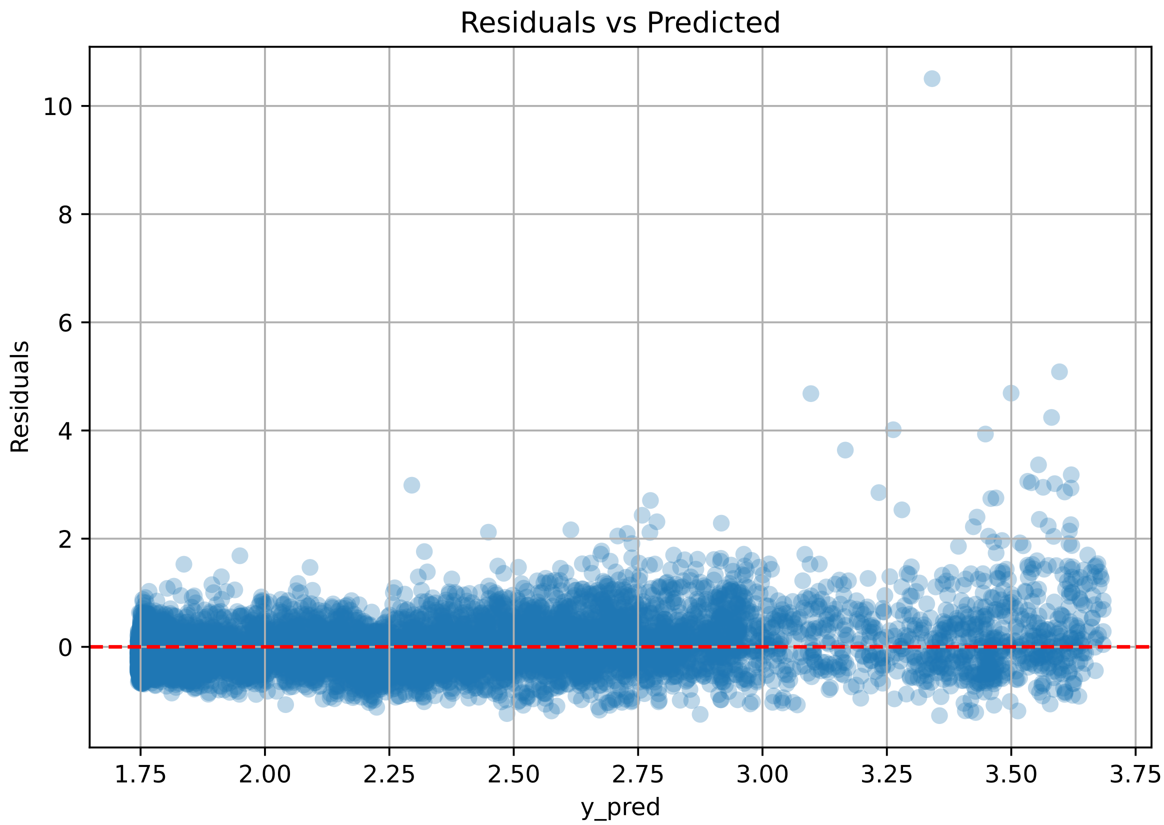
Rank 10: {'reg___n_estimators': 300, 'reg___max_depth': 3, 'reg___learning_rate': 0.014930213993682995, 'reg___num_leaves': 33, 'reg___subsample': 0.6692804353781493, 'reg___colsample_bytree': 0.5663739894738183, 'reg___min_child_samples': 47, 'reg___reg_alpha': 6.354119489088445, 'reg___reg_lambda': 5.37233820827722, 'reg___min_split_gain': 2.652731247113964, 'reg___path_smooth': 1.7583115263443252, 'reg___min_child_weight': 0.25493239165605036, 'reg___feature_fraction': 0.8663049902343432, 'reg___bagging_fraction': 0.5408814619545597, 'reg___bagging_freq': 7, 'reg___max_bin': 114, 'reg___min_data_per_group': 162}

3.5.2 Scatter



Predicted vs True
perfect prediction
14
12
10
8
6
4
2
y_pred
2
4
6
8
10
12
14
y_true

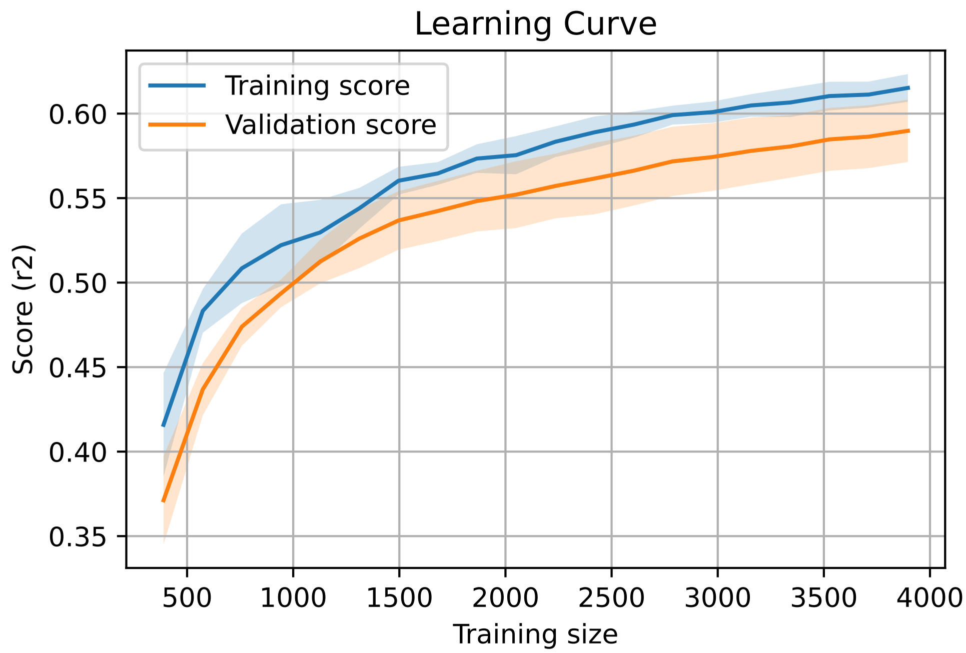
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7953, **Len(training)** : 5568,

Len(training_real) : 3898, **Len(test_of_training)** : 1670, **Len(test)** : 2385

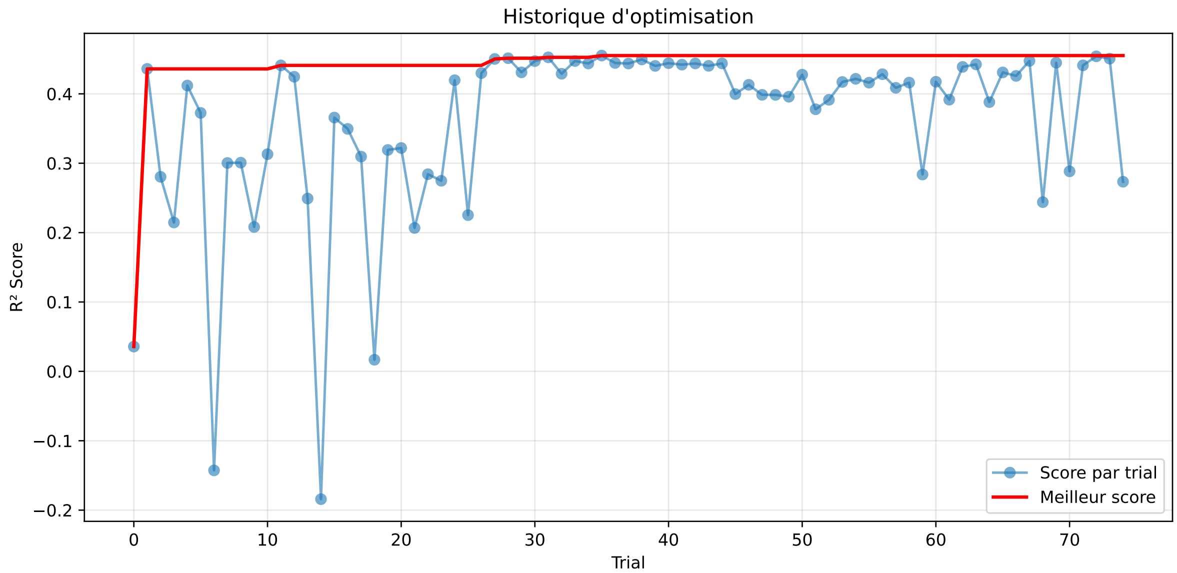


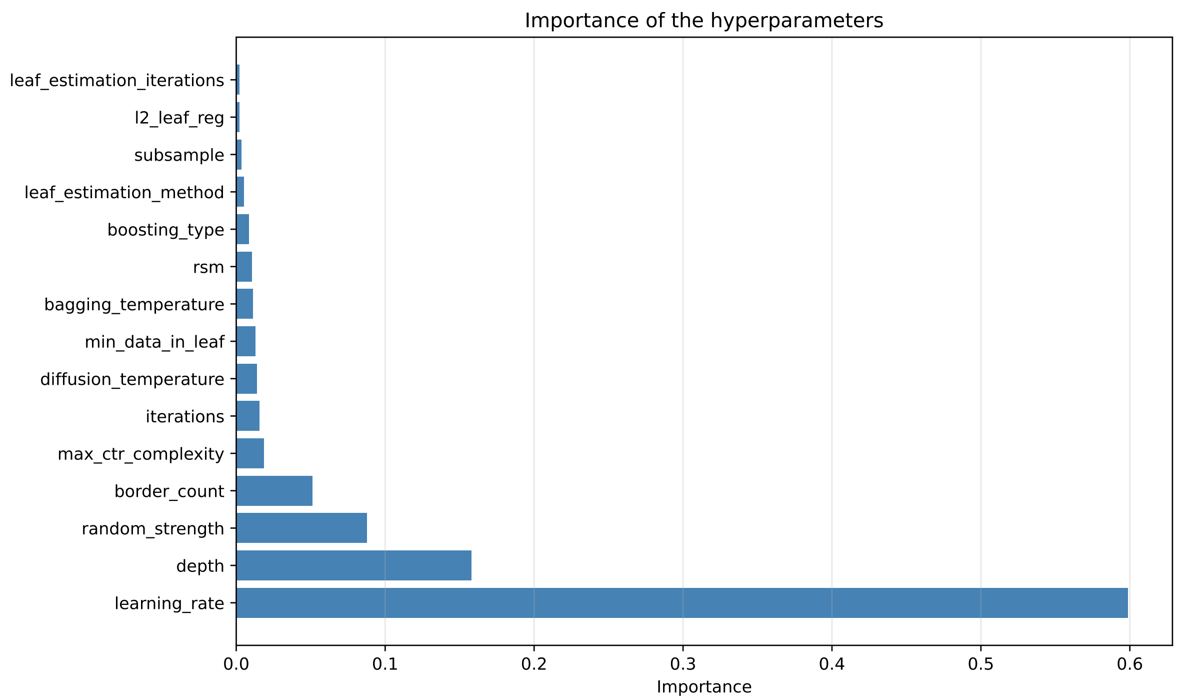
3.6 Model : CatBoost

3.6.1 Gridsearch

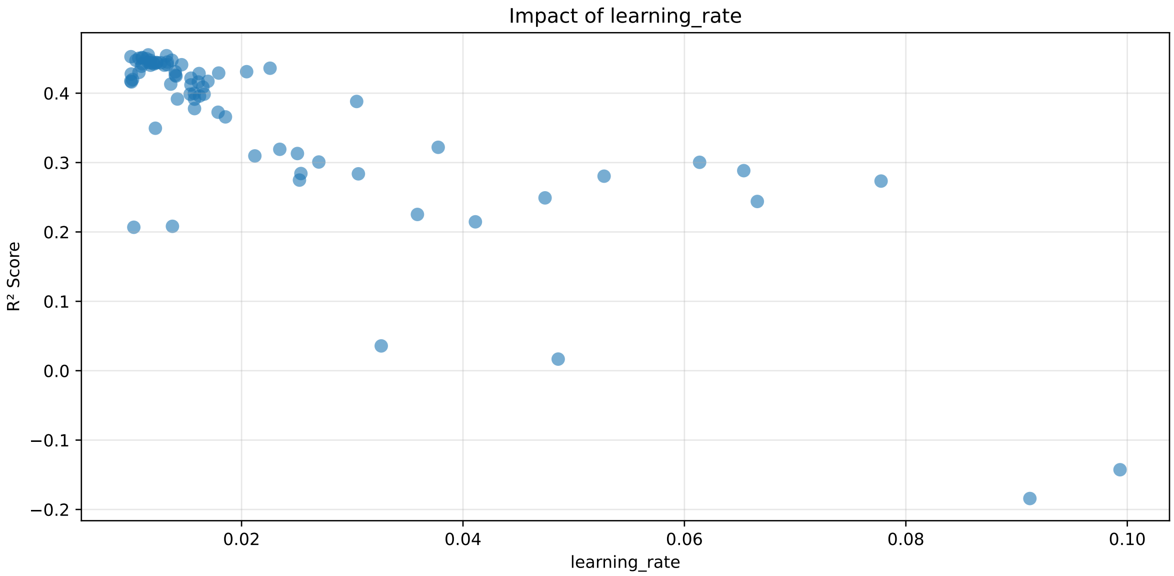
Distribution y_train vs y_test: y_train: mean=2.351, std=0.790, min=0.959, max=8.682 y_test: mean=2.340, std=0.815, min=1.041, max=13.845

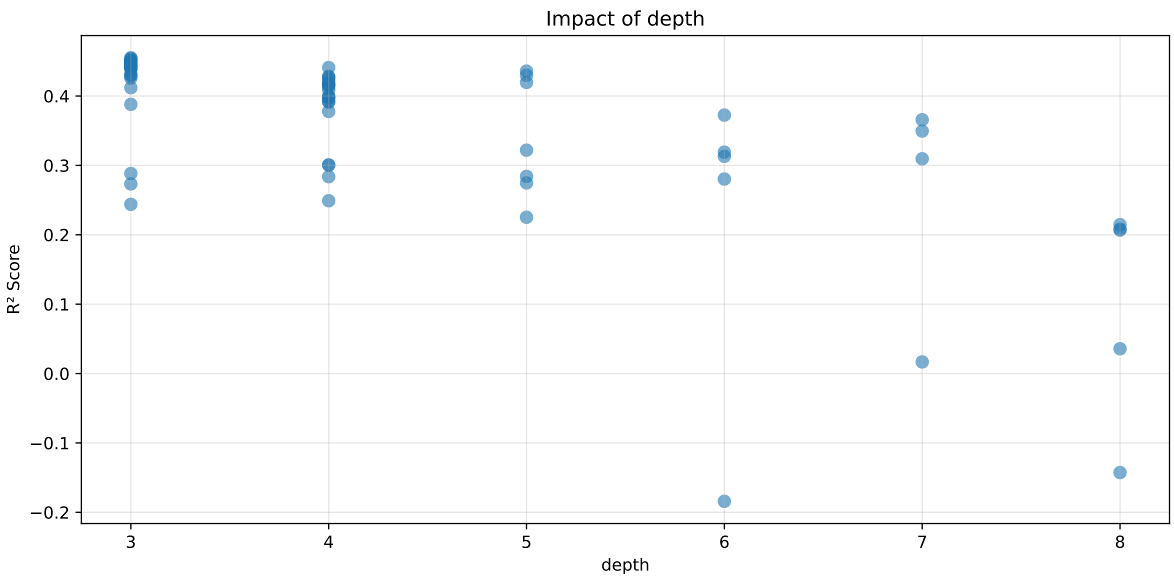
===================== TOP Importance FEATURES ===================== feature importance r_umidity 22.346883 r_temperature_wl_3h_A_rms 15.712027 r_temperature_wl_2h_A_rms 8.812945 r_temperature_wl_3h_A_energy_rel 6.682444 r_temperature 6.088041 r_umidity_wl_3h_A_energy_rel 4.270333 r_CO2 4.173454 r_temperature_wl_2h_D1_energy_rel 4.103738 r_NH3 3.001471 r_NH3_wl_4h_D2_energy 2.456690 r_NH3_wl_4h_A_energy 1.736444 r_umidity_wl_3h_A_energy 1.567003 r_NH3_wl_3h_D2_energy 1.455082 r_NH3_wl_2h_A_energy 1.444667 r_NH3_wl_3h_A_energy 1.441388 r_THI 1.375587 r_NH3_wl_4h_D1_energy 1.257438 r_umidity_wl_4h_A_energy 0.937977 r_umidity_wl_2h_A_energy 0.883464 r_NH3_wl_3h_D1_energy 0.851212 r_umidity_wl_4h_D2_energy 0.785770 r_NH3_wl_2h_D1_energy 0.691757 r_umidity_wl_2h_A_rms 0.679041 r_temperature_wl_4h_A_energy 0.664418 r_umidity_wl_4h_A_rms 0.632338 r_umidity_wl_3h_D2_energy 0.594714 r_temperature_wl_2h_A_energy 0.592018 r_NH3_wl_4h_D1_energy_rel 0.591832 r_umidity_wl_4h_D1_energy_rel 0.553626 r_NH3_wl_3h_D1_energy_rel 0.539339 r_NH3_wl_3h_A_rms 0.489971 r_umidity_wl_4h_D1_energy 0.465501 r__NH3__wl__3h__D2__energy__rel 0.441083 r__NH3__wl__2h__A__energy__rel 0.404913 r__NH3__wl__2h__A__rms 0.378600 r__umidity__wl__2h__D1__energy 0.369980 r__temperature__wl__4h__D1__energy 0.268612 r__NH3__wl__4h__A__rms 0.258199

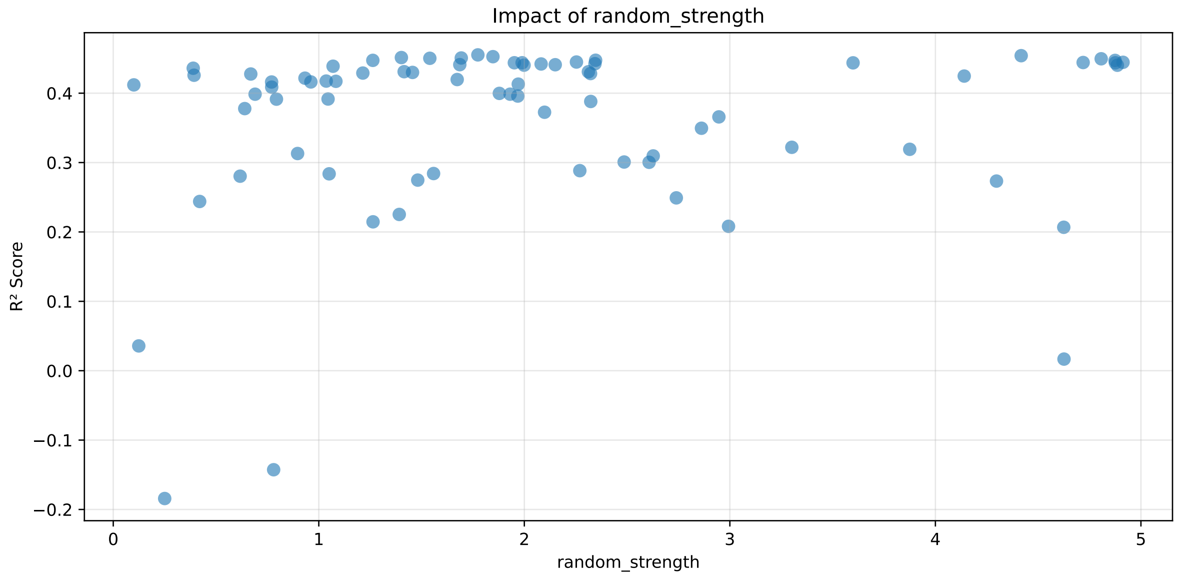


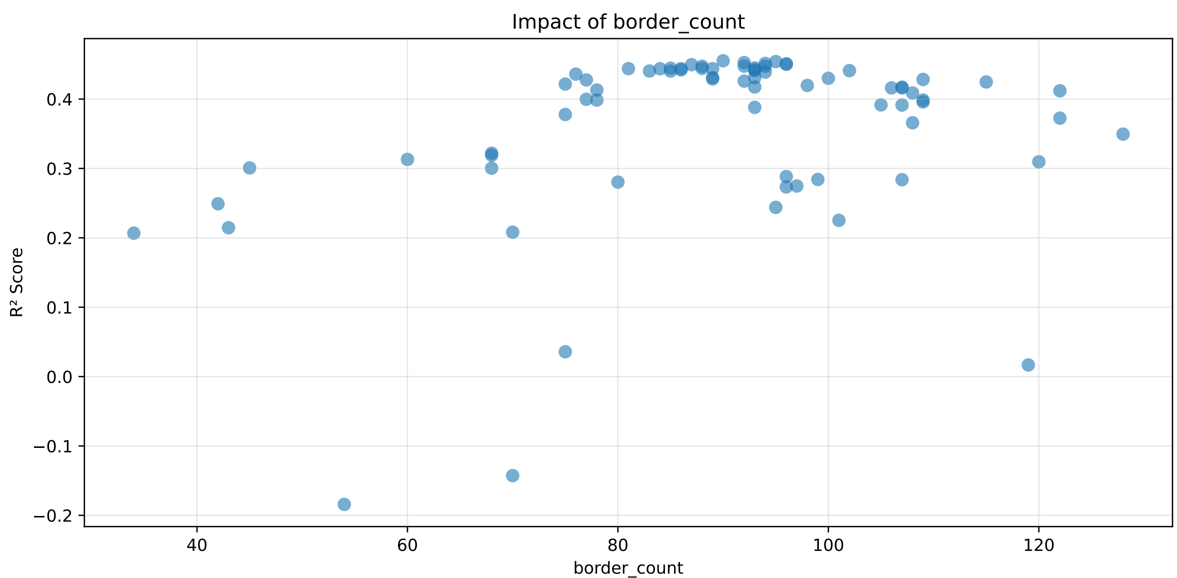


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'random_strength', 'border_count']


Impact of learning_rate
R² Score
learning_rate
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0.02
0.04
0.06
0.08
0.10

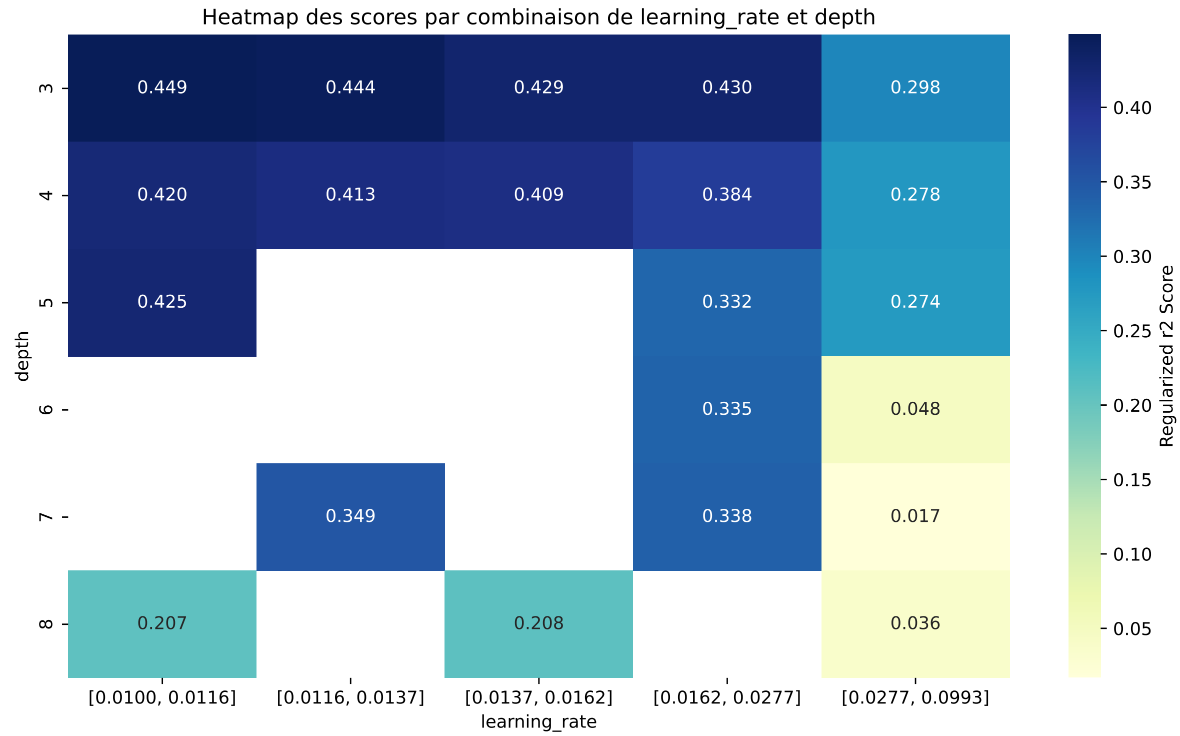

Impact of depth
R² Score
depth
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
3
4
5
6
7
8






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6728	0.6365	0.596	0.0232	0.4552	0.5983	-0.0383	0.9399	1.057
2	0.6753	0.6384	0.5968	0.0235	0.4541	0.5981	-0.0403	0.9369	1.0578
3	0.6703	0.6341	0.5928	0.0232	0.4526	0.5962	-0.0379	0.9403	1.0572
4	0.6721	0.6353	0.5948	0.0221	0.4513	0.5994	-0.0359	0.9434	1.0579
5	0.6814	0.6429	0.6029	0.0222	0.4507	0.6057	-0.0372	0.9421	1.0598
6	0.6672	0.631	0.5905	0.0226	0.4503	0.5939	-0.0371	0.9412	1.0573
7	0.6426	0.6104	0.5686	0.0238	0.4496	0.573	-0.0374	0.9388	1.0527
8	0.6962	0.6547	0.613	0.0235	0.4475	0.6163	-0.0384	0.9413	1.0633
9	0.6939	0.6528	0.612	0.022	0.4473	0.6148	-0.038	0.9418	1.063
10	0.6347	0.6034	0.5626	0.0233	0.4472	0.567	-0.0364	0.9397	1.0518
---	---	---	---	---	---	---	---	---	---
75	0.9789	0.7851	0.7404	0.0156	-0.1842	0.7613	-0.0238	0.9697	1.2469

Hyperparameters:

Rank 1: {’reg___iterations’: 700, ’reg___depth’: 3, ’reg___learning_rate’: 0.011574240955599977, ’reg___l2_leaf_reg’: 22.61401511974515, ’reg___bagging_temperature’: 2.7164075654682596, ’reg___random_strength’: 1.7740129404991656, ’reg___subsample’: 0.6056632907880106, ’reg___min_data_in_leaf’: 32, ’reg___leaf_estimation_iterations’: 9, ’reg___rsm’: 0.547990430403736, ’reg___border_count’: 90, ’reg___leaf_estimation_method’: ‘Gradient’, 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9859.129488659642}

Rank 2: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.013223359591096651, 'reg___l2_leaf_reg': 27.108779218549774, 'reg___bagging_temperature': 3.2739158854059296, 'reg___random_strength': 4.417390167953171, 'reg___subsample': 0.5879944768835283, 'reg___min_data_in_leaf': 28, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.6254789073636753, 'reg___border_count': 95, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9135.596714419547}

Rank 3: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.010011088439387037, 'reg___l2_leaf_reg': 28.168669630884047, 'reg___bagging_temperature': 2.944480824229246, 'reg___random_strength': 1.84726414194915, 'reg___subsample': 0.6088991318126217, 'reg___min_data_in_leaf': 29, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.6289370484791399, 'reg___border_count': 92, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9132.068490675032}

Rank 4: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.011018749018928485, 'reg___l2_leaf_reg': 28.10628983941099, 'reg___bagging_temperature': 2.870351131213899, 'reg___random_strength': 1.4018520987572414, 'reg___subsample': 0.5924692042277938, 'reg___min_data_in_leaf': 10, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.5049514099556686, 'reg___border_count': 94, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 7651.638699012935}

Rank 5: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.011079217668889628, 'reg___l2_leaf_reg': 27.004362256260784, 'reg___bagging_temperature': 0.1366536875035007, 'reg___random_strength': 1.6932292532940512, 'reg___subsample': 0.5847069576865491, 'reg___min_data_in_leaf': 28, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.5042830969729453, 'reg___border_count': 96, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9107.346997140055}

Rank 6: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.0107219754164643, 'reg___l2_leaf_reg': 30.182438749226492, 'reg___bagging_temperature': 1.0904104460803645, 'reg___random_strength': 1.5402435088026092, 'reg___subsample': 0.6652219321145787, 'reg___min_data_in_leaf': 29, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.5090182899915232, 'reg___border_count': 96, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 8238.016497751472}

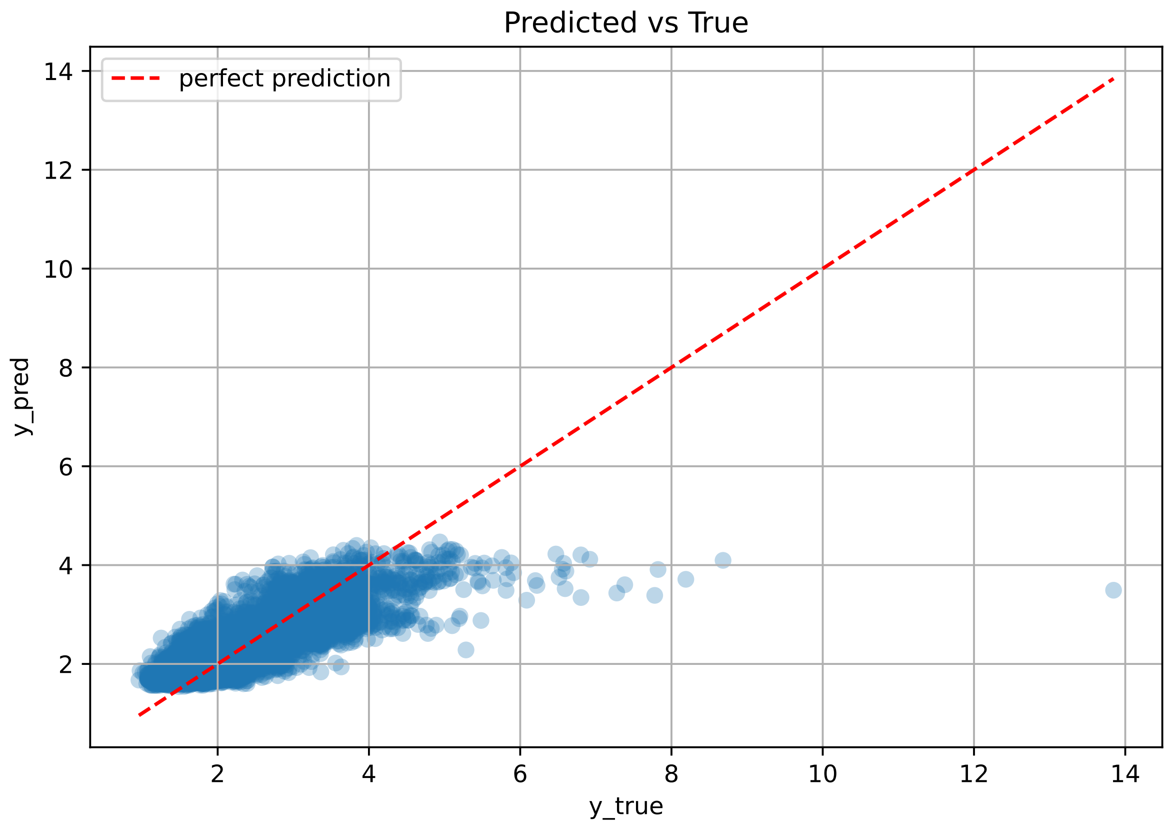
Rank 7: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.01143028480760276, 'reg___l2_leaf_reg': 24.186496604338714, 'reg___bagging_temperature': 1.455515633656058, 'reg___random_strength': 4.806668306384068, 'reg___subsample': 0.5590678618648719, 'reg___min_data_in_leaf': 14, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.632197617057784, 'reg___border_count': 87, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9563.050639554105}

Rank 8: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.01371005252834206, 'reg___l2_leaf_reg': 26.669719135584053, 'reg___bagging_temperature': 1.970683797857205, 'reg___random_strength': 2.346846208324597, 'reg___subsample': 0.5854650854499693, 'reg___min_data_in_leaf': 13, 'reg___leaf_estimation_iterations': 8, 'reg___rsm': 0.5761663621705339, 'reg___border_count': 92, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9078.089831877613}

Rank 9: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.011757717486891767, 'reg___l2_leaf_reg': 29.3941566016661, 'reg___bagging_temperature': 2.7038109350415196, 'reg___random_strength': 1.2629373061146665, 'reg___subsample': 0.6000425768697897, 'reg___min_data_in_leaf': 11, 'reg___leaf_estimation_iterations': 9, 'reg___rsm': 0.5326518751116245, 'reg___border_count': 88, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9367.867705169829}

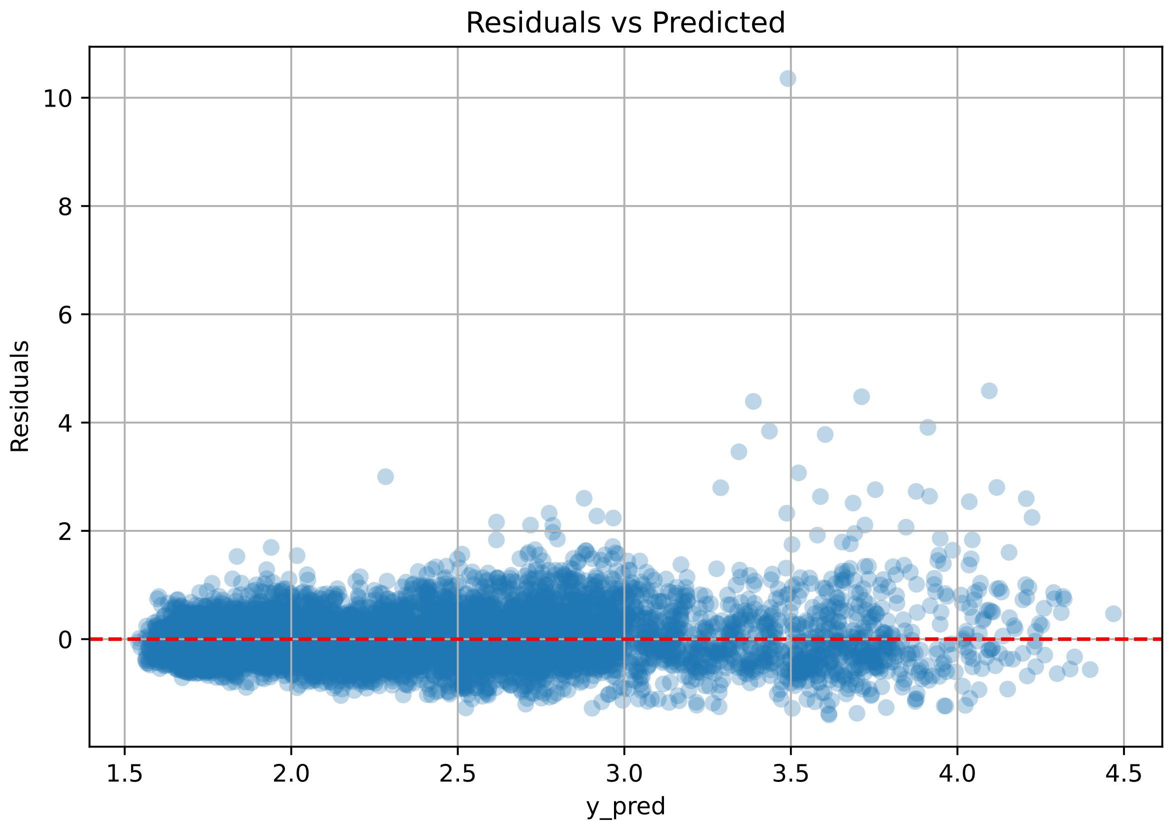
Rank 10: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.010463357930691683, 'reg___l2_leaf_reg': 29.128993909029035, 'reg___bagging_temperature': 2.8694181071078972, 'reg___random_strength': 4.874254035608674, 'reg___subsample': 0.5995640566847471, 'reg___min_data_in_leaf': 27, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.6461997920249913, 'reg___border_count': 94, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 8841.038471532258}

3.6.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

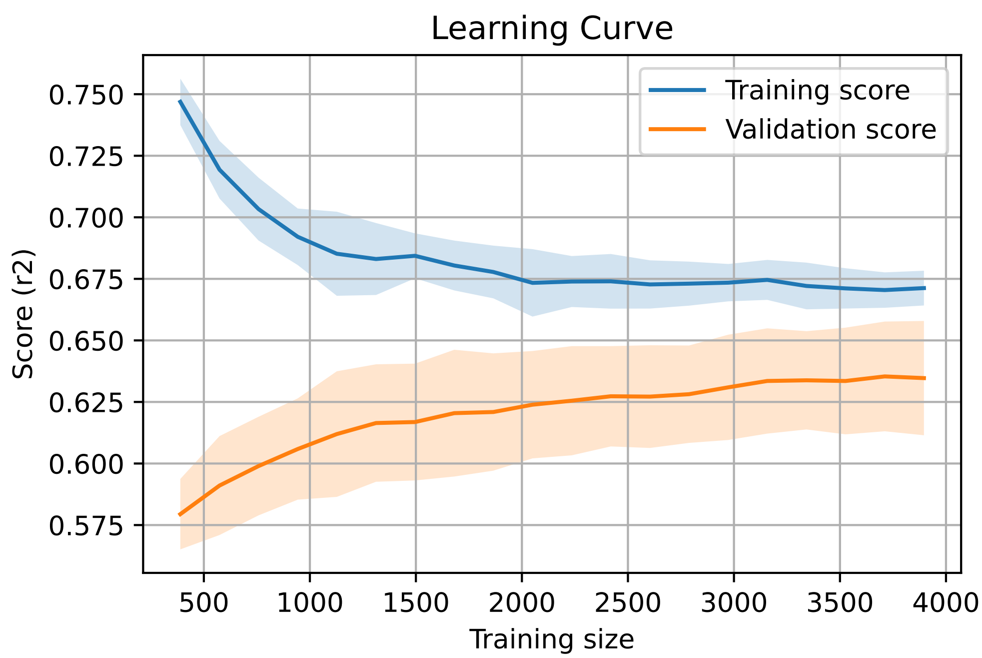
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 7953, **Len(training)** : 5568,

Len(training_real) : 3898, **Len(test_of_training)** : 1670, **Len(test)** : 2385



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.603	0.582	0.022	0.543	0.543	-0.039	0.933	1.035	nan
LGBM	0.614	0.588	0.018	0.544	0.565	-0.023	0.962	1.045	0.5878
CatBoost	0.673	0.636	0.023	0.596	0.598	-0.038	0.94	1.057	0.4552

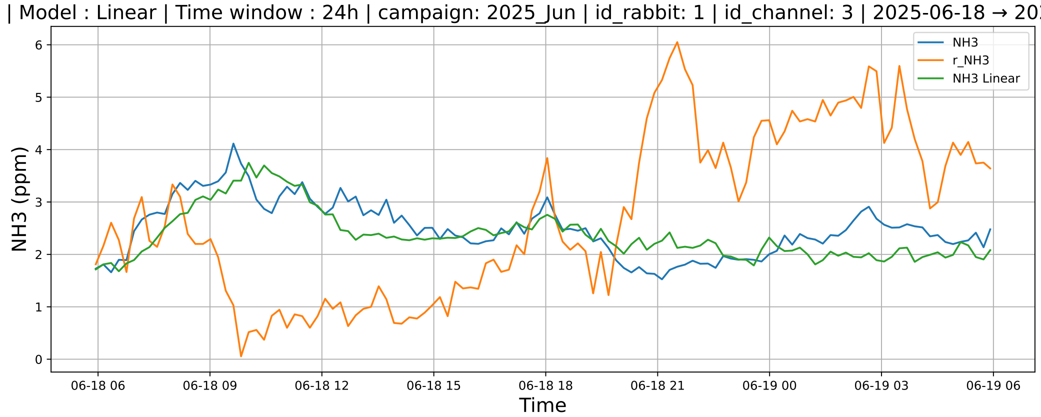
4 Plot

4.1 Standalone plots

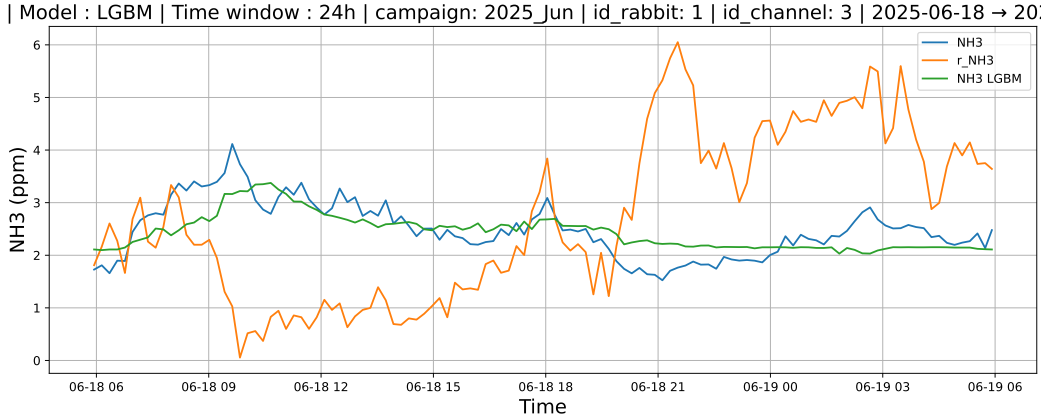
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

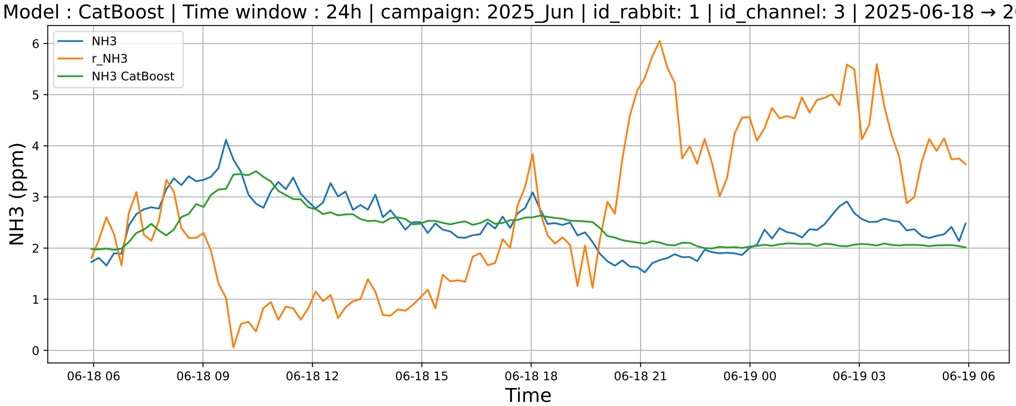
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

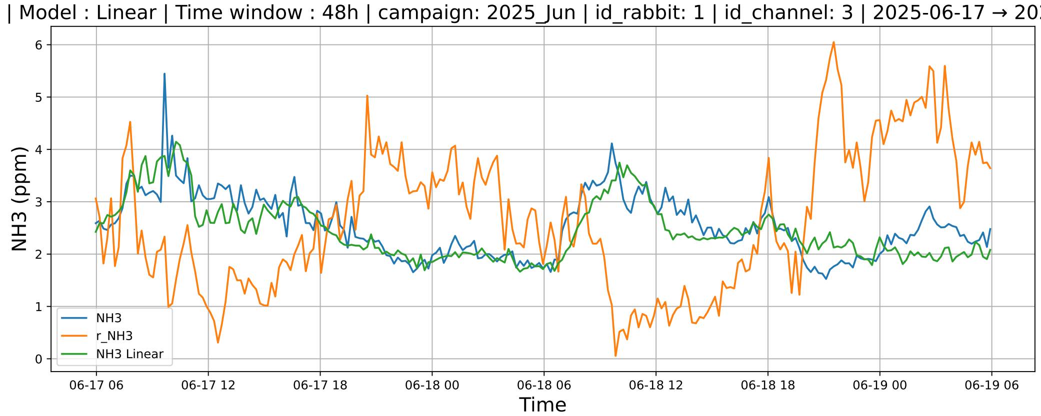


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

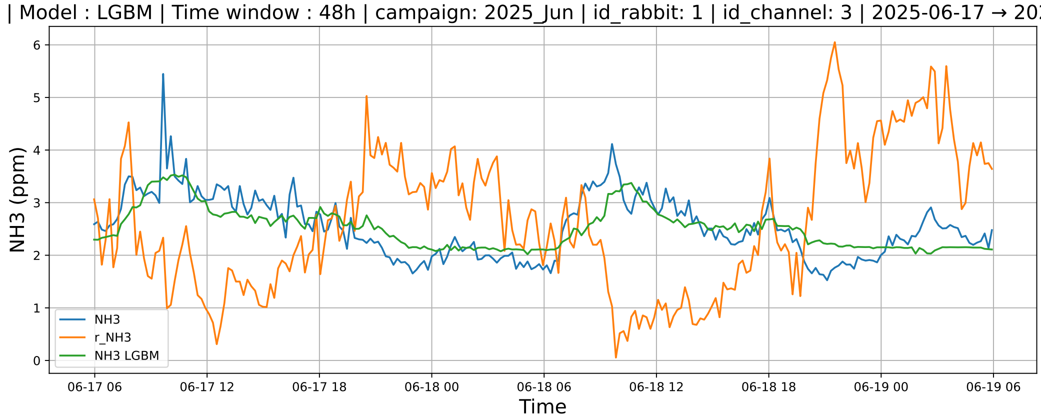


4.1.1.2 Time window : 48

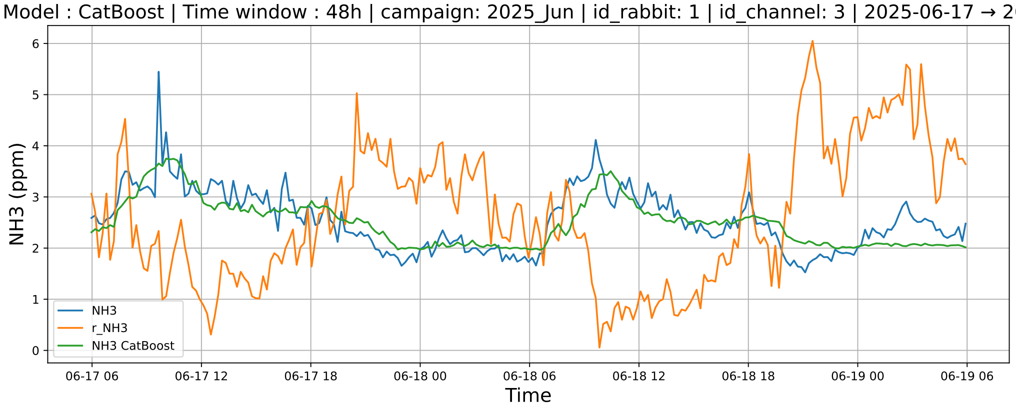
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

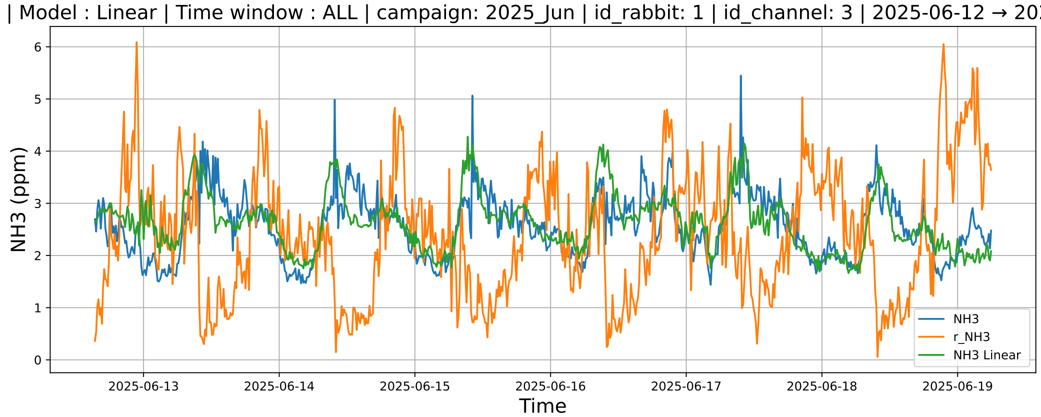


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

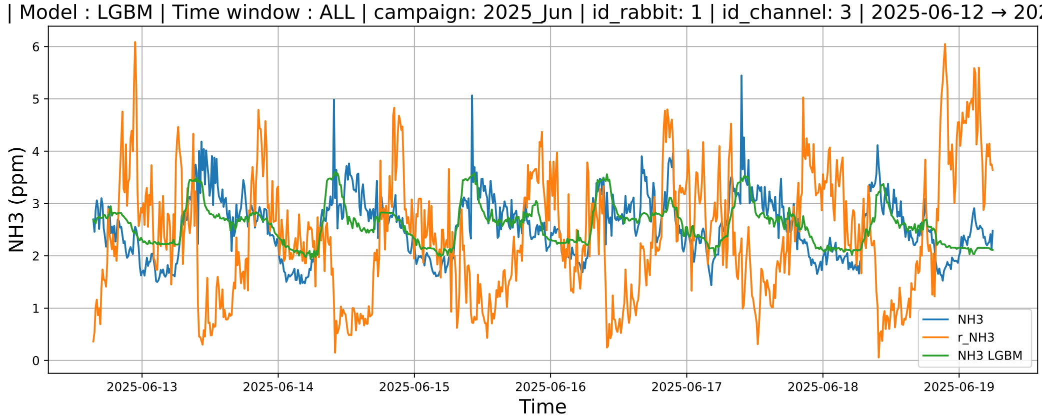


4.1.1.3 Time window : ALL

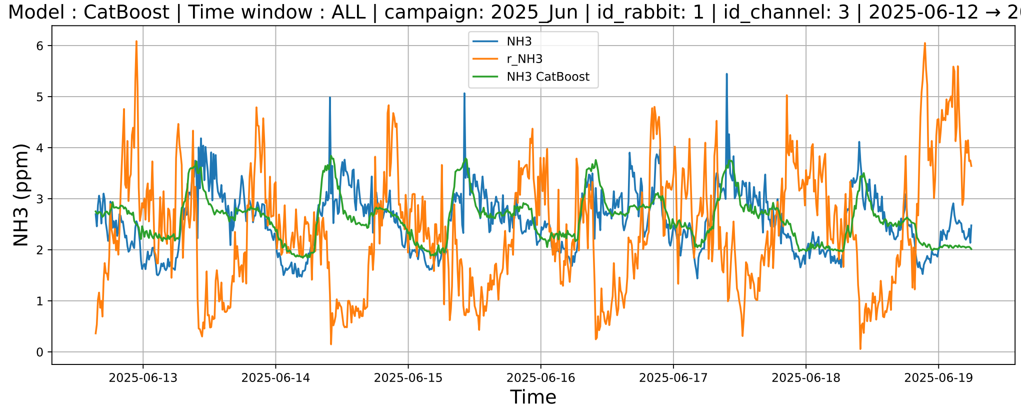
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



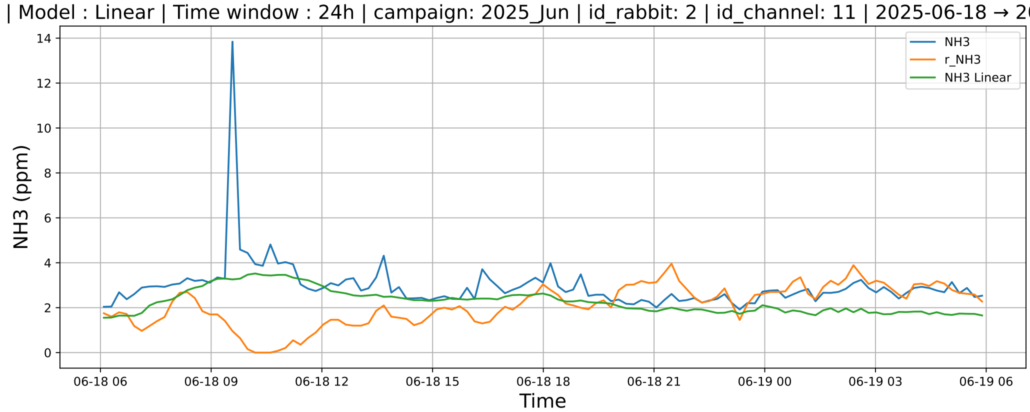
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



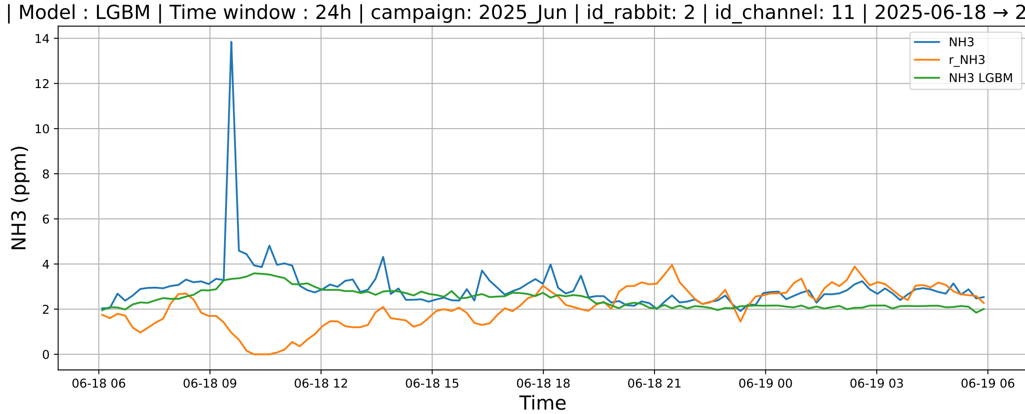
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

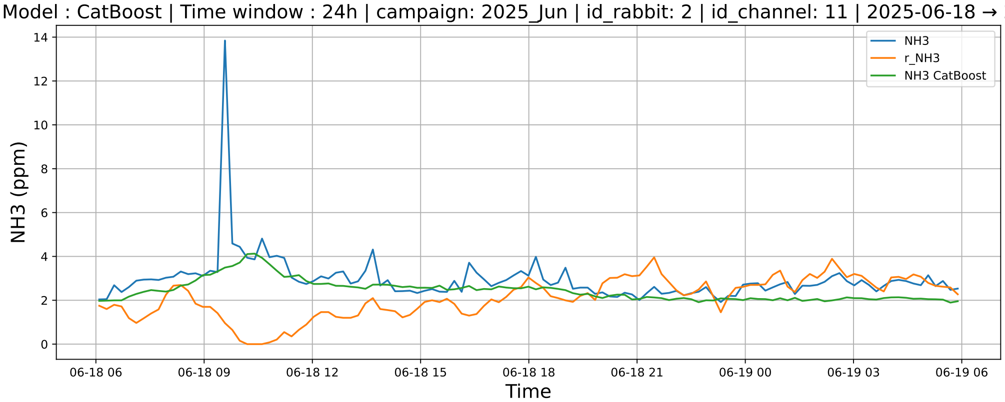
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



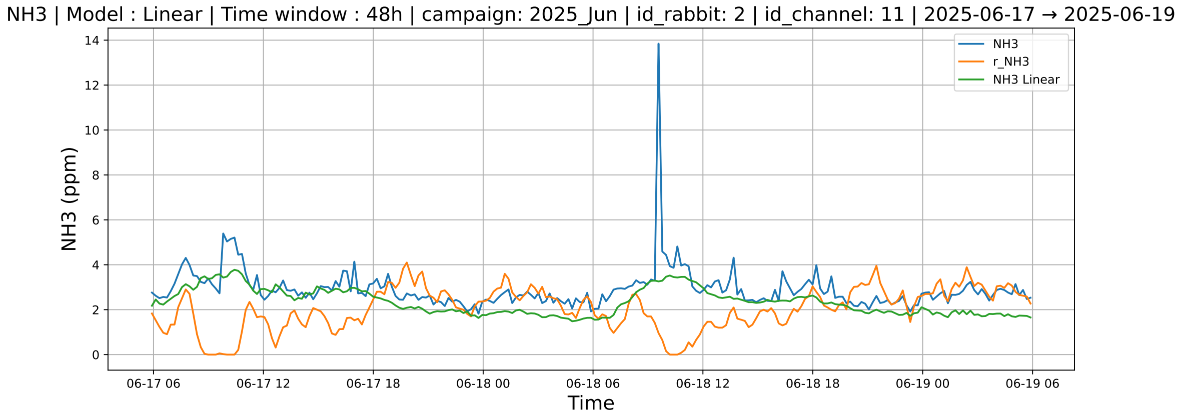
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

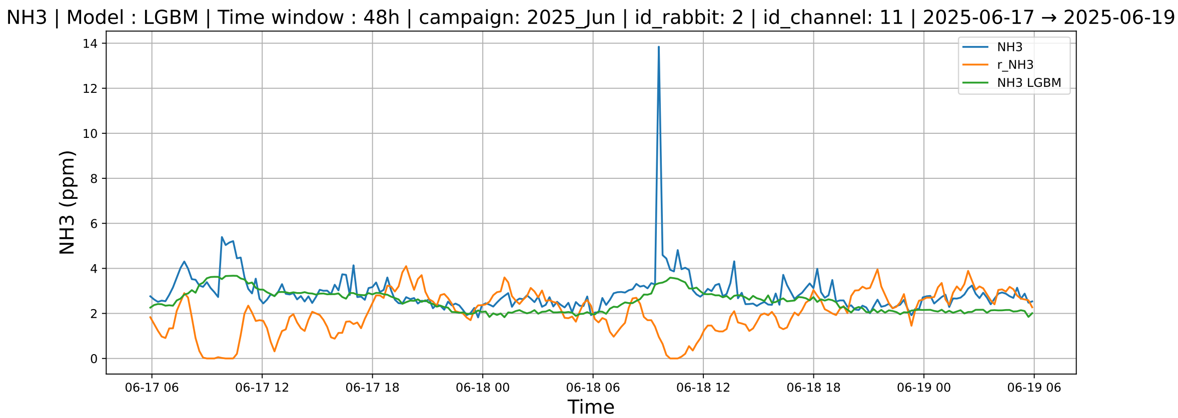


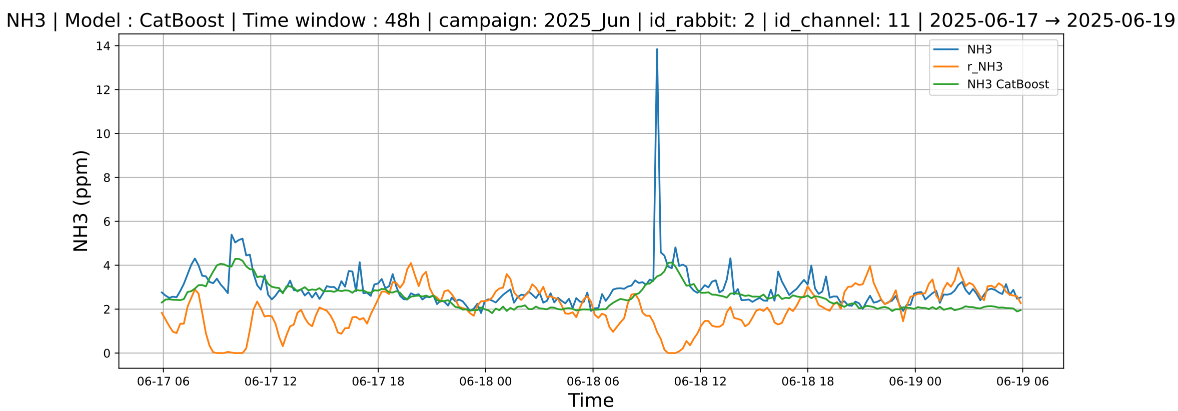
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



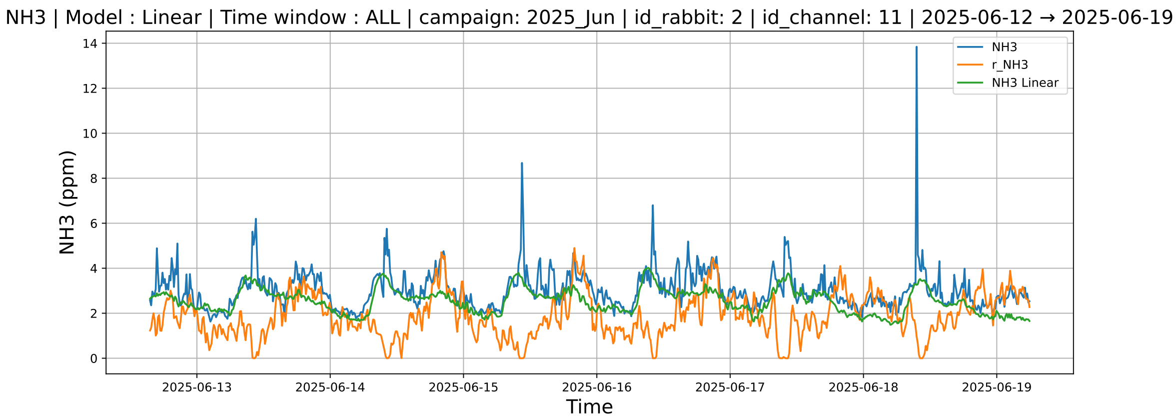
4.1.2.2 Time window : 48

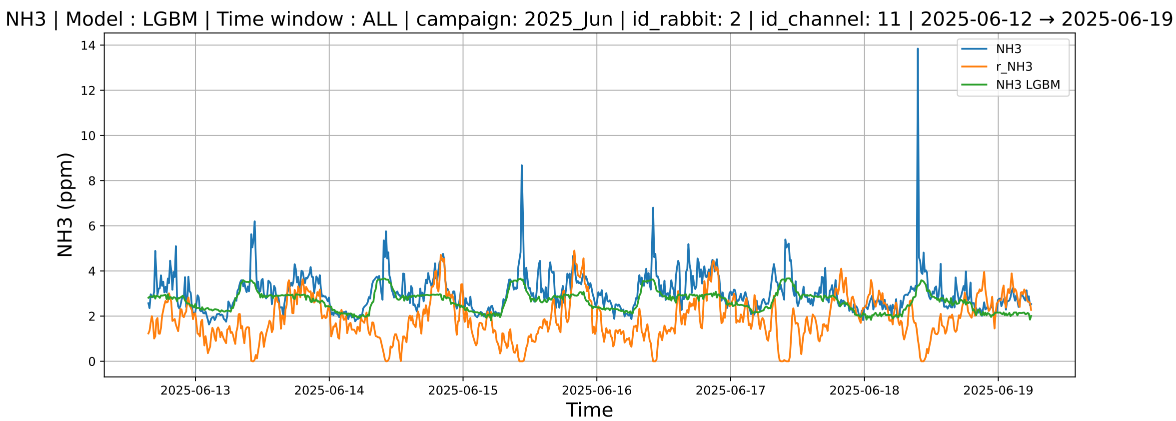


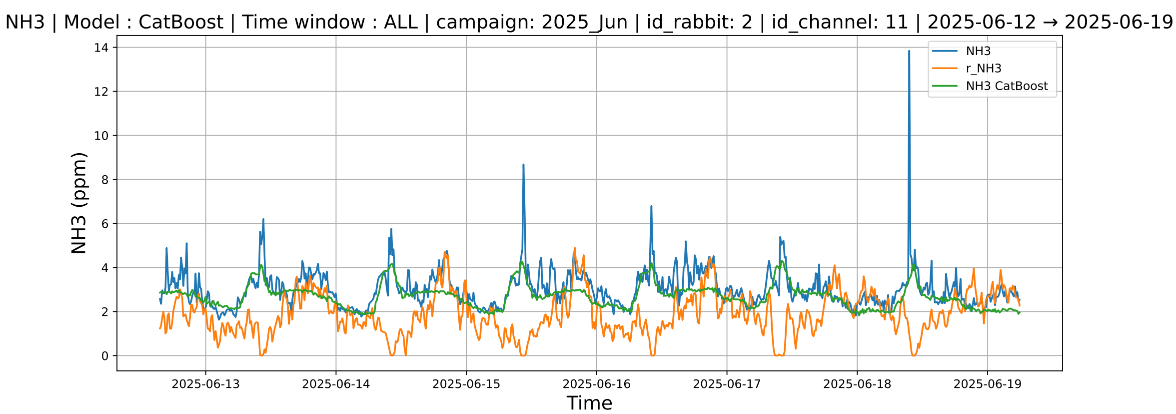




4.1.2.3 Time window : ALL



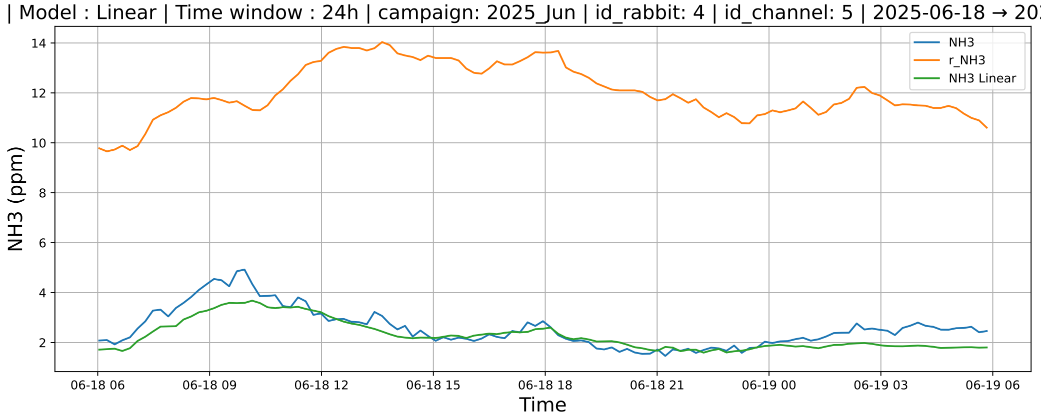




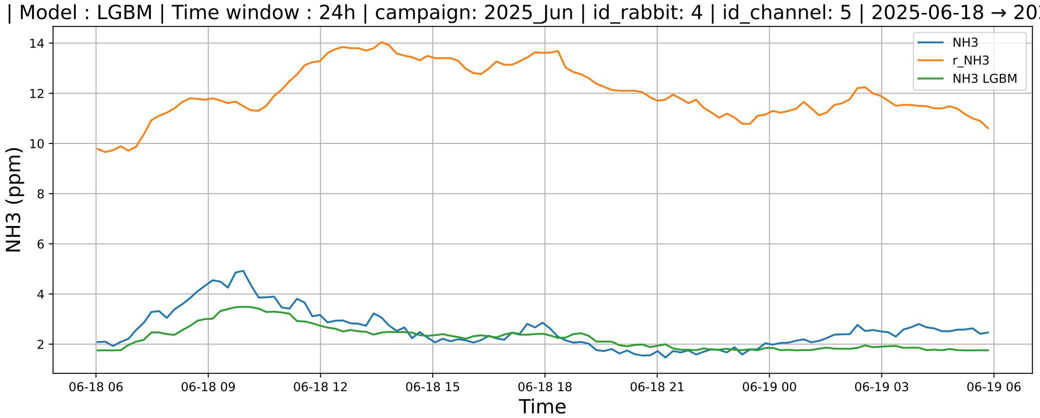
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

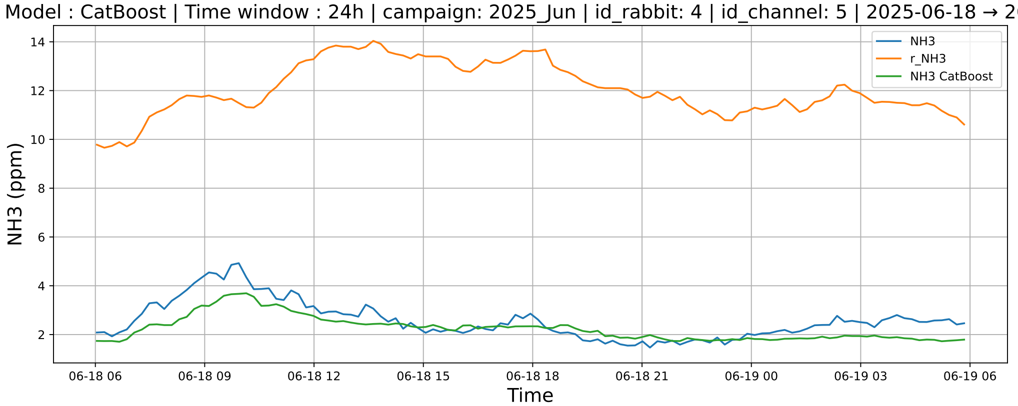
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



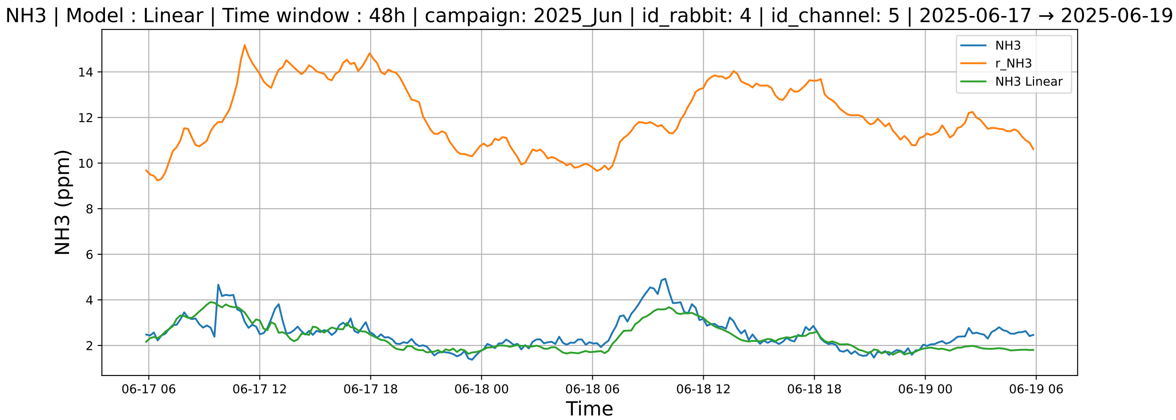
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

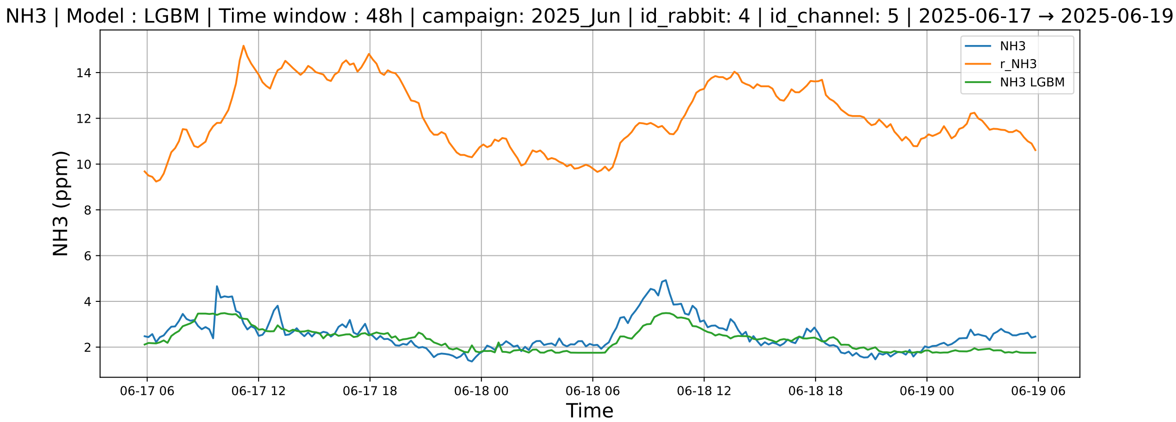


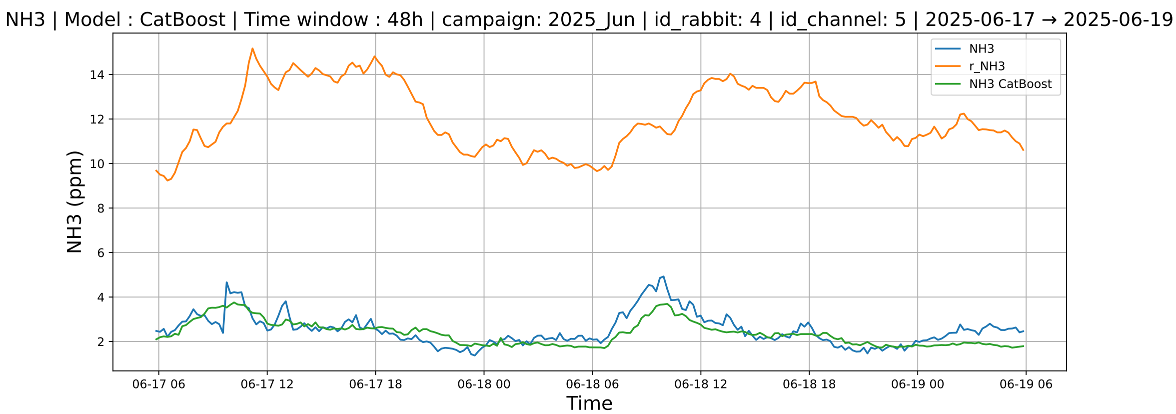
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



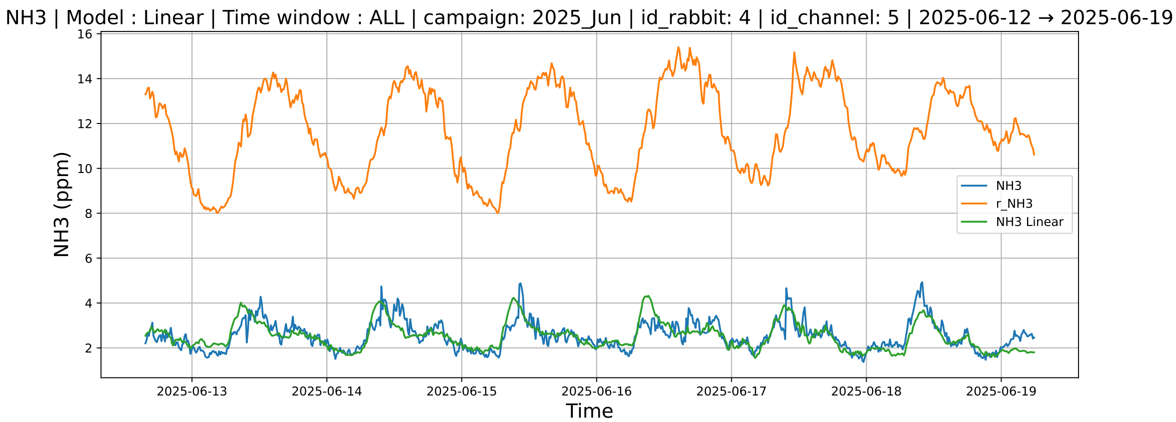
4.1.3.2 Time window : 48

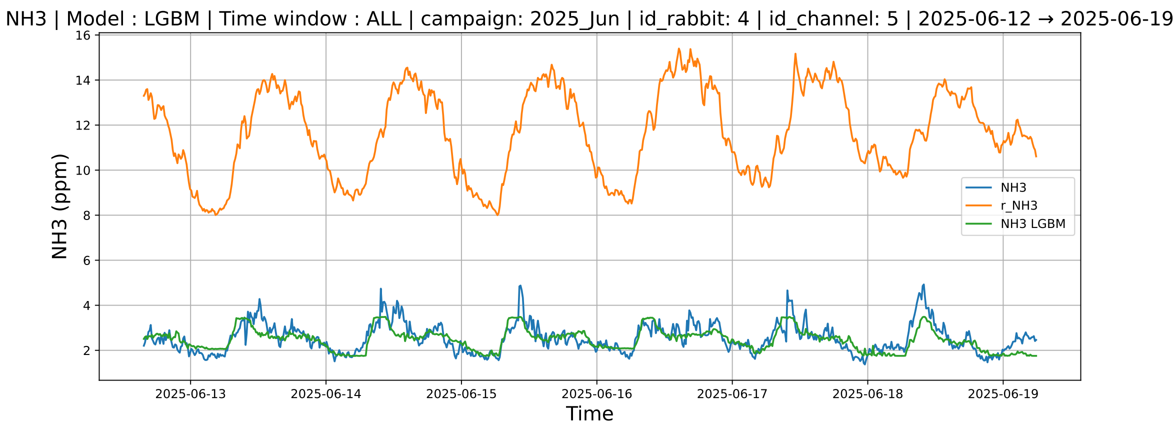


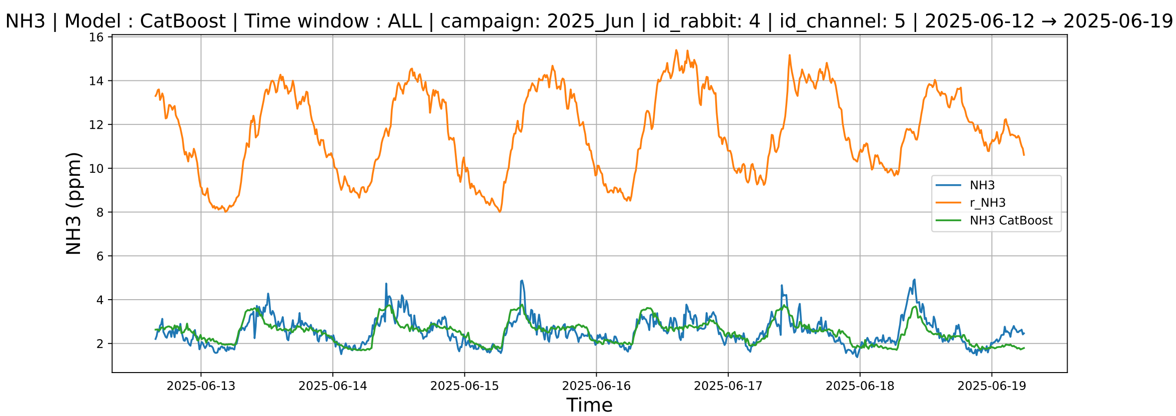




4.1.3.3 Time window : ALL



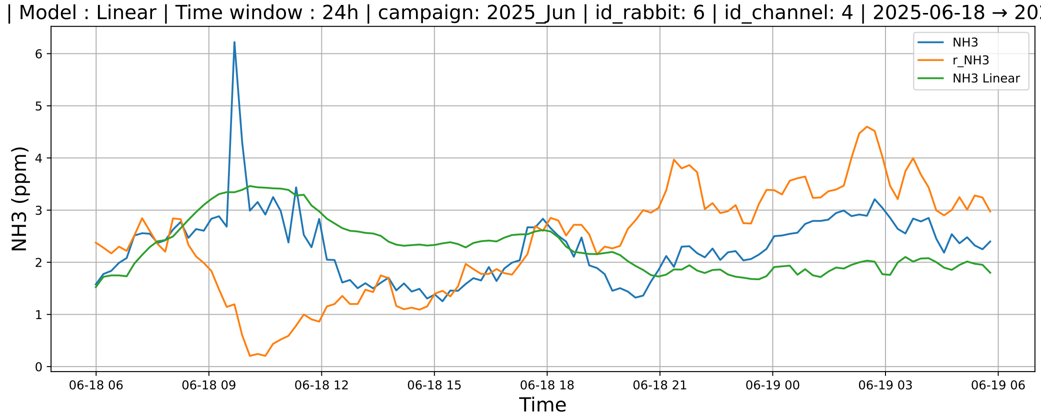




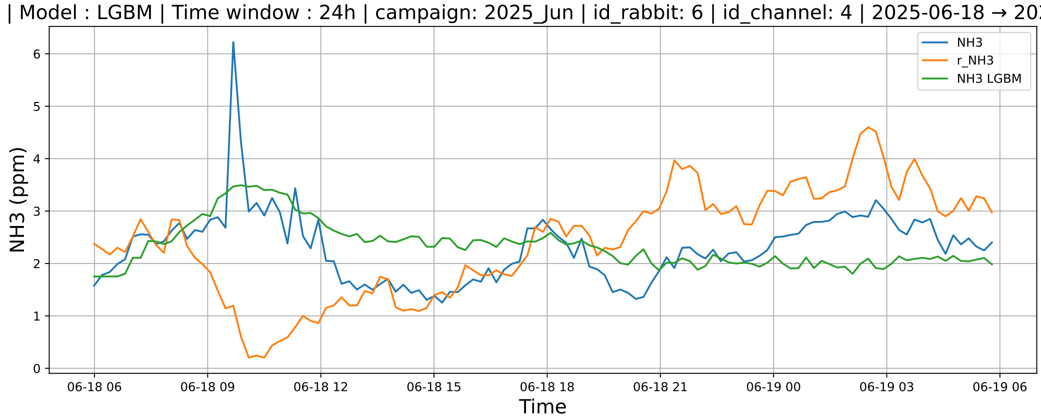
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

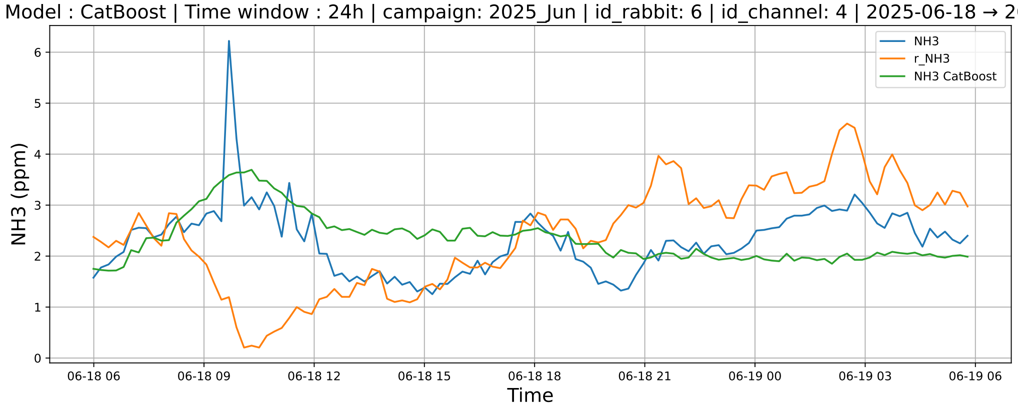
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



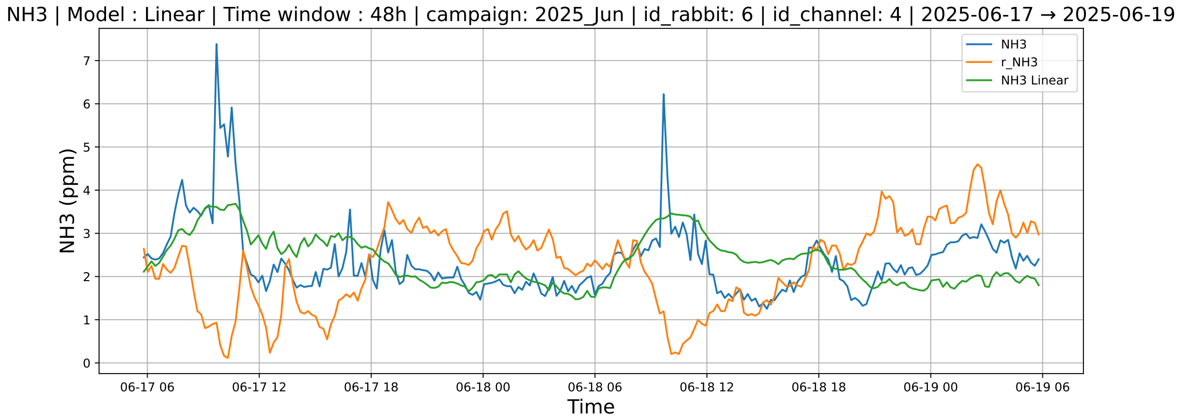
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

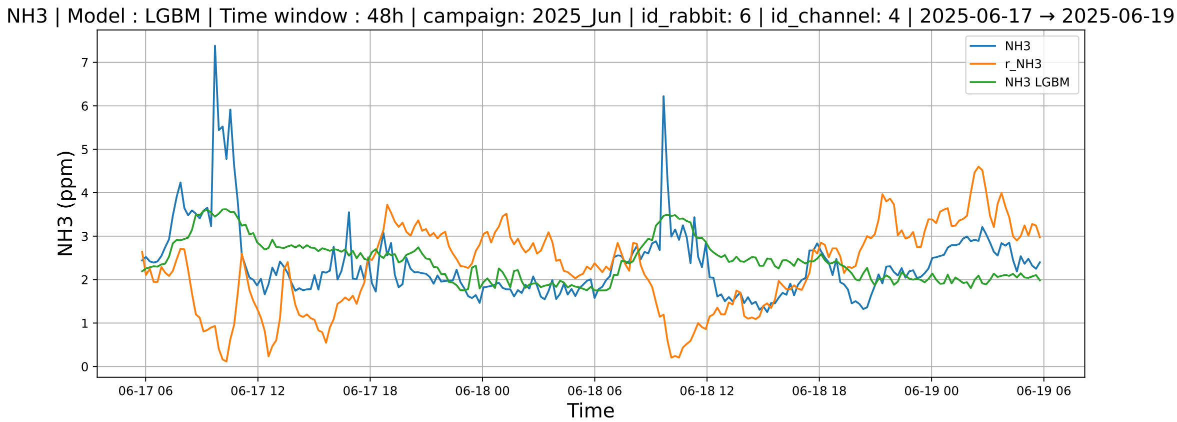


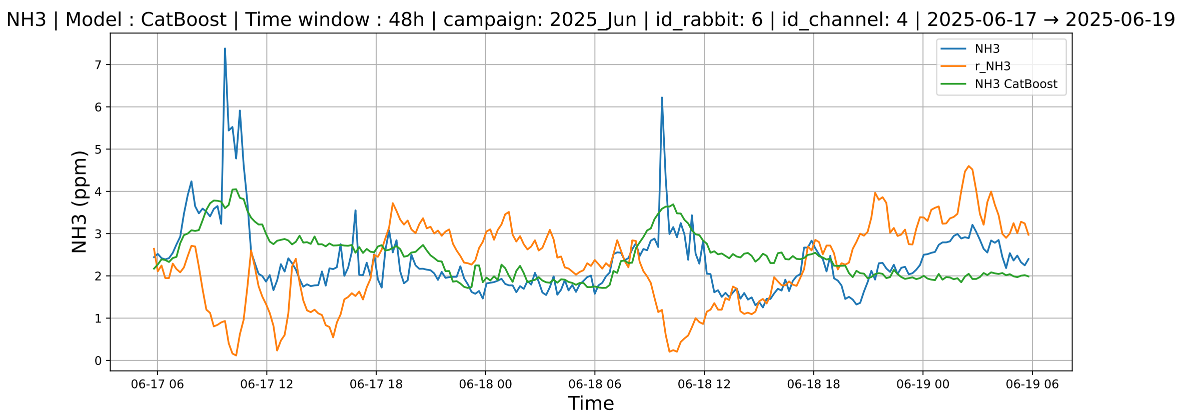
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



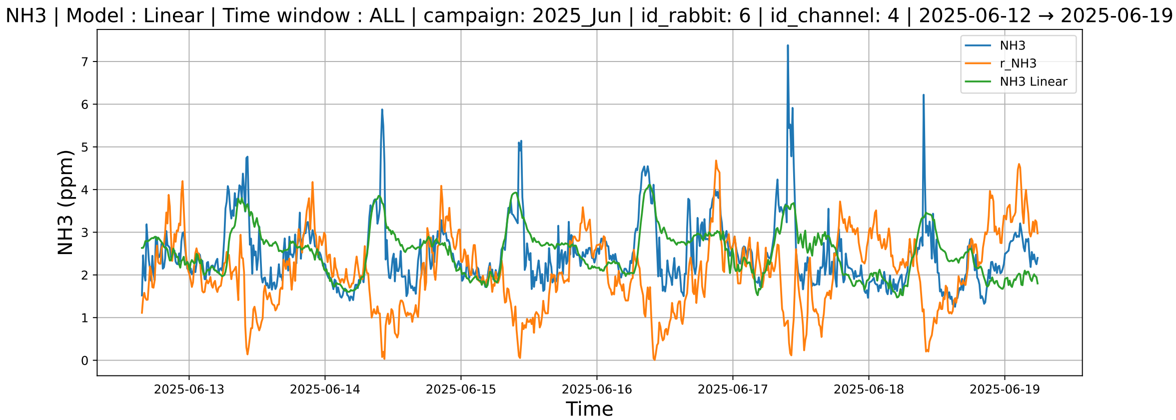
4.1.4.2 Time window : 48

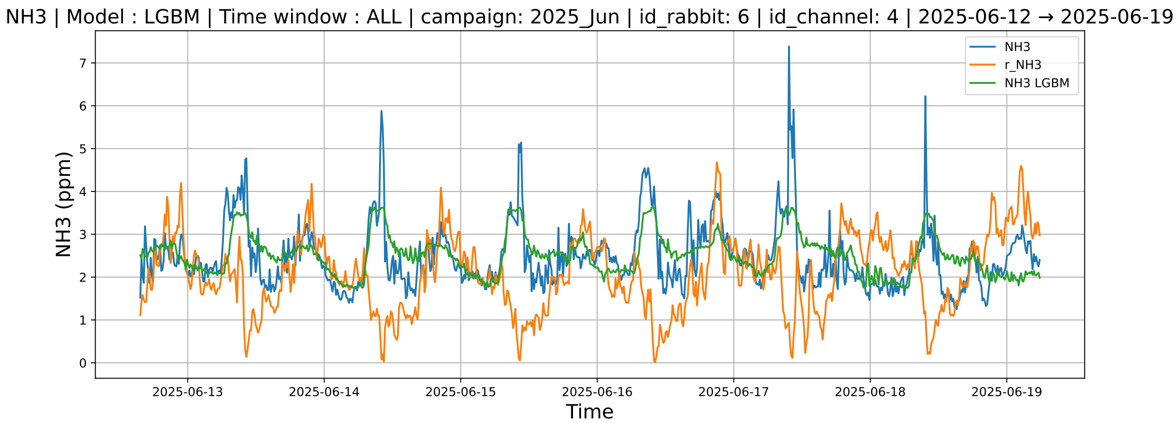


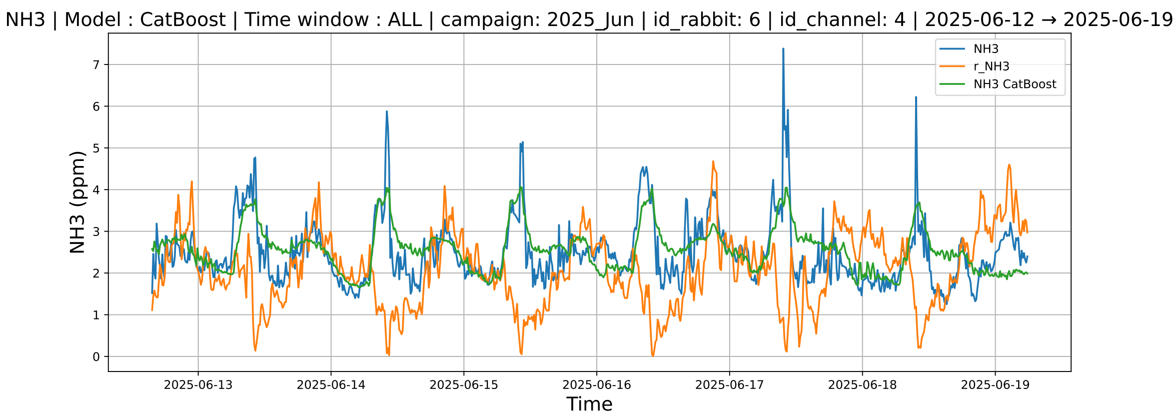




4.1.4.3 Time window : ALL



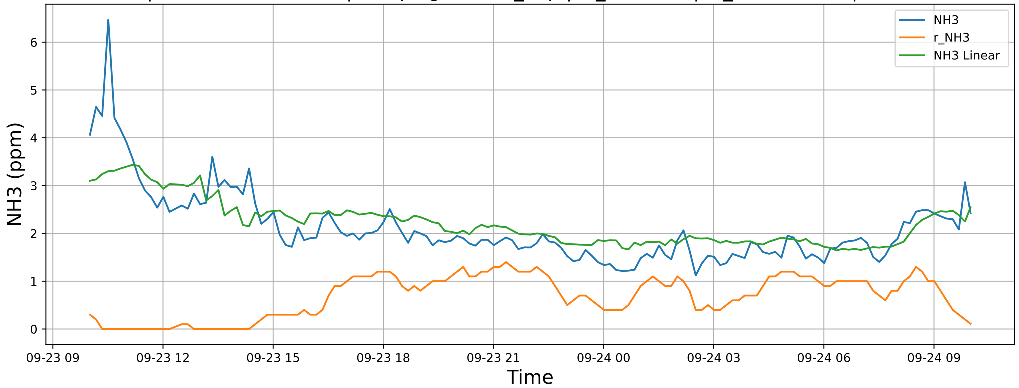




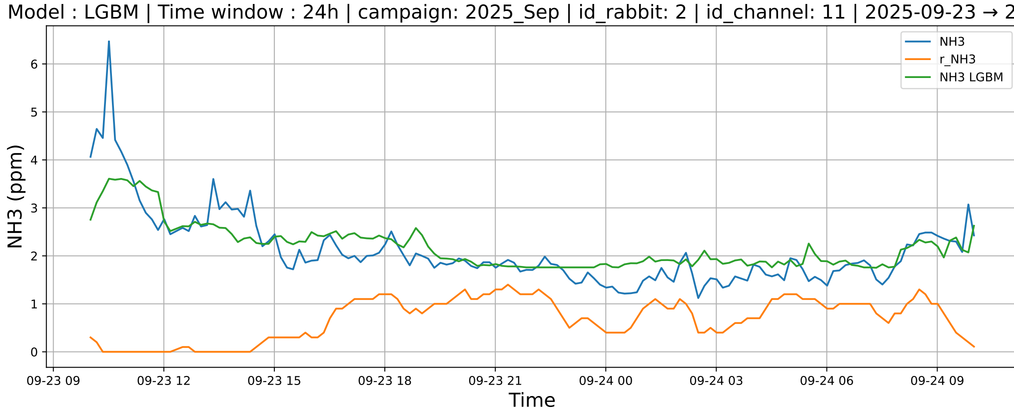
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

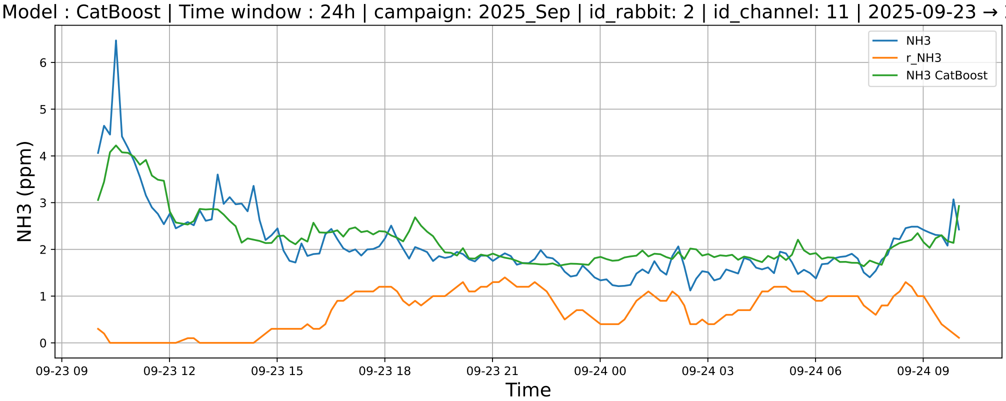
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



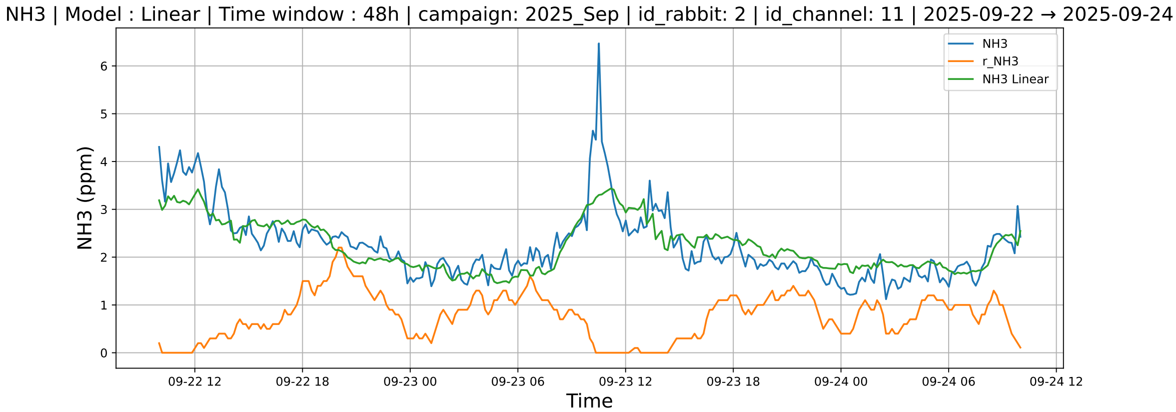
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

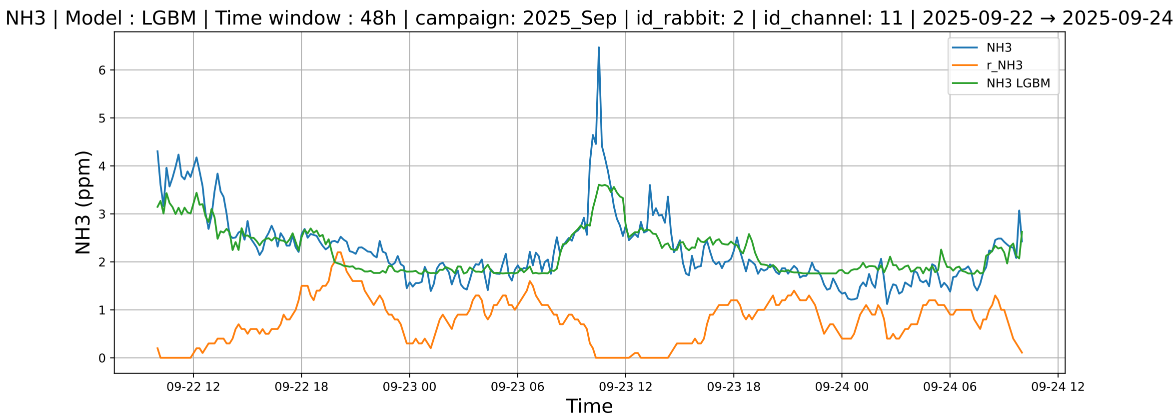


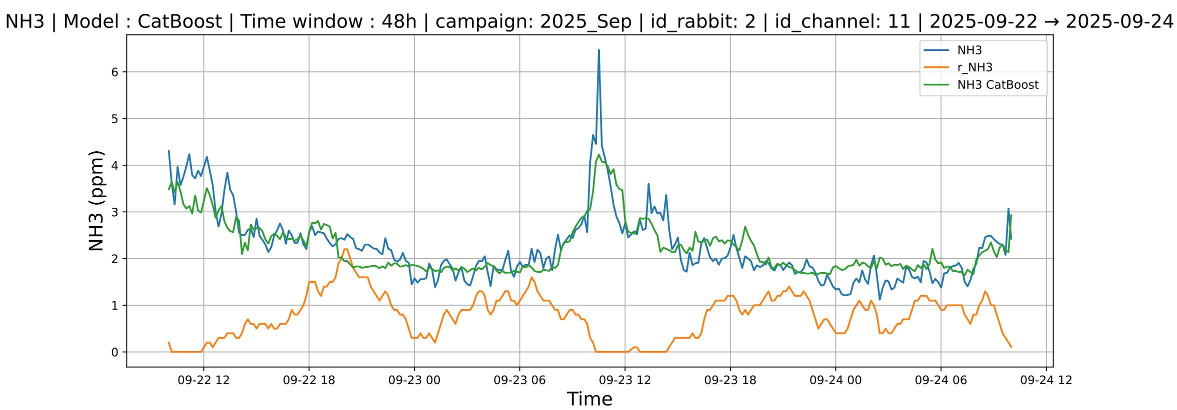
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



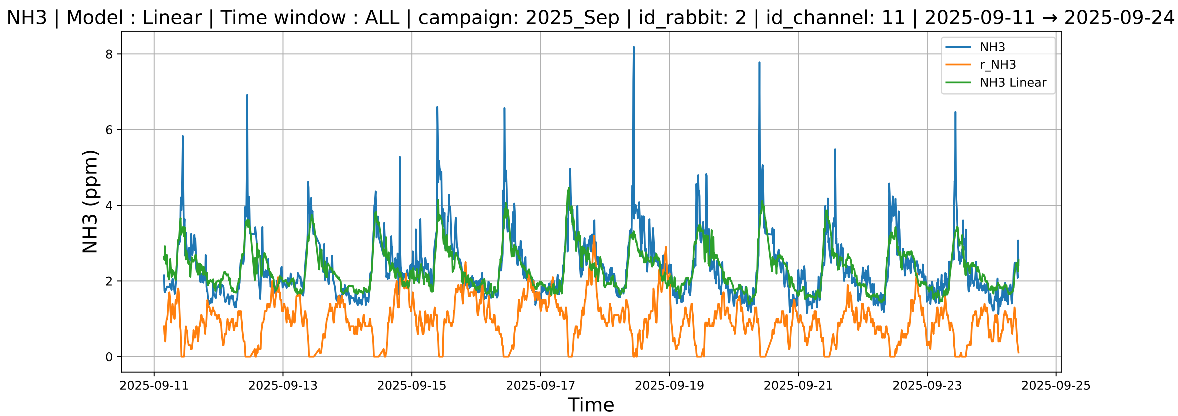
4.1.5.2 Time window : 48

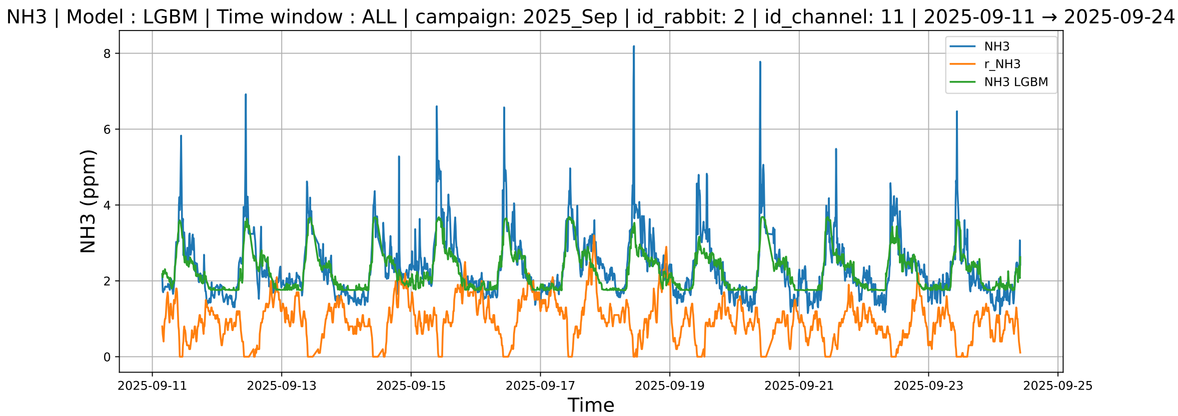


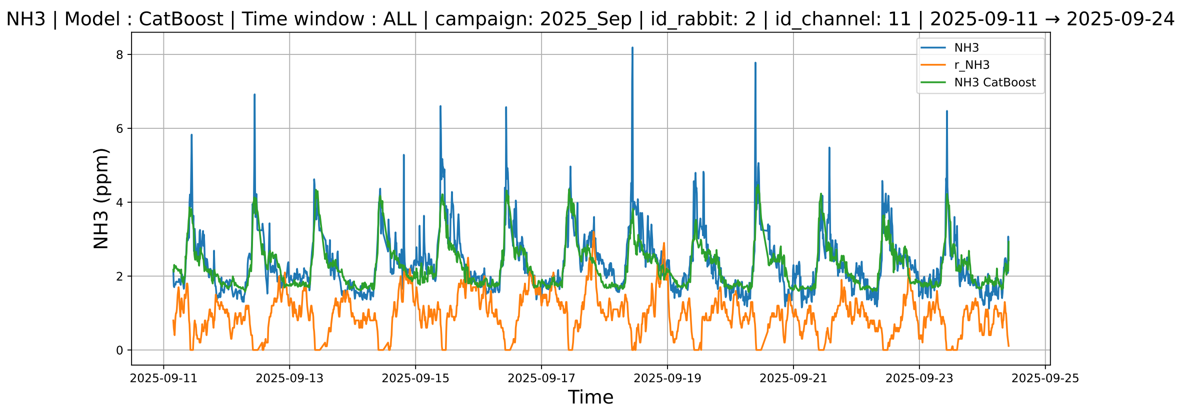




4.1.5.3 Time window : ALL



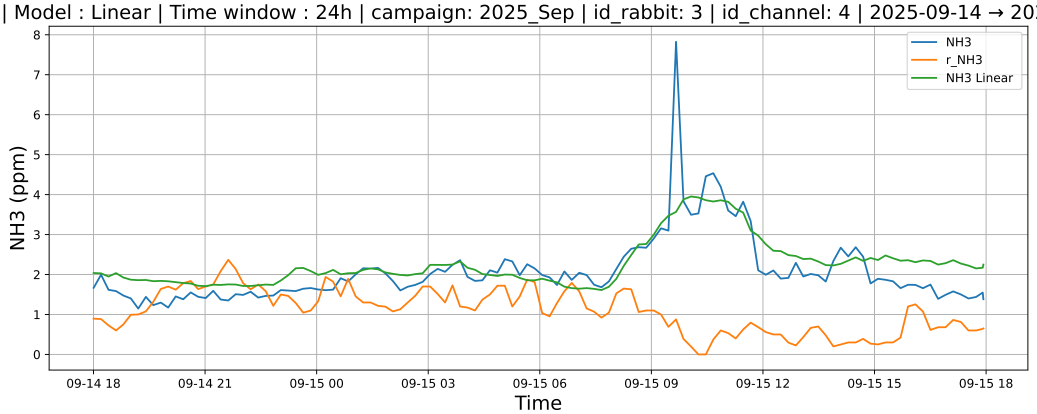




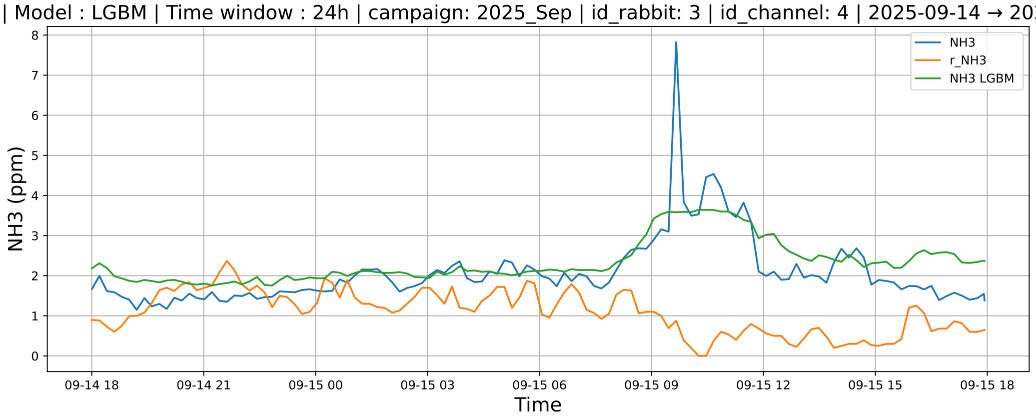
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

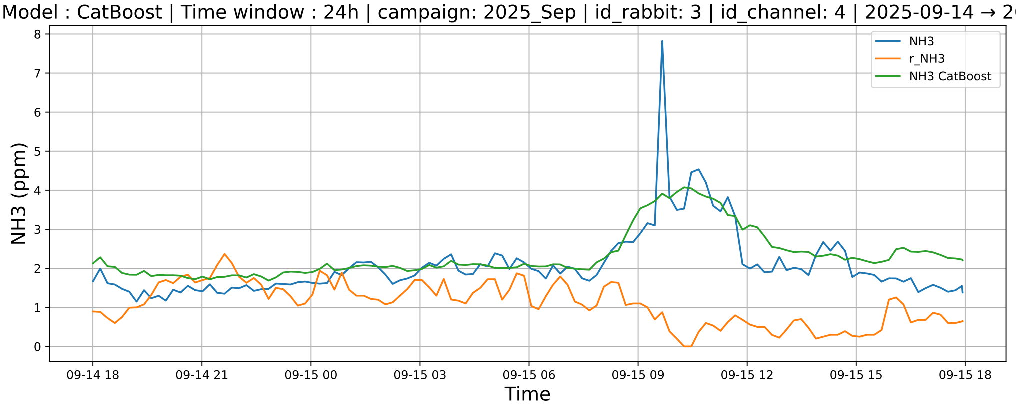
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

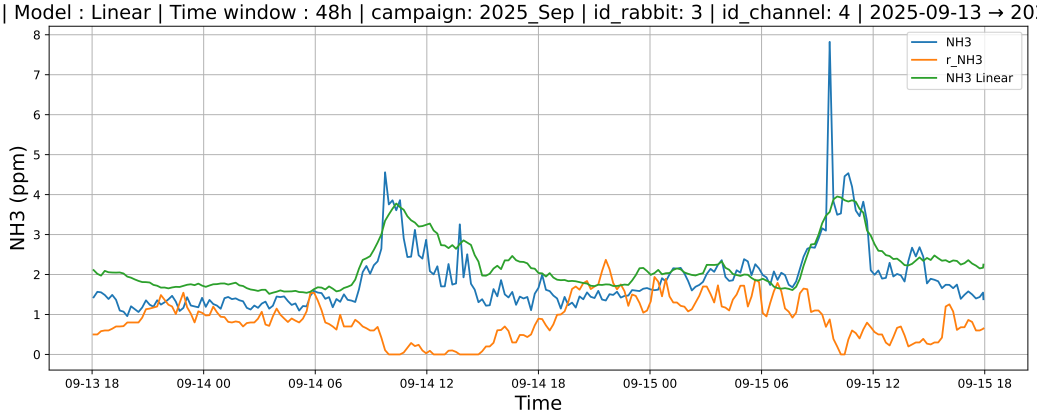


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

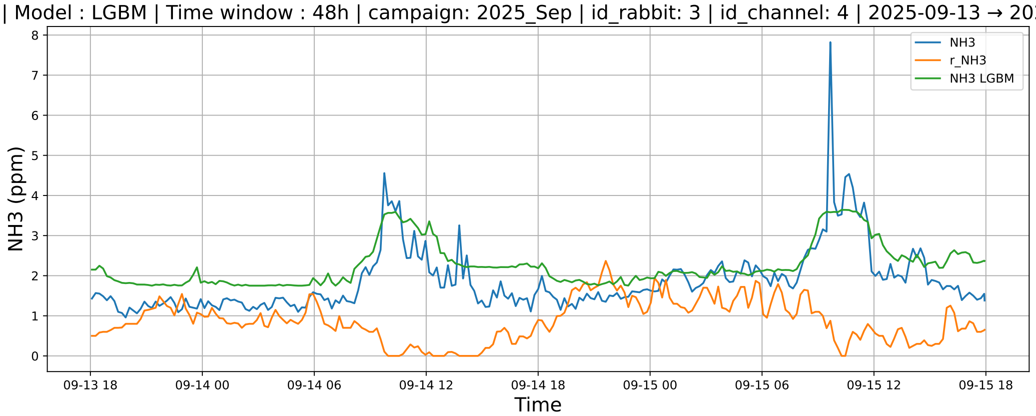


4.1.6.2 Time window : 48

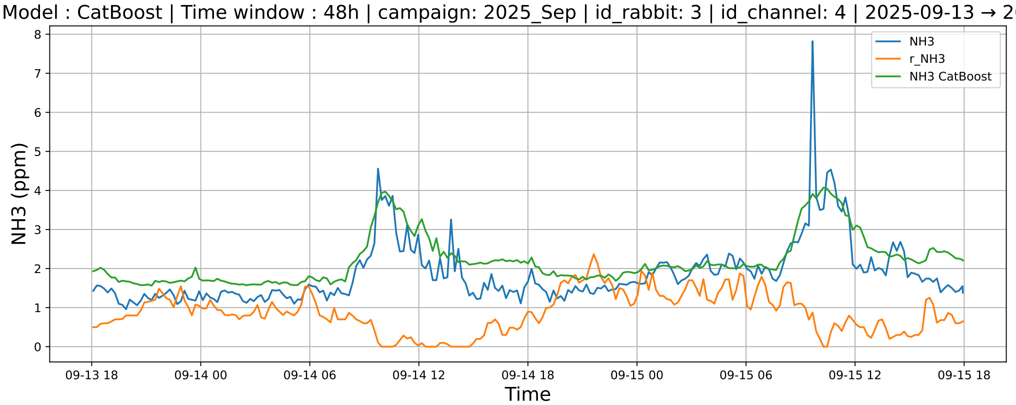
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



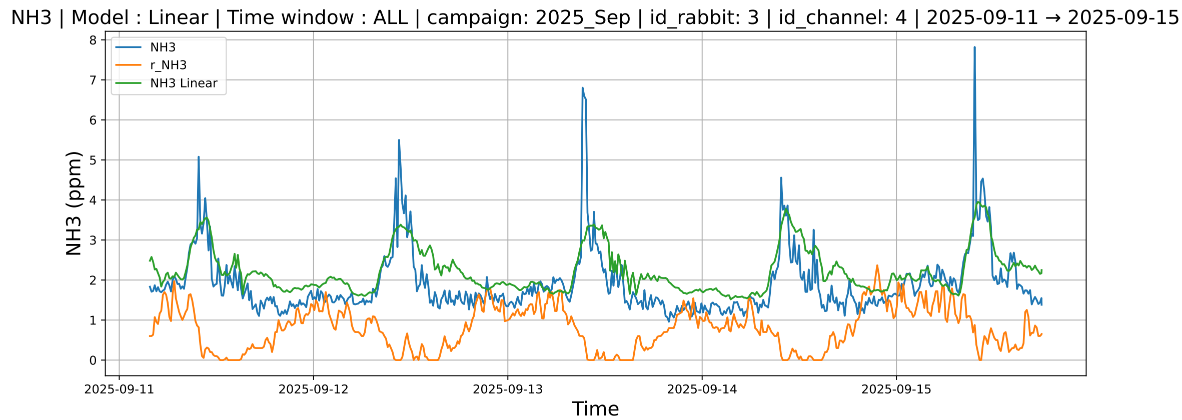
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

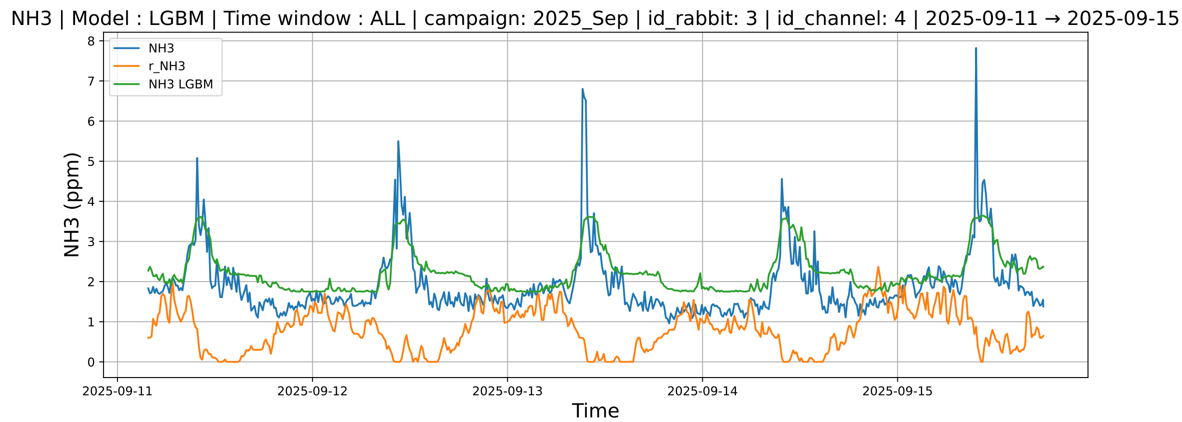


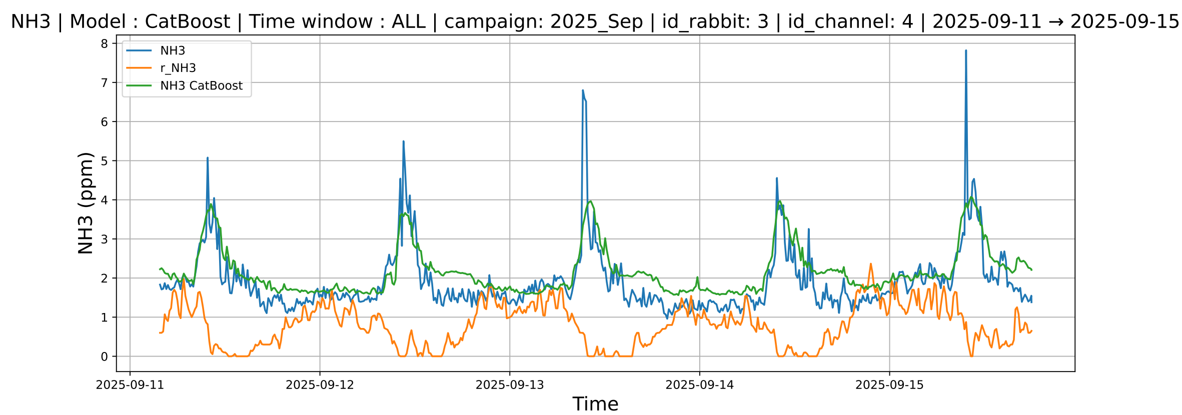
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.1.6.3 Time window : ALL

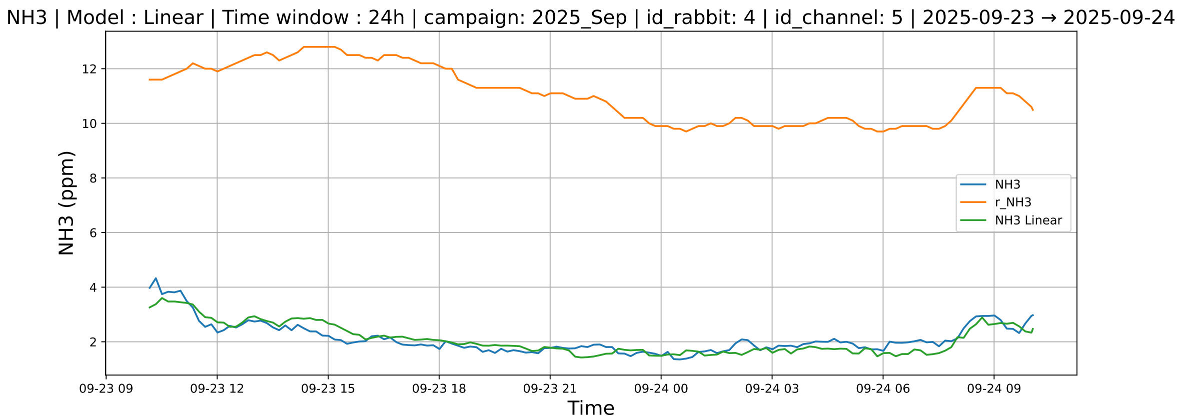


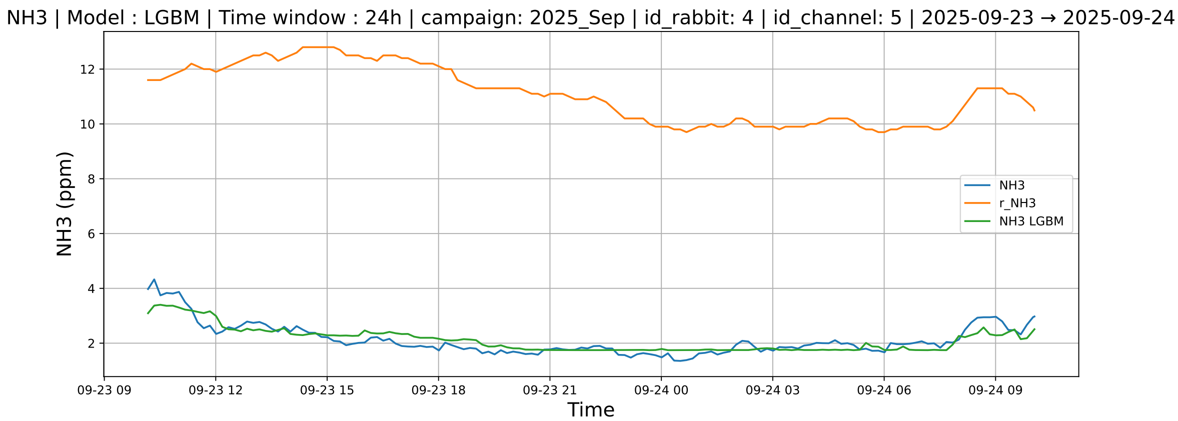


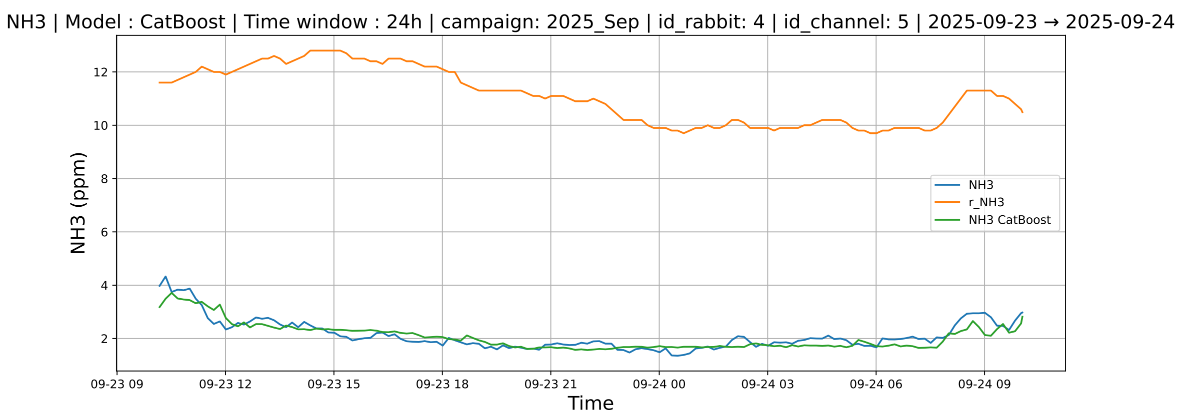


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

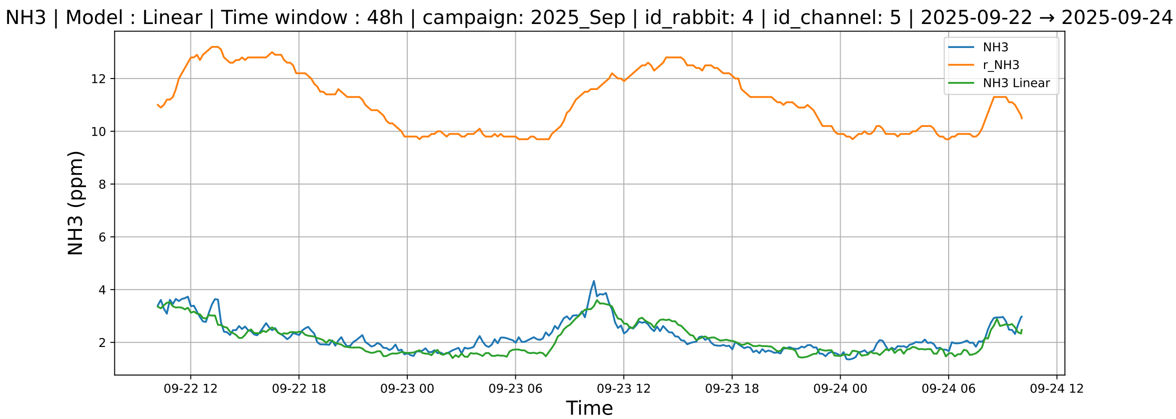
4.1.7.1 Time window : 24

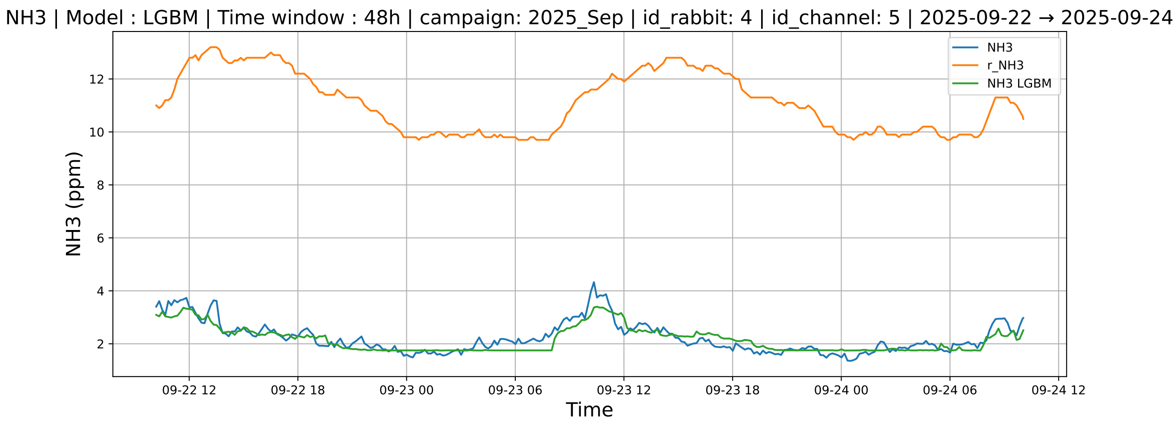


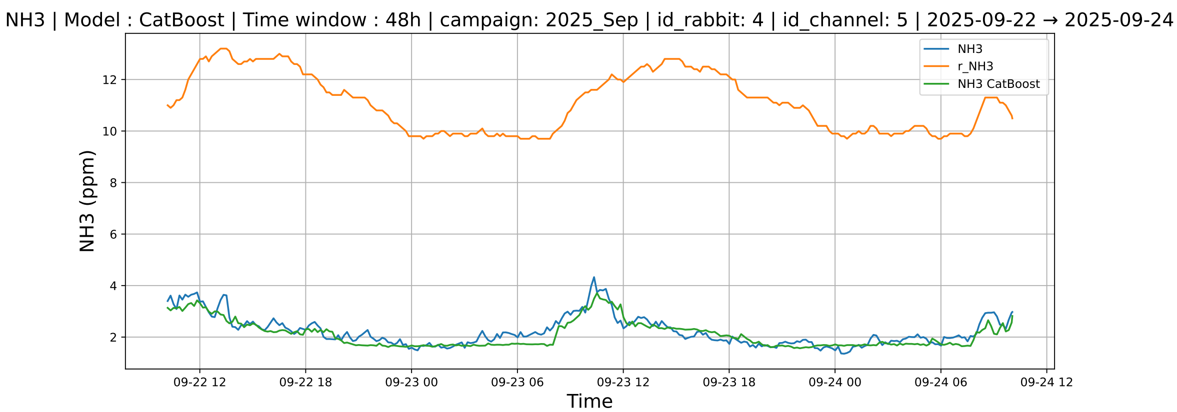




4.1.7.2 Time window : 48

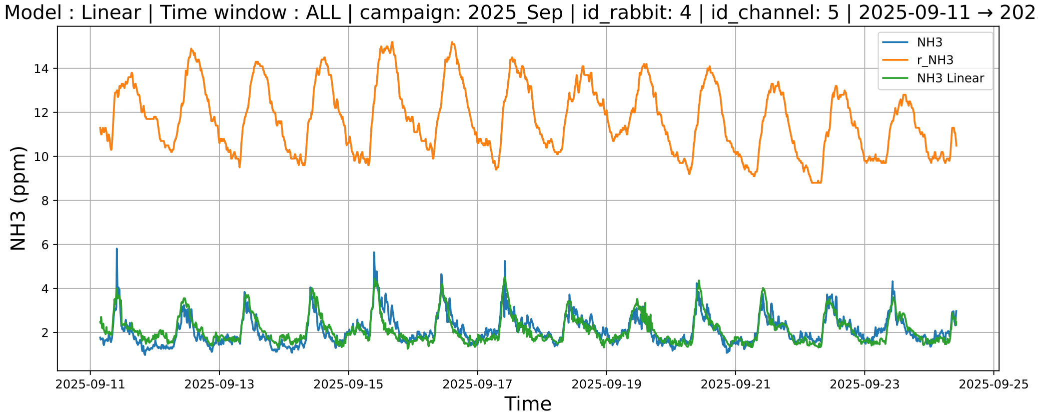




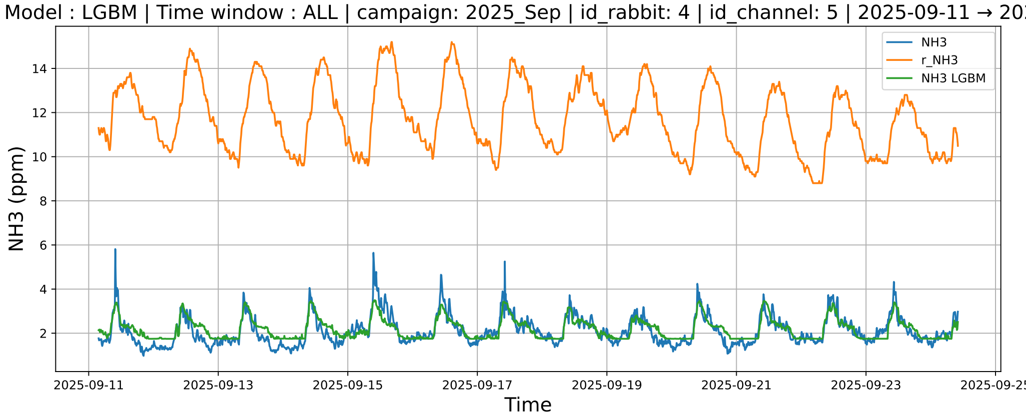


4.1.7.3 Time window : ALL

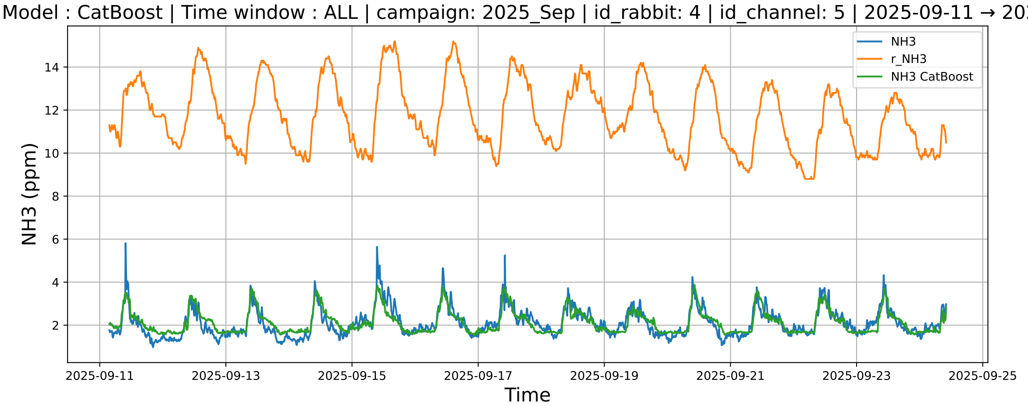
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



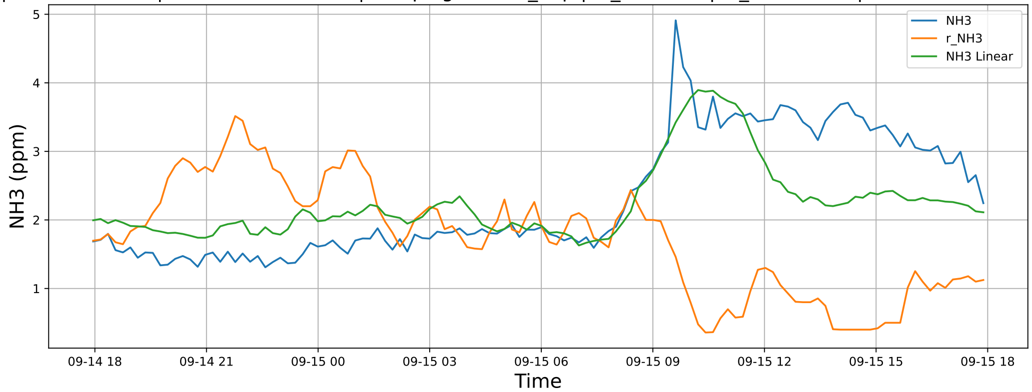
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



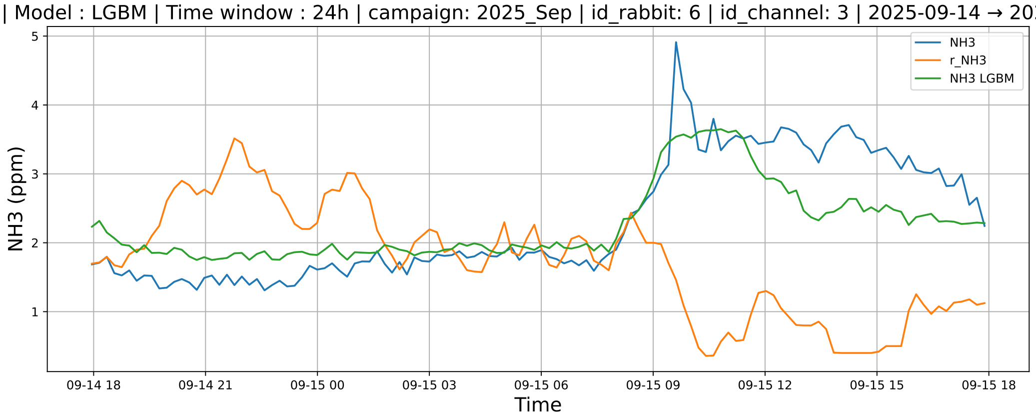
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

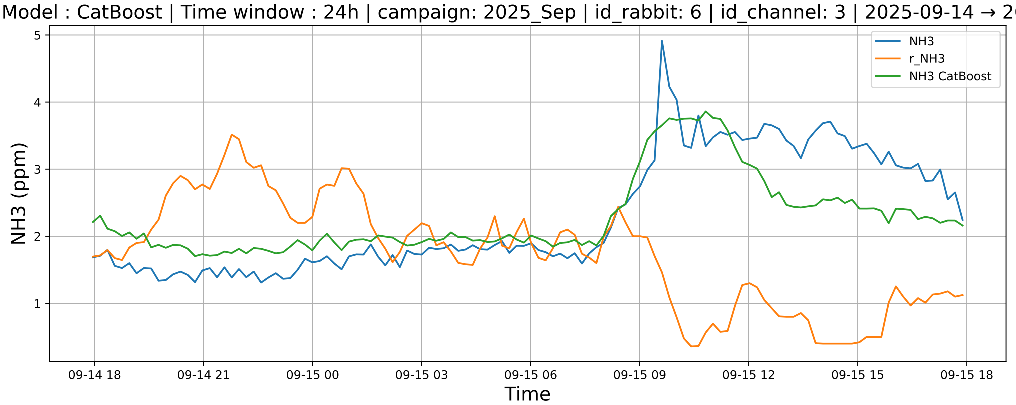
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



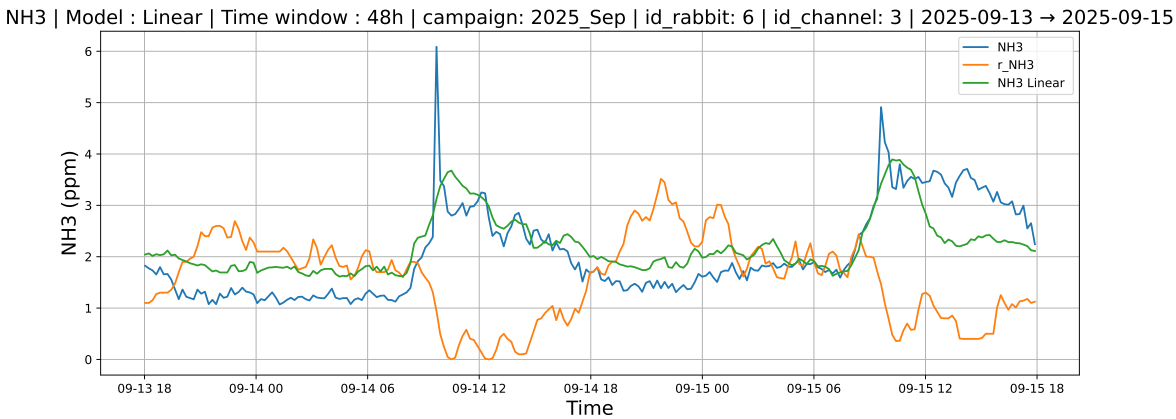
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

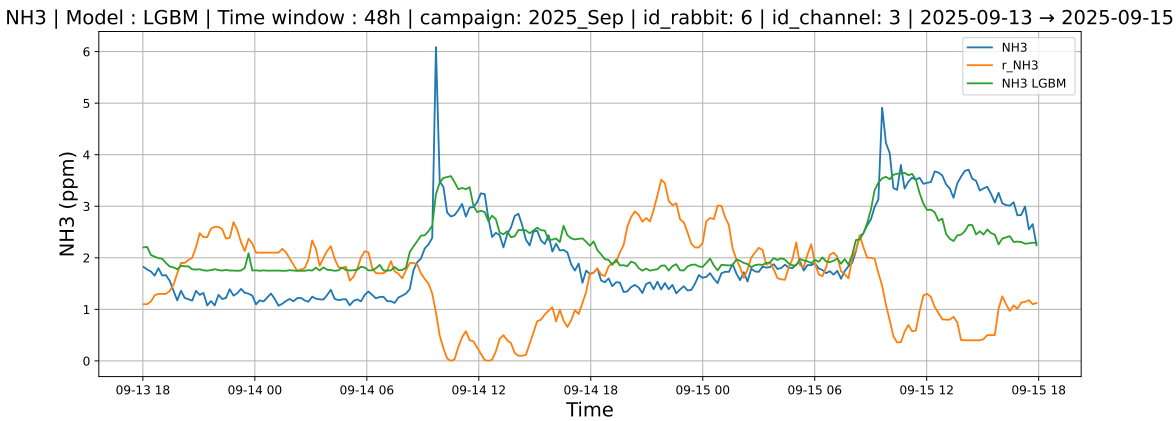


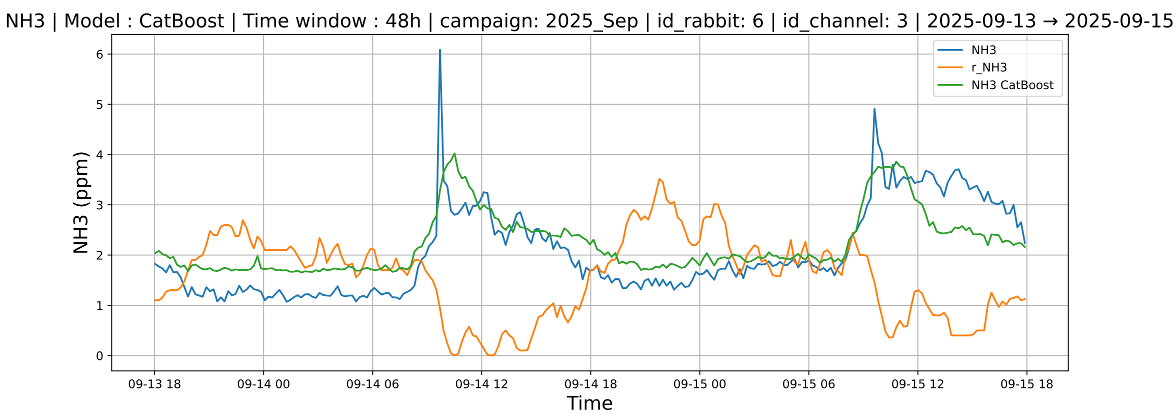
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.1.8.2 Time window : 48

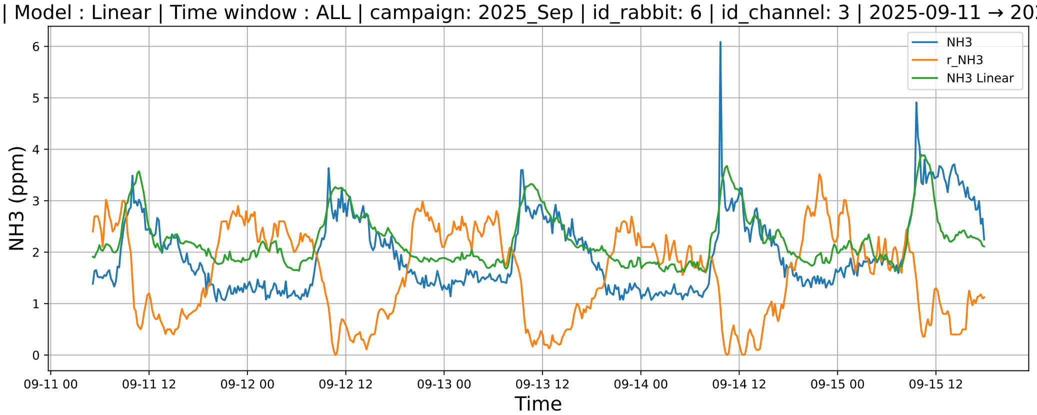




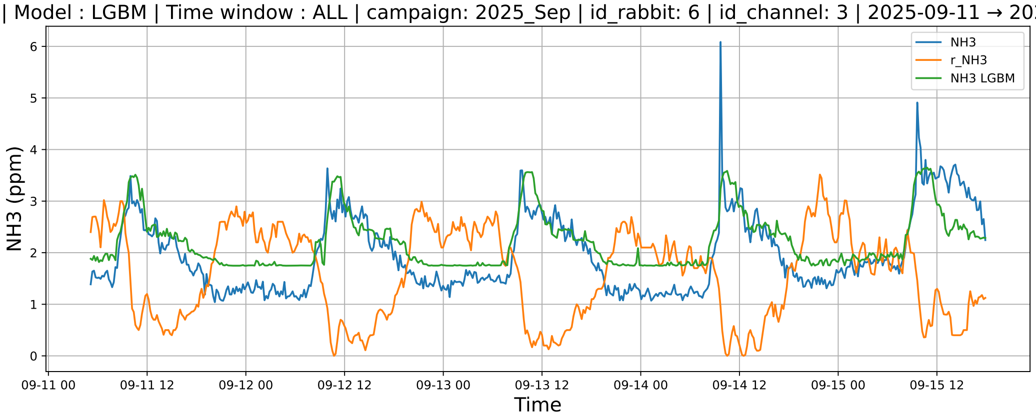


4.1.8.3 Time window : ALL

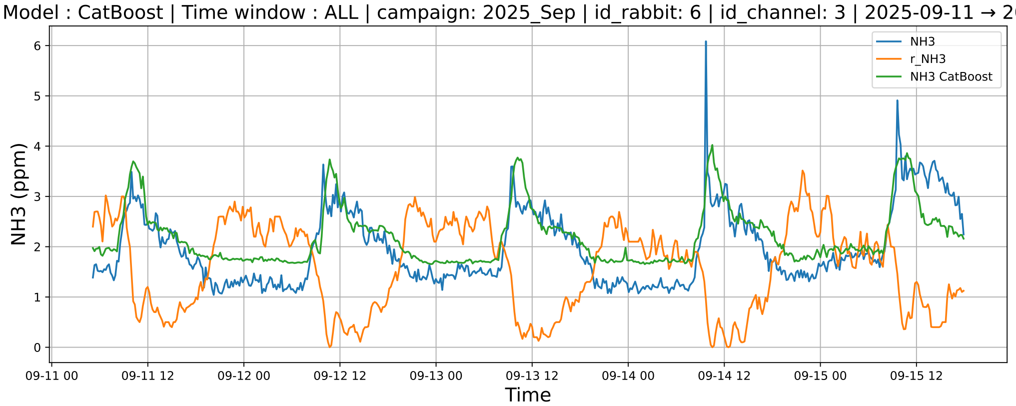
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



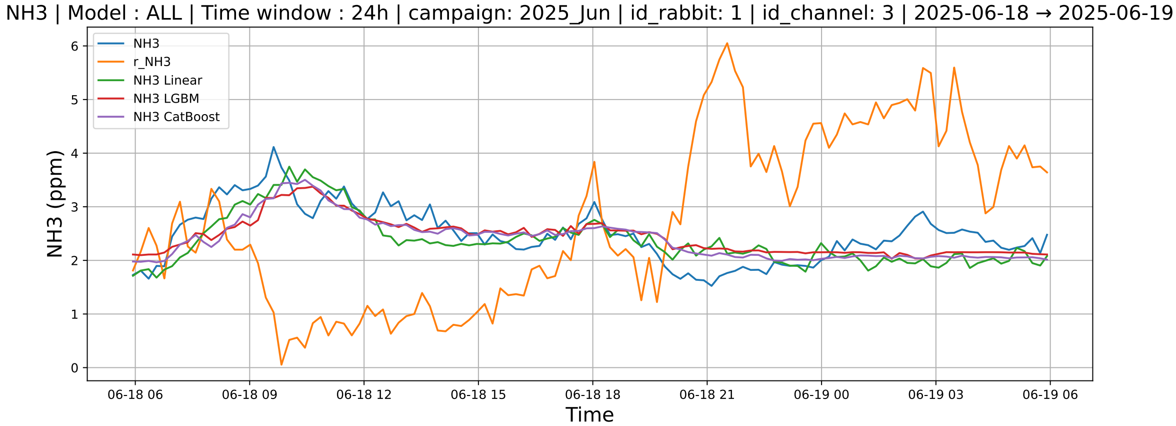
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



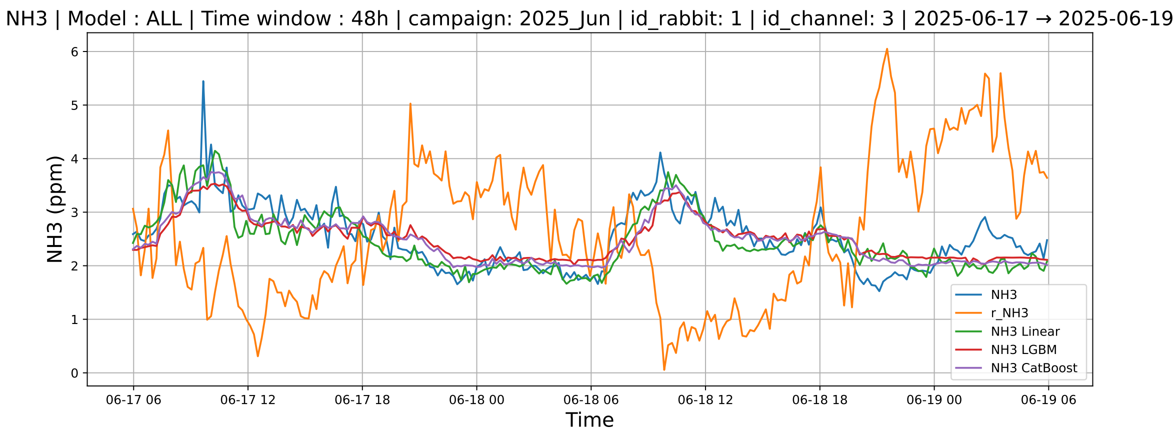
4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

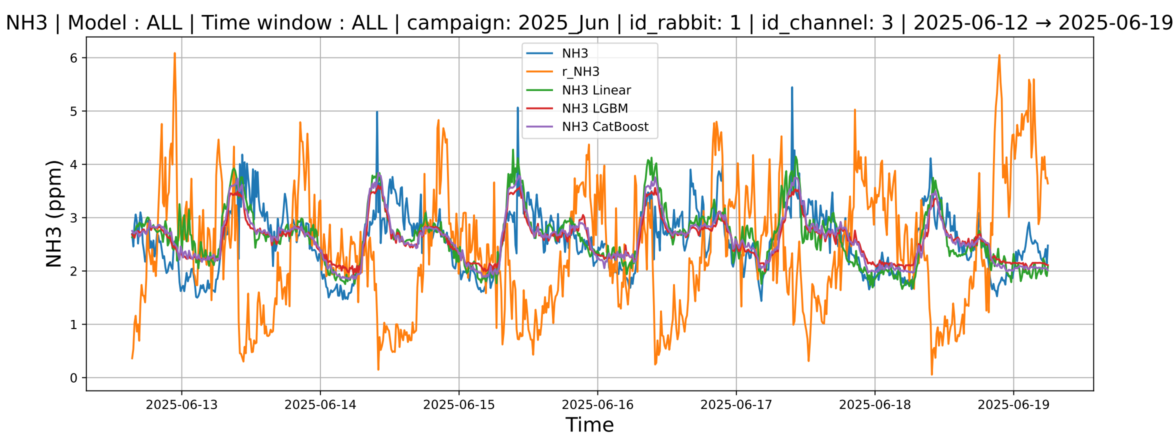
4.2.1.1 Time window : 24



4.2.1.2 Time window : 48

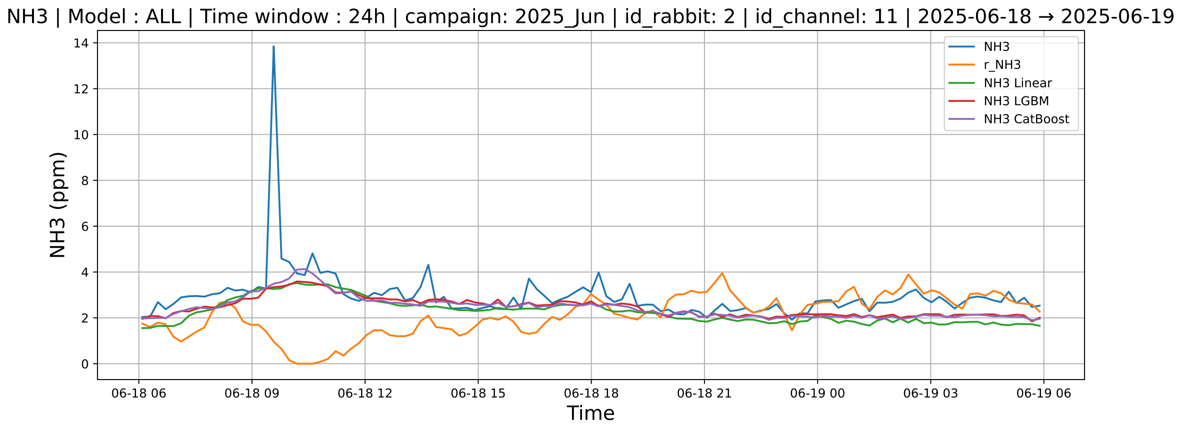


4.2.1.3 Time window : ALL

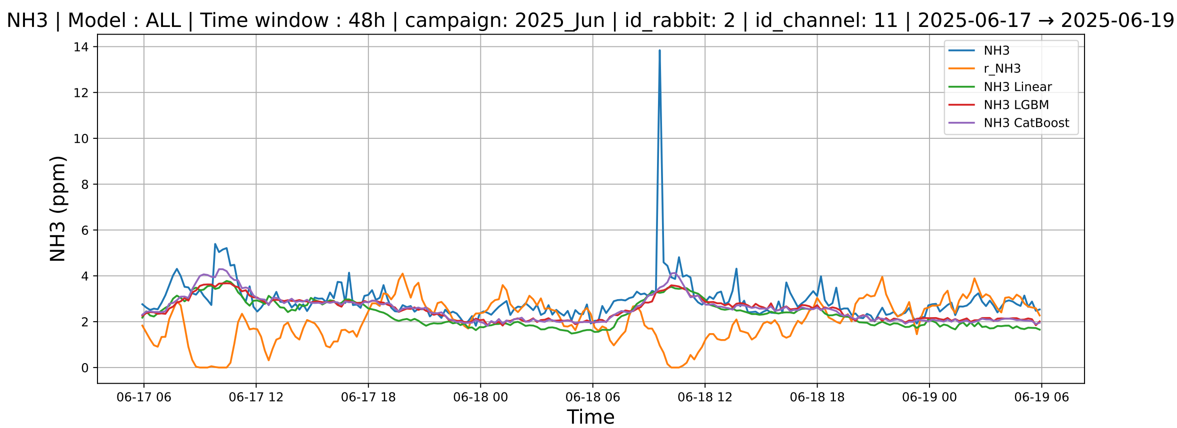


4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

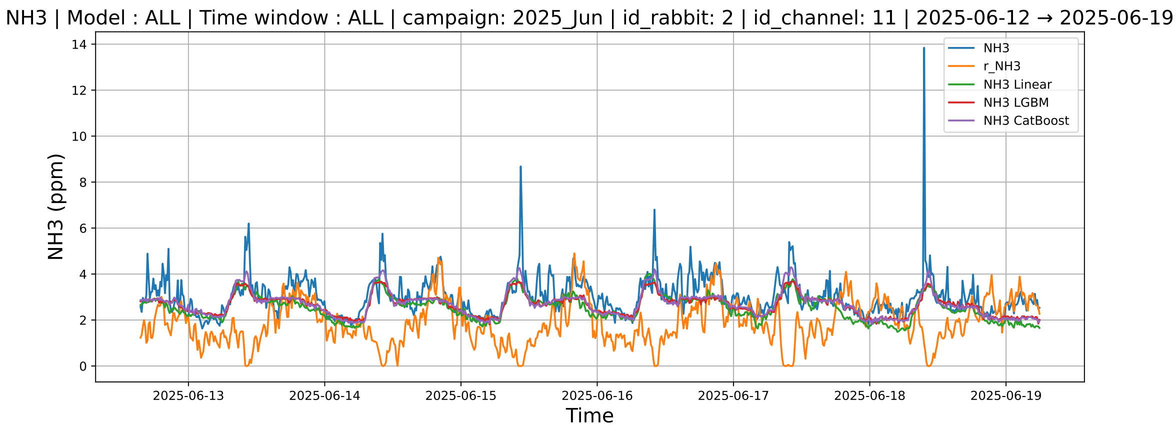
4.2.2.1 Time window : 24



4.2.2.2 Time window : 48

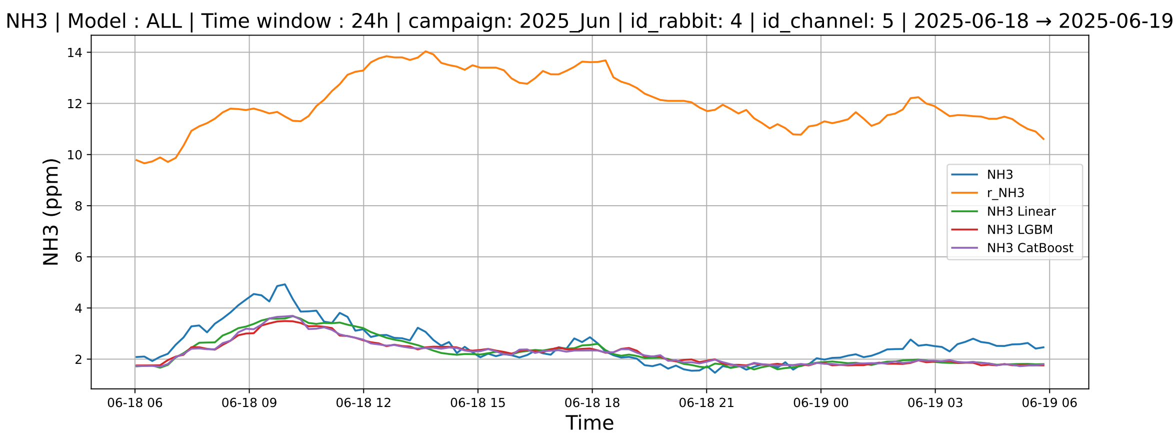


4.2.2.3 Time window : ALL



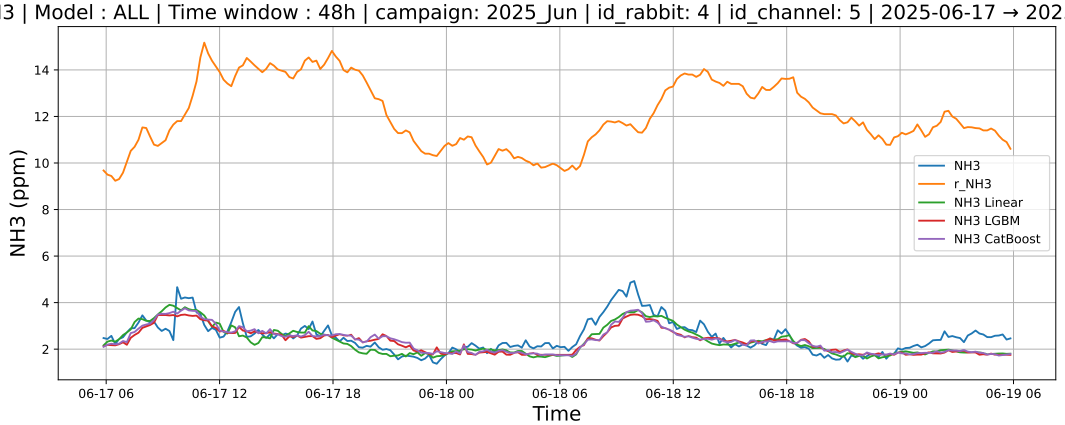
4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.2.3.1 Time window : 24



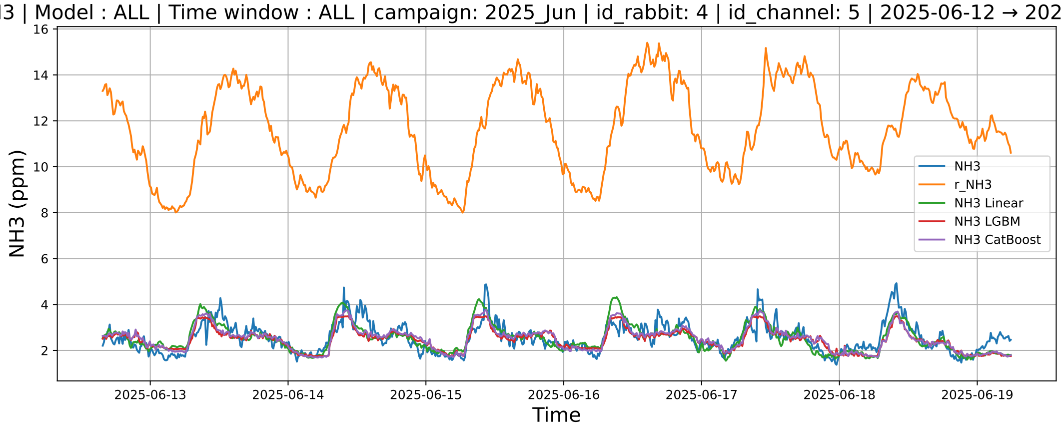
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



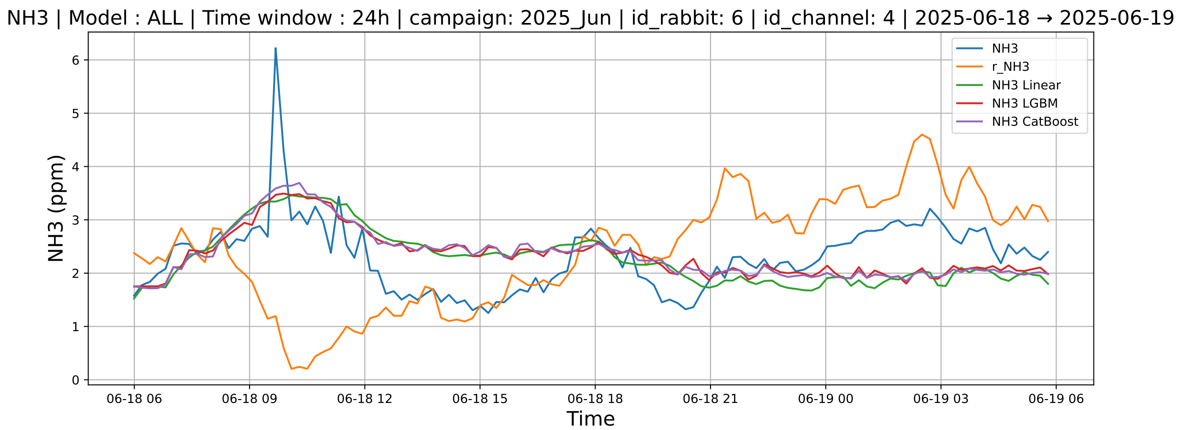
4.2.3.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

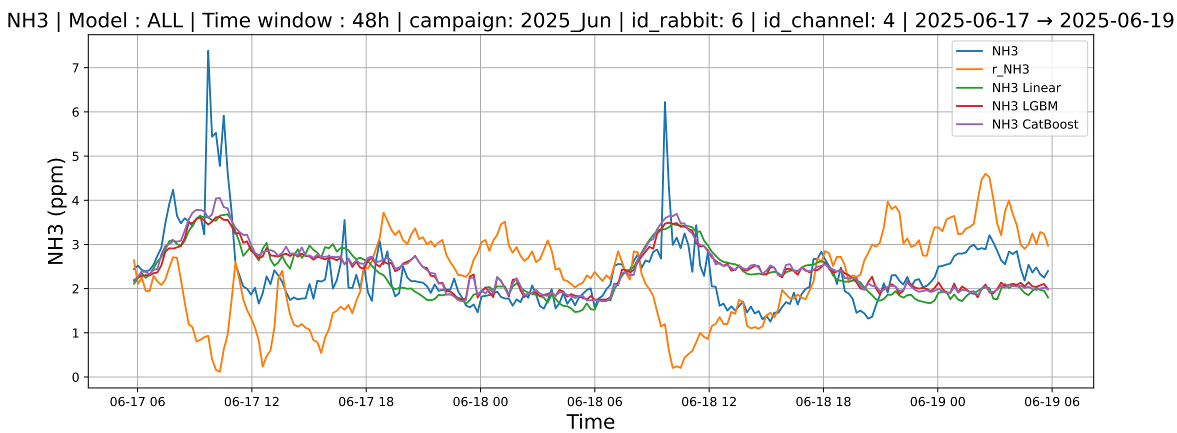


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

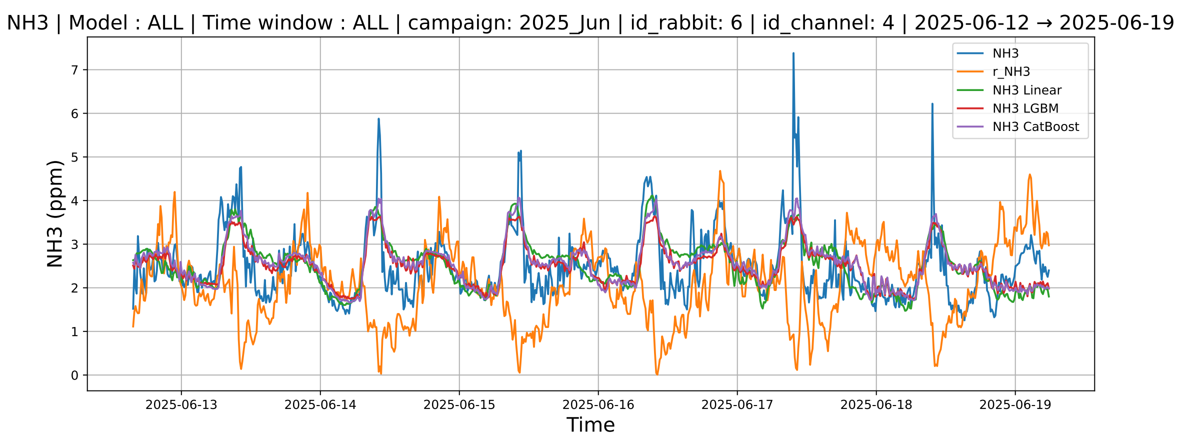
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



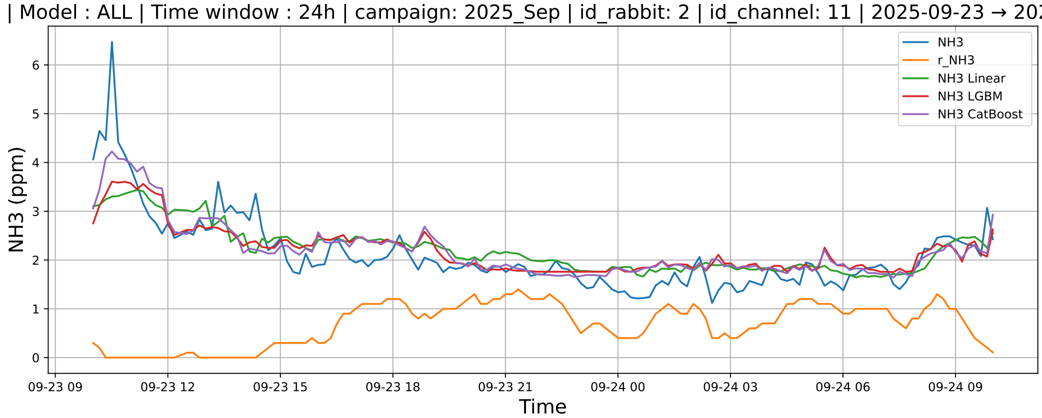
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

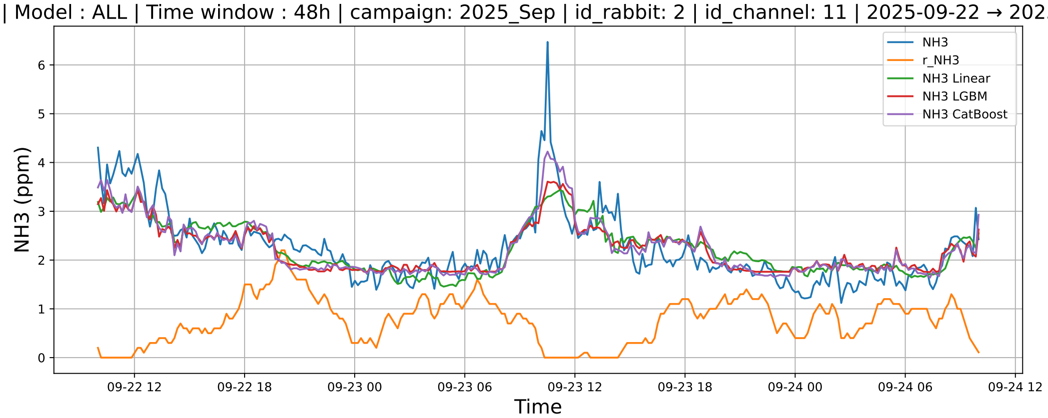
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

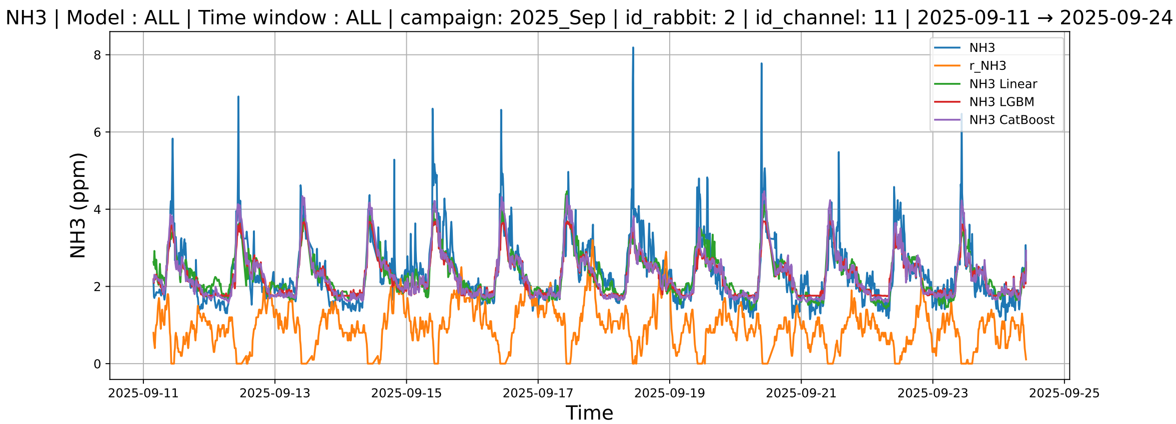


4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

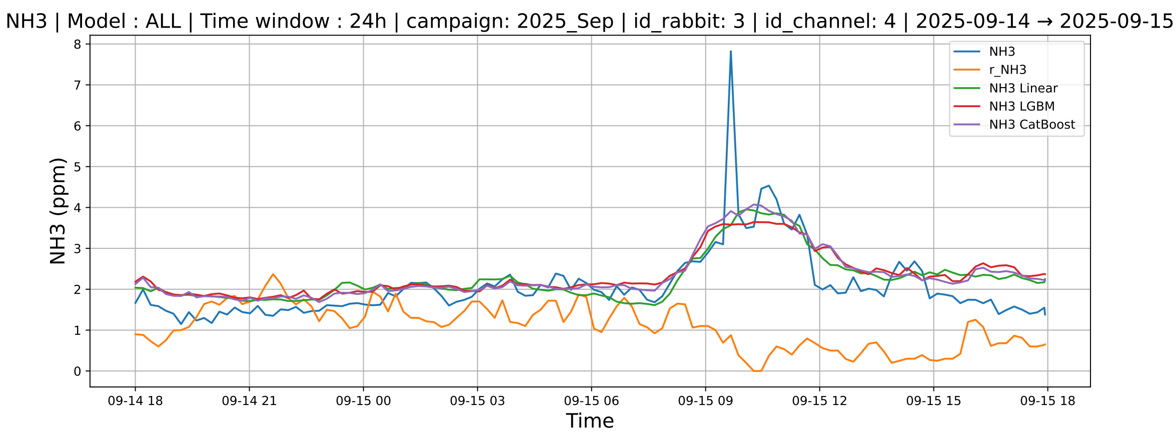


4.2.5.3 Time window : ALL

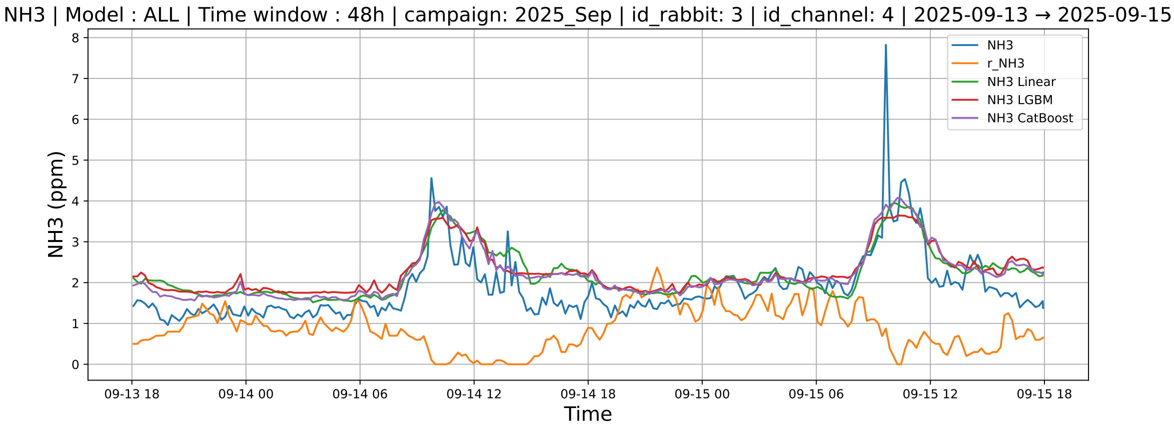


4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

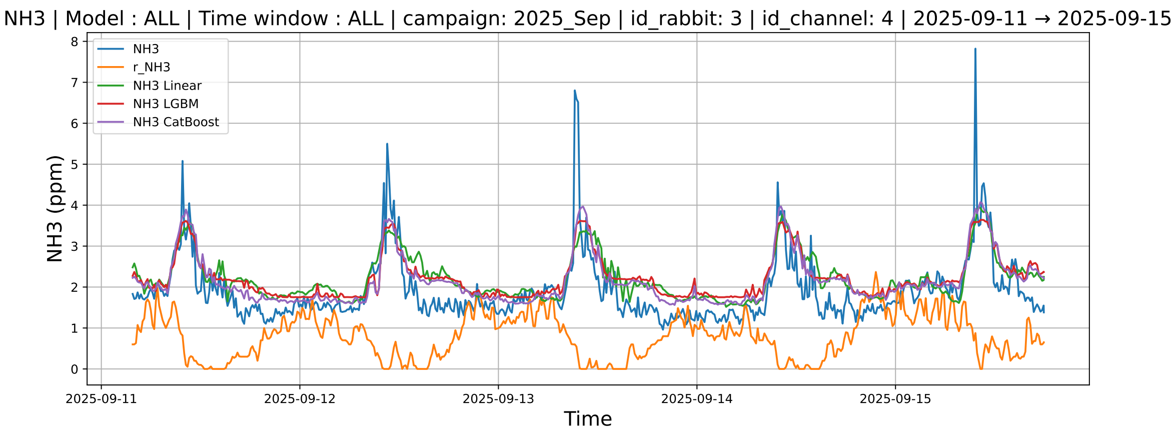
4.2.6.1 Time window : 24



4.2.6.2 Time window : 48



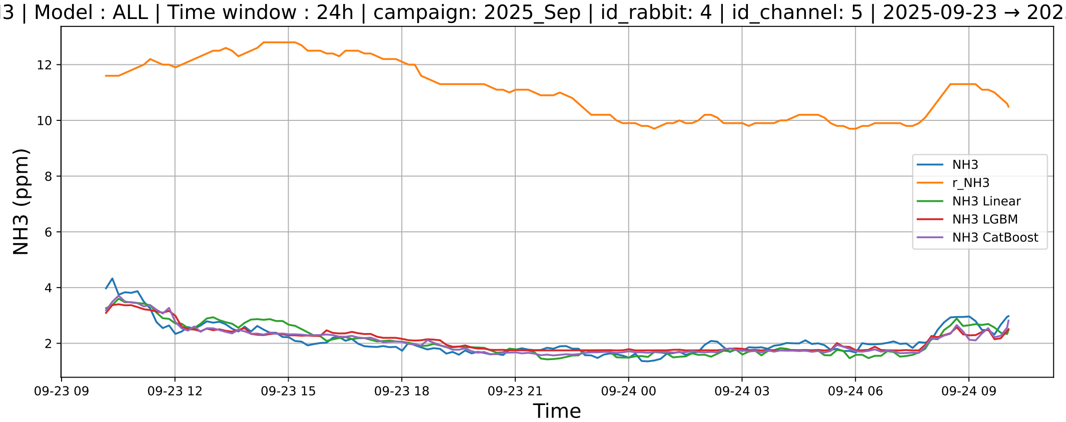
4.2.6.3 Time window : ALL



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

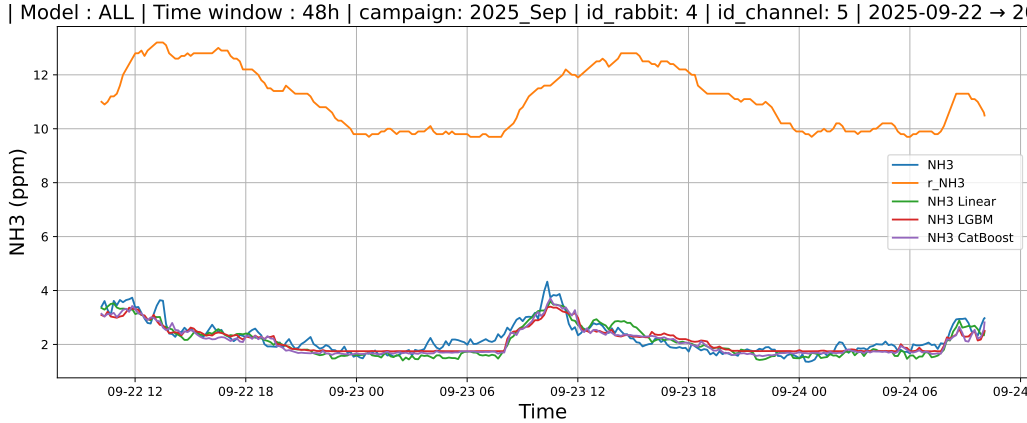
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



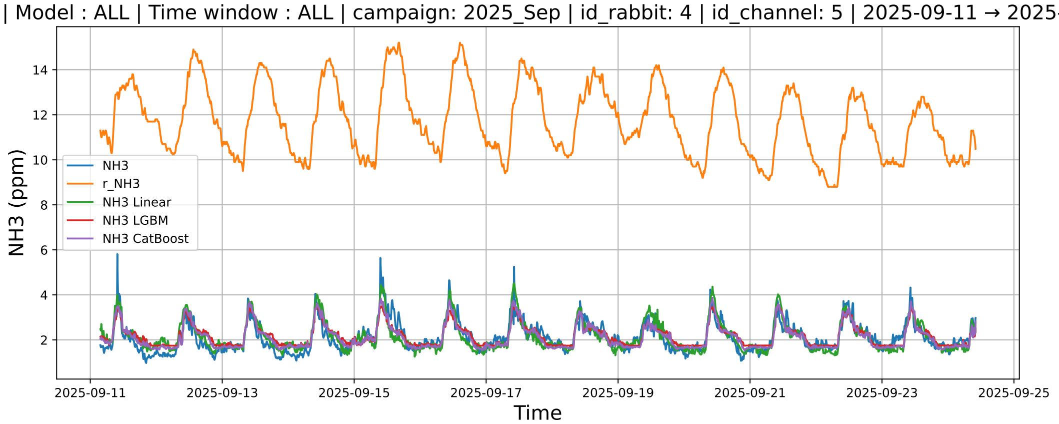
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



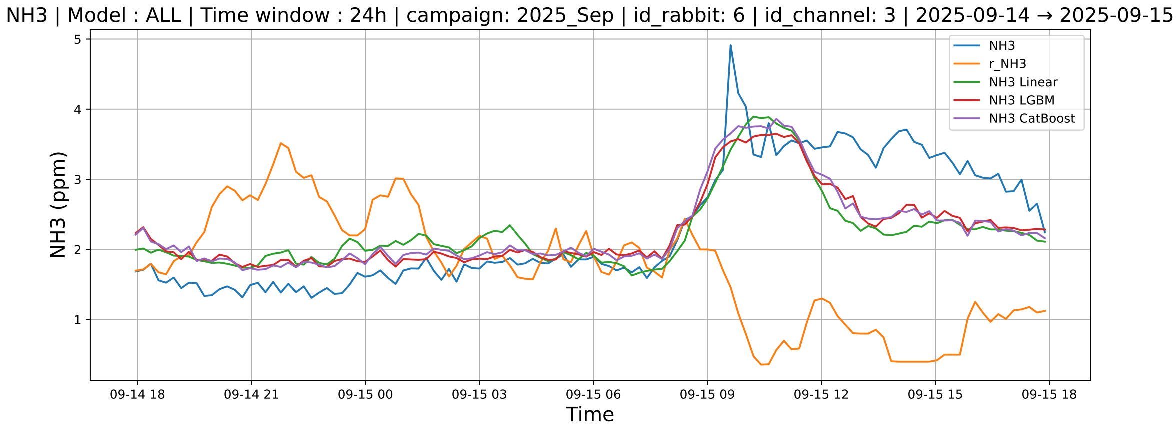
4.2.7.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

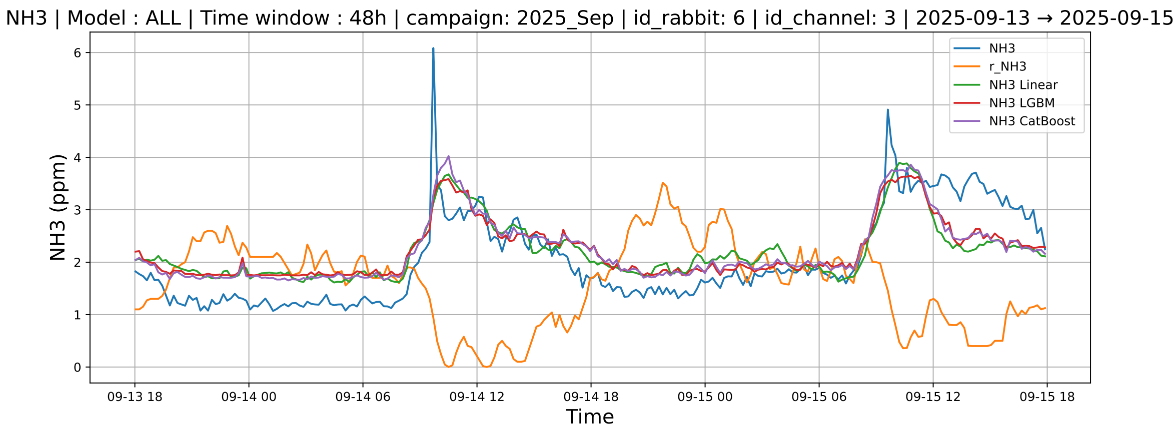


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

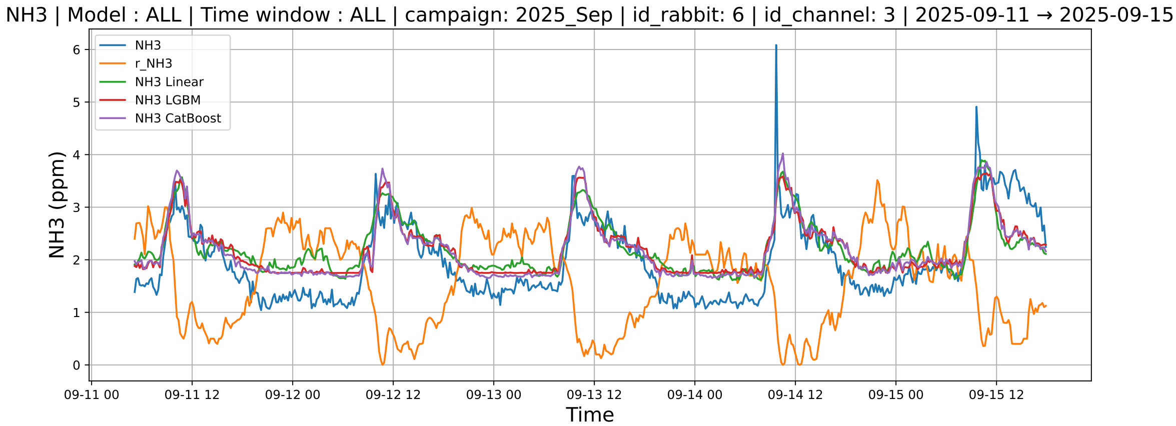
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
Time
09-11 00
09-11 12
09-12 00
09-12 12
09-13 00
09-13 12
09-14 00
09-14 12
09-15 00
09-15 12